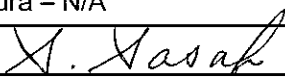
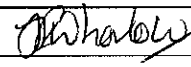
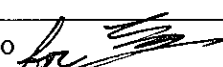
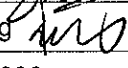


**DUBAI RAPID LINK CONSORTIUM
DUBAI METRO PROJECT OFFICE**

Contract No.: DM001		CDRL No.:	
Project Title: DUBAI METRO		CDRL Title:	
Document Title: Structural Design Calculation & Drawings Main Power Substation MPS1 DCP3 – SUBMISSION			
Revision History			
A5			
A4			
A3			
A2	09 July'06	Second Issue	Signed below
A1	13 May'06	First Issue	Signed below
MARK	DATE	DESCRIPTION	RAIL
APPROVED		Project Director	S. Shimamura – N/A
		Deputy Project Director	S. Sasaki 
		Checked By (QA/QC Manager)	J Whorlow 
		Checked By (Safety Manager)	N/A
Checked By	N/A	Checked By (Project Manager)	N/A
Checked By	N/A	Checked By (Engineering Manager)	M. Makino 
Prepared By	N/A	Issued By	James Ng 
DATE	-	DATE	09 July 2006
RAIL SYSTEM		CIVIL JV	
CONTRACTOR'S DOCUMENT No.:		DOCUMENT No.:	REVISION
		DM001-E-MP1-ISW-DR-DCC-309502	A2

MPS1

SUPPLEMENTARY CALCULATIONS

- | | |
|--------|---|
| Item 1 | Design of Concrete Staircase |
| Item 2 | Analysis & Design of Roof Truss for Main Transformer Room |
| Item 3 | Analysis & Design of Overhead Crane Girder 5Tons Capacity |
| Item 4 | Analysis & Design of Columns & Walls |
| Item 5 | Foundation Analysis & Design |
| Item 6 | Seismic Analysis |

STAIRCASE

DESIGN OF STAIRCASE SLAB

JOB No. : 1225	DATE : March 15, 2006
STRUCTURE : Typ. Flight : GF - FF	Design ref. : B S 8110 : Part 1 : 1985 &
SLAB REFERENCE	B S 6399 : Part 1 : 1984

Fcu 35 N/mm² Fy 415 N/mm² Cover 30 mm

Length of Flight (Length in Slope)	4.66 m	End Condition (C=1, D=2)
Width of Flight	1.20 m	LHS 2
Thickness of Waist slab (D)	200 mm	RHS 1
Finishes with Concrete steps	4.00 kn/m ²	
Imposed load	5.00 kn/m ²	

Load calculation

Dead load (Gk)	
Self weight of waist slab	5.00 kn/m
Finishes with Concrete steps	4.00 kn/m
Total	9.00 kn/m
Live load (Qk)	5.00 kn/m
Ultimate load (Pu) 1.4*Gk + 1.6*Qk	20.60 kn/m

Bending Moment and Shear force

Maximum +ve bending moment	0.1*Pu*Lx ²	44.73 knm
Maximum -ve bending moment	0.111*Pu*Lx ²	49.65 knm
Maximum shear force	0.6*Pu*Lx	57.60 kn

Area of Steel

Effective depth 170 mm

$$A_{st} = M \cdot 10^6 / (0.87 \cdot F_y \cdot Z)$$

	K	Z	Ast	Provide				Ast _{pro}	
		mm	mm ²					mm ²	
Span	0.044	161.2	769	Y	16	@	175 mm c/c	1149	OK
Support	0.049	160.2	859	Y	16	@	175 mm c/c	1149	OK
Span - Effective Depth Ratio				Limit	33.6	Actual	27.4	33.6 > 27.4	Safe

Shear check v = 0.34 N/mm² vc = 0.70 N/mm² 0.34 < 0.70 **Safe**

Distribution steel (0.13%)

260 mm²

Provide

	Ast _{pro}	
	mm ²	
Y 10 mm @ 150 c/c	524	OK

Programme created by Eng. M. I. SADIQ., B.E (Struct.)

DESIGN OF STAIRCASE SLAB

JOB No. : 1225

DATE : March 15, 2006

STRUCTURE :

Typ. Flight : BM - GF

Design ref. : B S 8110 : Part 1 : 1985 &

SLAB REFERENCE

B S 6399 : Part 1 : 1984

Fcu	35	N/mm²	Fy	415	N/mm²	Cover	30	mm
Length of Flight (Length in Slope)			5.04			End Condition (C=1, D=2)		
Width of Flight			1.20			LHS		
Thickness of Waist slab (D)			200			RHS		
Finishes with Concrete steps			4.00					
Imposed load			5.00					

Load calculation

Dead load (Gk)

Self weight of waist slab	5.00	kn/m
Finishes with Concrete steps	4.00	kn/m
Total	9.00	kn/m

Live load (Qk) 5.00 kn/m

Ultimate load (Pu) 1.4*Gk + 1.6*Qk 20.60 kn/m

Bending Moment and Shear force

Maximum +ve bending moment	0.1*Pu*Lx ²	52.33	knm
Maximum -ve bending moment	0.111*Pu*Lx ²	58.08	knm
Maximum shear force	0.6*Pu*Lx	62.29	kn

Area of Steel

Effective depth 170 mm

$$A_{st} = M \cdot 10^6 / (0.87 \cdot F_y \cdot Z)$$

	K	Z	Ast	Provide				Ast _{pro}	
		mm	mm ²					mm ²	
Span	0.052	159.6	908	Y	16	@	150 mm c/c	1340	OK
Support	0.057	158.4	1016	Y	16	@	150 mm c/c	1340	OK
Span - Effective Depth Ratio			Limit	31.7	Actual	29.6	31.7 > 29.6	Safe	

Shear check v = 0.37 N/mm² vc = 0.74 N/mm² 0.37 < 0.74 Safe

Distribution steel (0.13%)

260 mm²

Provide

				Ast _{pro}	
				mm ²	
Y	10	mm @	150 c/c	524	OK

Programme created by Eng. M. I. SADIQ., B.E (Struct.)

MAIN TRANSFORMER ROOM TRUSS

BEL-YOAHAAH Architects & Eng. Consultants

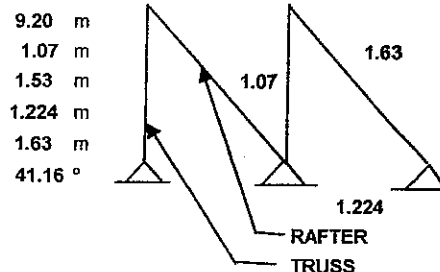
P.O Box : 827, Dubai, United Arab Emirates. Ph : 04 - 2232311 Fax : 04 - 2216422 E.mail : belyoha@emirates.net.ae

ROOF TRUSS DESIGN

(Steel Structures)

JOB No. : 1225**DATE : May 2, 2006****STRUCTURE : MAIN TRANS. ROOM****Design ref. : B S 5950 : Part 1 : 1990****TRUSS REFERENCE : T1****CP 3 : Chapter V : Part 2 : 1972****Truss on Grid : I'-G'/8'**

Span, L
 Height of Truss, H
 Panel Length, L_{RS} (Rafter Spacing)
 Plan Length, L_P (Truss Spacing)
 Slope Length, L_S (Rafter Length)
 Slope angle of Diagonal Support, θ

**Load Calculation****Dead Load**

Sheeting, W_1	0.05	Kn/m^2
Purlins, W_4	0.05	Kn/m^2
Steel Work, W_5	0.40	Kn/m^2
Services, W_6	0.25	Kn/m^2
Total	0.75	Kn/m^2
Dead Load on Plan, W_{DL}	1.00	Kn/m^2

Imposed Load

Roof Imposed Load on Plan, W_{IL}	0.75	Kn/m^2
-------------------------------------	------	-----------------

Load Factors

Dead Load Factor	1.4
Live Load Factor	1.6
Wind Load Factor	1.2

Wind Loads

CP 3 : Chapter V : Part 2 : 1972	
Height above Ground Level	11.40 m
Basic Wind Speed	45.0 m/sec
Topography factor 'S1'	Case a S1 = 1.0
Factor 'S2' :- Ground roughness :	Class 2
Cladding and Buildg size :	Class B S2 = 0.900
Statistical factor 'S3'	1.0
Design Wind Speed, V_s	40.5 m/sec
Dynamic Pressure, q	$0.613 \cdot k \cdot V_s^2$ But Min. 1 Kn/m^2 1.0 Kn/m^2

Design of Purlins

Dead Load, W_{TDL}	$(W_1 + W_4) \cdot L_P / 2$	0.06	Kn/m
Live Load, W_{TLL}	$W_{IL} \cdot L_P / 2$	0.46	Kn/m
Wind Load, W_{TWL}	$q \cdot L_P / 2$	0.62	Kn/m
Factored Load Downward, W_D	$1.4 \cdot W_{TDL} + 1.6 \cdot W_{TLL}$	0.82	Kn/m
Factored Load Upward, W_U	$1.0 \cdot W_{TDL} - 1.4 \cdot W_{TWL}$	0.80	Kn/m
Max Factored Load, W		0.82	Kn/m
Max Shear Force, V	$0.6 \cdot W \cdot L_{RS}$	0.75	Kn
Max +ve Bending Moment, M	$W \cdot L_{RS}^2 / 8$	0.24	Knm

Provide RSA 60X60X5 @ 4.57Kg/m

Depth (D)	60	mm
Thickness (t)	5	mm
Flange width bottom (F2)	60	mm

Elastic modulus (Zx)	4.45	cm ³
Moment of Inertia (Ix)	19.4	cm ⁴
Youngs modulus (E)	205	Kn/mm ²
Radius of gyration	1.82	cm
Area (Ag)	5.82	cm ²
Design strength Py from Table 6 is	275	N/mm ²

Elastic Modulus

Elastic Modulus of purlin about its axis parallel to the plane of cladding (Zx)	4.45	cm ³
From BS 5950, Table 29		
$Z_{min} = W_p \cdot L_{RS} / 1800$		
$W_p = \text{Max. of Total unfactored load of } W_{TDL} + W_{TLL} \text{ or } W_{TWL} - W_{TDL}$	0.85	KN
Z_{min}	0.72	cm ³
Check, $Z_{min} < Z_x$	0.72	< 4.45 cm ³ Safe

Dimensions

The dimension of member perpendicular to the plan of the cladding (D)	60	mm
From BS 5950, Table 29		
$D_{min} = L_{RS} / 45$	34.0	
Check, $D_{min} < D$	34.0	< 60 mm Safe
The dimension of member parallel to the plan of the cladding (B)	60	mm
From BS 5950, Table 29		
$B_{min} = L_{RS} / 60$	25.5	mm
Check, $B_{min} < B$	25.5	< 60 mm Safe

CHECK SECTION FOR SHEAR BUCKLING

Since $d/t < 63\epsilon$ no check required for shear buckling

CHECK SECTION FOR COMBINED MOMENT AND SHEAR

1) Maximum moment and co-existent shear (at the centre)

Shear area (Av)	0.9 * A	524	mm ²
Shear capacity (Pv)	0.6 * Py * Av	86.43	kn
	0.6 * Pv	51.86	kn
Design shear force $F_v = F_{vc}$	Fv	0.75	kn
Check, $F_v < 0.6P_v$	0.75	< 51.86 kn	Safe
Moment capacity for compact section and $z = \text{elastic modulus} = 1.2 * P_y * Z$		1.47	knm
	Mc	1.47	knm
Design moment at centre, (Mm)		0.82	knm
Check, $M_m < M_c$	0.82	< 1.47 knm	Safe

2) Maximum shear and co-existent moment (at the end)

Shear capacity (Pv)		86.43	kn
Maximum shear force $F_v = F_{ve}$		0.75	kn
Check, $F_v < P_v$	0.75	< 86.43 kn	Safe
Moment capacity for compact section			
	Mc	1.47	knm
Design moment at end (Ms)		0.00	knm
Check, $M_m < M_c$	0.00	< 1.47 knm	---

CHECK FOR DEFLECTION UNDER SERVICEABILITY LOADS

This case will take the serviceability loads

as unfactored imposed loads

Actual deflection	$Udl = 5WL^4 / 384EI$	0.8	mm
Limiting deflection	Span/200	7.7	mm
Actual < Limiting	0.8	< 7.7 mm	Safe

Design of Diagonal Support

Self Weight of the Member		0.10	Kn/m		
Dead Load, W_{TDL}	$(W_1 + W_4) \cdot L_p / 2$	0.16	Kn/m		
Live Load, W_{TLL}	$W_{LL} \cdot L_p / 2$	0.61	Kn/m		
Wind Load, W_{TWL}	$q \cdot L_p / 2$	0.82	Kn/m		
Factored Load Downward, W_D	$1.4 \cdot W_{TDL} + 1.6 \cdot W_{TLL}$	1.20	Kn/m		
Factored Load Upward, W_U	$1.0 \cdot W_{TDL} - 1.4 \cdot W_{TWL}$	0.98	Kn/m		
Max Factored Load, W		1.20	Kn/m		
Max Shear Force, V	$0.6 \cdot W \cdot L_s$	1.17	Kn		
Max +ve Bending Moment, M	$W \cdot L_s^2 / 8$	0.40	Knm		
Provide RSA 60X60X5 @ 4.57Kg/m					
Depth (D)		60	mm		
Thickness (t)		5	mm		
Flange width bottom (F2)		60	mm		
Elastic modulus (Zx)		4.45	cm ³		
Moment of Inertia (Ix)		19.4	cm ⁴		
Youngs modulus (E)		205	Kn/mm ²		
Radius of gyration		1.82	cm		
Area (Ag)		5.82	cm ²		
Design strength P_y from Table 6 is		275	N/mm ²		
Elastic Modulus					
Elastic Modulus of purlin about its axis parallel to the plane of cladding (Zx)		4.45	cm ³		
From BS 5950, Table 29					
$Z_{min} = W_p \cdot L_{RS} / 1800$					
$W_p = \text{Max. of Total unfactored load of } W_{TDL} + W_{TLL} \text{ or } W_{TWL} - W_{TDL}$		1.26	KN		
Z_{min}		1.13	cm ³		
Check, $Z_{min} < Z_x$	1.13	<	4.45	cm ³	Safe
Dimensions					
The dimension of member perpendicular to the plan of the cladding (D)		60	mm		
From BS 5950, Table 29					
$D_{min} = L_s / 45$		36.1			
Check, $D_{min} < D$	36.1	<	60	mm	Safe
The dimension of member parallel to the plan of the cladding (B)		60	mm		
From BS 5950, Table 29					
$B_{min} = L_s / 60$		27.1	mm		
Check, $B_{min} < B$	27.1	<	60	mm	Safe
CHECK SECTION FOR SHEAR BUCKLING					
Since $d/t < 63\epsilon$ no check required for shear buckling					
CHECK SECTION FOR COMBINED MOMENT AND SHEAR					
1) Maximum moment and co-existent shear (at the centre)					
Shear area (A_v)	$0.9 \cdot A$	524	mm ²		
Shear capacity (P_v)	$0.6 \cdot P_y \cdot A_v$	86.43	kn		
	$0.6 \cdot P_v$	51.86	kn		
Design shear force $F_v = F_v$	F_v	1.17	kn		
Check, $F_v < 0.6 P_v$	1.17	<	51.86	kn	Safe
Moment capacity for compact section and $z = \text{elastic modulus} = 1.2 \cdot P_y \cdot Z$		1.47	knm		
	M_c	1.47	knm		
Design moment at centre, (Mm)		0.40	knm		
Check, $M_m < M_c$	0.40	<	1.47	knm	Safe
2) Maximum shear and co-existent moment (at the end)					

Shear capacity (P_v)			86.43	kn	
Maximum shear force $F_v = F_{ve}$			1.17	kn	
Check, $F_v < P_v$	1.17	<	86.43	kn	Safe
Moment capacity for compact section					
M_c			1.47	knm	
Design moment at end (M_s)			0.00	knm	
Check, $M_m < M_c$	0.00	<	1.47	knm	—

CHECK FOR DEFLECTION UNDER SERVICEABILITY LOADS

This case will take the serviceability loads					
as unfactored imposed loads				1.4	mm
Actual deflection	Udl- $5WL^4/384EI$			8.1	mm
Limiting deflection	Span/200				
Actual < Limiting	1.4	<	8.1	mm	Safe

Design of Truss

Truss Loadings

Point Load on Truss

Dead Load	$W_{DL} * L_P * L_{RS}$	1.4	Kn
Live Load	$W_{LL} * L_P * L_{RS}$	1.4	Kn
Wind Load	$Q * L_P * L_{RS}$	1.9	Kn

Please refer the Staad Analysis as attached

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*****
*
*          S T A A D - III
*      Revision 22.3
*      Proprietary Program of
*      Research Engineers, Inc.
*      Date=    15:31: 37006
*      Time=    15:31:34
*
*      USER ID: BEL-YOAHAH CONSULTANTS
*****

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1. STAAD PLANE TRUSS T1 FOR MAIN TRANSFORMER ROOM - MPS
2. INPUT WIDTH 72
3. UNIT METER KNS
4. JOINT COORDINATES
5.      1      .000      .000      .000
6.      2      9.500      .000      .000
7.      3      9.500      1.000      .000
8.      4      .000      1.000      .000
9.      5      1.583      .000      .000
10.     6      3.167      .000      .000
11.     7      4.750      .000      .000
12.     8      6.333      .000      .000
13.     9      7.917      .000      .000
14.    10      7.917      1.000      .000
15.    11      6.333      1.000      .000
16.    12      4.750      1.000      .000
17.    13      3.167      1.000      .000
18.    14      1.583      1.000      .000
19. MEMBER INCIDENCES
20.      1      1      5
21.      2      2      3
22.      3      3      10
23.      4      4      1
24.      5      5      6
25.      6      6      7
26.      7      7      8
27.      8      8      9
28.      9      9      2
29.     10     10     11
30.     11     11     12
31.     12     12     13
32.     13     13     14
33.     14     14      4
34.     15      4      5
35.     16      5     14
36.     17     14      6
37.     18      6     13
38.     19     13      7
39.     20      7     12
40.     21      7     11
41.     22     11      8

```

42. 23 8 10
 43. 24 10 9
 44. 25 9 3
 45. MEMBER PROPERTY BRITISH
 46. 1 TO 25 TABLE ST UA60X60X6
 47. CONSTANT
 48. E STEEL ALL
 49. DENSITY STEEL ALL
 50. POISSON STEEL ALL
 51. SUPPORT
 52. 1 2 PINNED
 53. LOAD 1 DL
 54. SELFWEIGHT Y -1.
 55. JOINT LOAD
 56. 5 TO 14 FY -1.4
 57. 1 TO 4 FY -.7
 58. LOAD 2 LL
 59. JOINT LOAD
 60. 5 TO 14 FY -1.4
 61. 1 TO 4 FY -.7
 62. LOAD 3 WL
 63. JOINT LOAD
 64. 5 TO 14 FY 1.9
 65. 1 TO 4 FY .95
 66. LOAD COMB 4 ULT DL+LL
 67. 1 1.4 2 1.6
 68. LOAD COMB 5 SERVICE DL+LL
 69. 1 1. 2 1.
 70. LOAD COMB 6 ULT DL+WL
 71. 1 1. 3 1.4
 72. LOAD COMB 7 ULT DL+LL+WL
 73. 1 1.2 2 1.2 3 1.2
 74. PERFORM ANALYSIS

PROBLEM STATISTICS

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 14/ 25/ 2
 ORIGINAL/FINAL BAND-WIDTH = 10/ 3
 TOTAL PRIMARY LOAD CASES = 3, TOTAL DEGREES OF FREEDOM = 38
 SIZE OF STIFFNESS MATRIX = 456 DOUBLE PREC. WORDS
 REQD/AVAIL. DISK SPACE = 12.04/ 2047.7 MB, EXMEM = 1969.2 MB

++ Processing Element Stiffness Matrix. 15:31:37
 ++ Processing Global Stiffness Matrix. 15:31:37
 ++ Processing Triangular Factorization. 15:31:37
 ++ Calculating Joint Displacements. 15:31:37
 ++ Calculating Member Forces. 15:31:37

□75. PRINT ANALYSIS RESULTS

SUPPORT REACTIONS -UNIT KNS METE STRUCTURE TYPE = PLANE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
□							
1	1	10.75	9.39	.00	.00	.00	.00
	2	9.62	8.40	.00	.00	.00	.00
	3	-13.05	-11.40	.00	.00	.00	.00
	4	30.44	26.58	.00	.00	.00	.00
	5	20.37	17.79	.00	.00	.00	.00
	6	-7.52	-6.57	.00	.00	.00	.00
	7	8.78	7.67	.00	.00	.00	.00
2	1	-10.75	9.39	.00	.00	.00	.00
	2	-9.62	8.40	.00	.00	.00	.00
	3	13.05	-11.40	.00	.00	.00	.00
	4	-30.44	26.58	.00	.00	.00	.00
	5	-20.37	17.79	.00	.00	.00	.00
	6	7.52	-6.57	.00	.00	.00	.00
	7	-8.78	7.67	.00	.00	.00	.00

TRUSS T1 FOR MAIN TRANSFORMER ROOM - MPS -- PAGE NO. 4
***** END OF LATEST ANALYSIS RESULT *****

76. PARAMETER
77. CODE BS5950
78. TRACK 1. ALL
79. CHECK CODE ALL

STAAD-III CODE CHECKING - (BSI)

PROGRAM CODE REVISION V2.1_5950_90/2

ALL UNITS ARE - KNS METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
1 ST UA60X60X6	PASS	BS-4.7 (C)	.390	4	
	30.30 C	.00	.06	.00	
CALCULATED CAPACITIES FOR MEMB 1 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 77.8 PT= 190.0 MB= 1.7 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .16 MRZ= 1.1 MRY= 2.4					
2 ST UA60X60X6	PASS	BS-4.7 (C)	.239	4	
	24.37 C	.00	.06	.00	
CALCULATED CAPACITIES FOR MEMB 2 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 101.8 PT= 190.0 MB= 1.9 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .13 MRZ= 1.1 MRY= 2.4					
3 ST UA60X60X6	PASS	BS-4.7 (C)	.450	4	
	34.97 C	.00	.05	.00	
CALCULATED CAPACITIES FOR MEMB 3 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 77.8 PT= 190.0 MB= 1.7 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .18 MRZ= 1.1 MRY= 2.4					
4 ST UA60X60X6	PASS	BS-4.7 (C)	.239	4	
	24.37 C	.00	-.06	1.00	
CALCULATED CAPACITIES FOR MEMB 4 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 101.8 PT= 190.0 MB= 1.9 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .13 MRZ= 1.1 MRY= 2.4					
5 ST UA60X60X6	PASS	BS-4.8.2	.053	4	
	4.65 T	.00	.03	.00	
CALCULATED CAPACITIES FOR MEMB 5 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 77.7 PT= 190.0 MB= 1.7 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .02 MRZ= 1.1 MRY= 2.4					
6 ST UA60X60X6	PASS	BS-4.8.2	.151	4	
	25.65 T	.00	.02	1.58	
CALCULATED CAPACITIES FOR MEMB 6 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 77.8 PT= 190.0 MB= 1.7 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .13 MRZ= 1.1 MRY= 2.4					

ALL UNITS ARE ~ KNS METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
7 ST UA60X60X6	PASS	BS-4.8.2	.151	4	
	25.65 T	.00	-.02	.00	
CALCULATED CAPACITIES FOR MEMB 7 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 77.8 PT= 190.0 MB= 1.7 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .13 MRZ= 1.1 MRY= 2.4					
8 ST UA60X60X6	PASS	BS-4.8.2	.053	4	
	4.65 T	.00	-.03	1.58	
CALCULATED CAPACITIES FOR MEMB 8 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 77.7 PT= 190.0 MB= 1.7 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .02 MRZ= 1.1 MRY= 2.4					
9 ST UA60X60X6	PASS	BS-4.7 (C)	.390	4	
	30.30 C	.00	-.03	.00	
CALCULATED CAPACITIES FOR MEMB 9 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 77.8 PT= 190.0 MB= 1.7 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .16 MRZ= 1.1 MRY= 2.4					
10 ST UA60X60X6	PASS	BS-4.7 (C)	.720	4	
	56.01 C	.00	.02	.00	
CALCULATED CAPACITIES FOR MEMB 10 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 77.7 PT= 190.0 MB= 1.7 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .29 MRZ= 1.1 MRY= 2.4					
11 ST UA60X60X6	PASS	BS-4.7 (C)	.810	4	
	63.01 C	.00	.01	.00	
CALCULATED CAPACITIES FOR MEMB 11 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 77.8 PT= 190.0 MB= 1.7 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .33 MRZ= 1.1 MRY= 2.4					
12 ST UA60X60X6	PASS	BS-4.7 (C)	.810	4	
	63.01 C	.00	-.01	.00	

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ALL UNITS ARE - KNS METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 12 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 77.8 PT= 190.0 MB= 1.7 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .33 MRZ= 1.1 MRY= 2.4					

13 ST UA60X60X6 PASS BS-4.7 (C) .720 4					
56.01 C .00 -.01 .00					

CALCULATED CAPACITIES FOR MEMB 13 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 77.7 PT= 190.0 MB= 1.7 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .29 MRZ= 1.1 MRY= 2.4					

14 ST UA60X60X6 PASS BS-4.7 (C) .450 4					
34.97 C .00 -.02 .00					

CALCULATED CAPACITIES FOR MEMB 14 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 77.8 PT= 190.0 MB= 1.7 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .18 MRZ= 1.1 MRY= 2.4					

15 ST UA60X60X6 PASS BS-4.7 (C) .265 3					
17.69 C .00 -.01 .00					

CALCULATED CAPACITIES FOR MEMB 15 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 66.7 PT= 190.0 MB= 1.7 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .09 MRZ= 1.1 MRY= 2.4					

16 ST UA60X60X6 PASS BS-4.7 (C) .173 4					
17.66 C .00 -.06 .00					

CALCULATED CAPACITIES FOR MEMB 16 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 101.8 PT= 190.0 MB= 1.9 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .09 MRZ= 1.1 MRY= 2.4					

17 ST UA60X60X6 PASS BS-4.7 (C) .160 3					
10.63 C .00 .00 .00					

CALCULATED CAPACITIES FOR MEMB 17 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 66.6 PT= 190.0 MB= 1.7 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .06 MRZ= 1.1 MRY= 2.4					

□

ALL UNITS ARE - KNS METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					
18 ST UA60X60X6	PASS	BS-4.7 (C)	.087	4	
	8.84 C	.00	-.04	.00	

CALCULATED CAPACITIES FOR MEMB 18 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 101.8 PT= 190.0 MB= 1.9 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .05 MRZ= 1.1 MRY= 2.4					

19 ST UA60X60X6	PASS	BS-4.8.2	.054	4	
	8.24 T	.00	.01	.00	

CALCULATED CAPACITIES FOR MEMB 19 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 66.7 PT= 190.0 MB= 1.7 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .04 MRZ= 1.1 MRY= 2.4					

20 ST UA60X60X6	PASS	BS-4.7 (C)	.043	4	
	4.36 C	.00	.00	.00	

CALCULATED CAPACITIES FOR MEMB 20 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 101.8 PT= 190.0 MB= 1.9 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .02 MRZ= 1.1 MRY= 2.4					

21 ST UA60X60X6	PASS	BS-4.8.2	.054	4	
	8.24 T	.00	-.01	1.87	

CALCULATED CAPACITIES FOR MEMB 21 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 66.7 PT= 190.0 MB= 1.7 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .04 MRZ= 1.1 MRY= 2.4					

22 ST UA60X60X6	PASS	BS-4.7 (C)	.087	4	
	8.84 C	.00	.04	1.00	

CALCULATED CAPACITIES FOR MEMB 22 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 101.8 PT= 190.0 MB= 1.9 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .05 MRZ= 1.1 MRY= 2.4					

23 ST UA60X60X6	PASS	BS-4.7 (C)	.160	3	
	10.63 C	.00	.01	.00	

□

ALL UNITS ARE - KNS METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 23 UNIT - kN,m SECTION CLASS 3					
MCZ=	1.1	MCY= 2.4	PC= 66.6	PT= 190.0	MB= 1.7 PV= 35.6
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ=	190.02	FX/PZ = .06	MRZ= 1.1	MRY= 2.4	

24 ST UA60X60X6	PASS	BS-4.7 (C)	.173	4	
	17.66 C	.00	.06	1.00	

CALCULATED CAPACITIES FOR MEMB 24 UNIT - kN,m SECTION CLASS 3					
MCZ=	1.1	MCY= 2.4	PC= 101.8	PT= 190.0	MB= 1.9 PV= 35.6
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ=	190.02	FX/PZ = .09	MRZ= 1.1	MRY= 2.4	

25 ST UA60X60X6	PASS	BS-4.7 (C)	.265	3	
	17.69 C	.00	.01	.00	

CALCULATED CAPACITIES FOR MEMB 25 UNIT - kN,m SECTION CLASS 3					
MCZ=	1.1	MCY= 2.4	PC= 66.7	PT= 190.0	MB= 1.7 PV= 35.6
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ=	190.02	FX/PZ = .09	MRZ= 1.1	MRY= 2.4	

***** END OF TABULATED RESULT OF DESIGN *****

80. FINISH

***** END OF STAAD-III *****

**** DATE= MAY 23,2006 TIME= 15:31:37 ****

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**SHUNT REACTOR ROOM
& EARTHING
TRANSFORMER ROOM
TRUSS**

BEL-YOAHAH Architects & Eng. Consultants

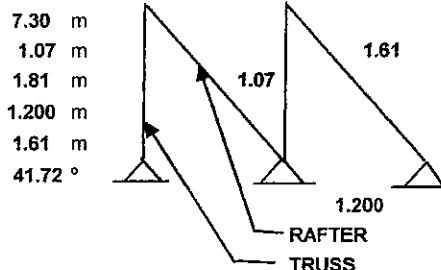
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ROOF TRUSS DESIGN

(Steel Structures)

JOB No. : 1225**DATE : April 25, 2006****STRUCTURE : SHUNT REACTOR ROOM****Design ref. : B S 5950 : Part 1 : 1990****TRUSS REFERENCE : T2****CP 3 : Chapter V : Part 2 : 1972****Truss on Grid : 1/7'-8'**

Span, L
 Height of Truss, H
 Panel Length, L_{RS} (Rafter Spacing)
 Plan Length, L_P (Truss Spacing)
 Slope Length, L_S (Rafter Length)
 Slope angle of Diagonal Support, $\emptyset$

**Load Calculation****Dead Load**

Sheeting, W_1	0.05	Kn/m^2
Purlins, W_4	0.05	Kn/m^2
Steel Work, W_5	0.40	Kn/m^2
Services, W_6	0.25	Kn/m^2
Total	0.75	Kn/m^2
Dead Load on Plan, W_{DL}	1.00	Kn/m^2

Imposed Load

Roof Imposed Load on Plan, W_{IL}	0.75	Kn/m^2
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Load Factors

Dead Load Factor	1.4
Live Load Factor	1.6
Wind Load Factor	1.2

Wind Loads

CP 3 : Chapter V : Part 2 : 1972	
Height above Ground Level	11.40 m
Basic Wind Speed	45.0 m/sec
Topography factor 'S1'	Case a S1 = 1.0
Factor 'S2' :- Ground roughness :	Class 2
Cladding and Buildg size :	Class B S2 = 0.900
Statistical factor 'S3'	1.0
Design Wind Speed, V_s	40.5 m/sec
Dynamic Pressure, q	$0.613 \cdot k \cdot V_s^2$ But Min. 1 Kn/m^2
	1.0 Kn/m^2

Design of Purlins

Dead Load, W_{TDL}	$(W_1 + W_4) \cdot L_P / 2$	0.06	Kn/m
Live Load, W_{TLL}	$W_{IL} \cdot L_P / 2$	0.45	Kn/m
Wind Load, W_{TWL}	$q \cdot L_P / 2$	0.60	Kn/m
Factored Load Downward, W_D	$1.4 \cdot W_{TDL} + 1.6 \cdot W_{TLL}$	0.80	Kn/m
Factored Load Upward, W_U	$1.0 \cdot W_{TDL} - 1.4 \cdot W_{TWL}$	0.78	Kn/m
Max Factored Load, W		0.80	Kn/m
Max Shear Force, V	$0.6 \cdot W \cdot L_{RS}$	0.87	Kn
Max +ve Bending Moment, M	$W \cdot L_{RS}^2 / 8$	0.33	Knm

Provide RSA 60X60X5 @ 4.57Kg/m

Depth (D)	60	mm
Thickness (t)	5	mm
Flange width bottom (F2)	60	mm

Elastic modulus (Zx)	4.45	cm ³
Moment of Inertia (Ix)	19.4	cm ⁴
Youngs modulus (E)	205	Kn/mm ²
Radius of gyration	1.82	cm
Area (Ag)	5.82	cm ²
Design strength Py from Table 6 is	275	N/mm ²

Elastic Modulus

Elastic Modulus of purlin about its axis parallel to the plane of cladding (Zx)	4.45	cm ³		
From BS 5950, Table 29				
$Z_{min} = W_p \cdot L_{RS} / 1800$				
Wp = Max. of Total unfactored load of $W_{TDL} + W_{TLL}$ or $W_{TWL} - W_{TDL}$	0.98	KN		
Z_{min}	0.99	cm ³		
Check, $Z_{min} < Z_x$	0.99	<	4.45	cm ³ Safe

Dimensions

The dimension of member perpendicular to the plan of the cladding (D)	60	mm		
From BS 5950, Table 29				
$D_{min} = L_{RS} / 45$	40.2			
Check, $D_{min} < D$	40.2	<	60	mm Safe
The dimension of member parallel to the plan of the cladding (B)	60	mm		
From BS 5950, Table 29				
$B_{min} = L_{RS} / 60$	30.2	mm		
Check, $B_{min} < B$	30.2	<	60	mm Safe

CHECK SECTION FOR SHEAR BUCKLING

Since $d/t < 63\epsilon$ no check required for shear buckling

CHECK SECTION FOR COMBINED MOMENT AND SHEAR

1) Maximum moment and co-existent shear (at the centre)

Shear area (Av)	0.9*A	524	mm ²	
Shear capacity (Pv)	0.6*Py*Av	86.43	kn	
	0.6*Pv	51.86	kn	
Design shear force $F_v = F_{vc}$	Fv	0.87	kn	
Check, $F_v < 0.6P_v$	0.87	<	51.86	kn Safe
Moment capacity for compact section and $z =$ elastic modulus = $1.2*P_y*Z$		1.47	knm	
	Mc	1.47	knm	
Design moment at centre, (Mm)		0.80	knm	
Check, $M_m < M_c$	0.80	<	1.47	knm Safe

2) Maximum shear and co-existent moment (at the end)

Shear capacity (Pv)		86.43	kn	
Maximum shear force $F_v = F_{ve}$		0.87	kn	
Check, $F_v < P_v$	0.87	<	86.43	kn Safe
Moment capacity for compact section				
	Mc	1.47	knm	
Design moment at end (Ms)		0.00	knm	
Check, $M_m < M_c$	0.00	<	1.47	knm --

CHECK FOR DEFLECTION UNDER SERVICEABILITY LOADS

This case will take the serviceability loads as unfactored imposed loads

Actual deflection	$UdI = 5WL^4 / 384EI$	1.6	mm	
Limiting deflection	Span/200	9.1	mm	
Actual < Limiting	1.6	<	9.1	mm Safe

Design of Diagonal Support

Self Weight of the Member		0.10	Kn/m		
Dead Load, W_{TDL}	$(W_1 + W_4) * L_p / 2$	0.16	Kn/m		
Live Load, W_{TLL}	$W_{IL} * L_p / 2$	0.60	Kn/m		
Wind Load, W_{TWL}	$q * L_p / 2$	0.81	Kn/m		
Factored Load Downward, W_D	$1.4 * W_{TDL} + 1.6 * W_{TLL}$	1.19	Kn/m		
Factored Load Upward, W_U	$1.0 * W_{TDL} - 1.4 * W_{TWL}$	0.97	Kn/m		
Max Factored Load, W		1.19	Kn/m		
Max Shear Force, V	$0.6 * W * L_s$	1.15	Kn		
Max +ve Bending Moment, M	$W * L_s^2 / 8$	0.38	Knm		
Provide RSA 60X60X5 @ 4.57Kg/m					
Depth (D)		60	mm		
Thickness (t)		5	mm		
Flange width bottom (F2)		60	mm		
Elastic modulus (Zx)		4.45	cm ³		
Moment of Inertia (Ix)		19.4	cm ⁴		
Youngs modulus (E)		205	Kn/mm ²		
Radius of gyration		1.82	cm		
Area (Ag)		5.82	cm ²		
Design strength P_y from Table 6 is		275	N/mm ²		
Elastic Modulus					
Elastic Modulus of purlin about its axis parallel to the plane of cladding (Zx)		4.45	cm ³		
From BS 5950, Table 29					
$Z_{min} = W_p * L_{RS} / 1800$					
$W_p = \text{Max. of Total unfactored load of } W_{TDL} + W_{TLL} \text{ or } W_{TWL} - W_{TDL}$		1.23	KN		
Z_{min}		1.10	cm ³		
Check, $Z_{min} < Z_x$	1.10	<	4.45	cm ³	Safe
Dimensions					
The dimension of member perpendicular to the plan of the cladding (D)		60	mm		
From BS 5950, Table 29					
$D_{min} = L_s / 45$		35.7			
Check, $D_{min} < D$	35.7	<	60	mm	Safe
The dimension of member parallel to the plan of the cladding (B)		60	mm		
From BS 5950, Table 29					
$B_{min} = L_s / 60$		26.8	mm		
Check, $B_{min} < B$	26.8	<	60	mm	Safe
CHECK SECTION FOR SHEAR BUCKLING					
Since $d/t < 63\epsilon$ no check required for shear buckling					
CHECK SECTION FOR COMBINED MOMENT AND SHEAR					
1) Maximum moment and co-existent shear (at the centre)					
Shear area (A_v)	$0.9 * A$	524	mm ²		
Shear capacity (P_v)	$0.6 * P_y * A_v$	86.43	kn		
	$0.6 * P_v$	51.86	kn		
Design shear force $F_v = F_{vc}$	F_v	1.15	kn		
Check, $F_v < 0.6 P_v$	1.15	<	51.86	kn	Safe
Moment capacity for compact section and $z = \text{elastic modulus} = 1.2 * P_y * Z$		1.47	knm		
	M_c	1.47	knm		
Design moment at centre, (M_m)		0.38	knm		
Check, $M_m < M_c$	0.38	<	1.47	knm	Safe
2) Maximum shear and co-existent moment (at the end)					

Shear capacity (Pv)			86.43	kn	
Maximum shear force Fv = Fve			1.15	kn	
Check, $F_v < P_v$	1.15	<	86.43	kn	Safe
Moment capacity for compact section					
Mc			1.47	knm	
Design moment at end (Ms)			0.00	knm	
Check, $M_m < M_c$	0.00	<	1.47	knm	—

CHECK FOR DEFLECTION UNDER SERVICEABILITY LOADS

This case will take the serviceability loads
as unfactored imposed loads

Actual deflection	Udl- $5WL^4/384EI$		1.3	mm	
Limiting deflection	Span/200		8.0	mm	
Actual < Limiting	1.3	<	8.0	mm	Safe

Design of Truss

Truss Loadings

Point Load on Truss

Dead Load	$W_{DL} * L_P * L_{RS}$	1.6	Kn
Live Load	$W_{LL} * L_P * L_{RS}$	1.6	Kn
Wind Load	$q * L_P * L_{RS}$	2.2	Kn

Please refer the Staad Analysis as attached

BEL-YOAHAH Architects & Eng. Consultants

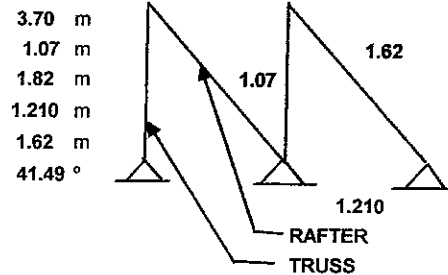
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ROOF TRUSS DESIGN

(Steel Structures)

JOB No. : 1225**DATE : May 2, 2006****STRUCTURE : EARTH TRANS. ROOM****Design ref. : B S 5950 : Part 1 : 1990****TRUSS REFERENCE : T3****CP 3 : Chapter V : Part 2 : 1972****Truss on Grid : K/4'-4"**

Span, L
 Height of Truss, H
 Panel Length, L_{RS} (Rafter Spacing)
 Plan Length, L_P (Truss Spacing)
 Slope Length, L_S (Rafter Length)
 Slope angle of Diagonal Support, θ

**Load Calculation****Dead Load**

Sheeting, W_1	0.05	Kn/m^2
Purlins, W_4	0.05	Kn/m^2
Steel Work, W_5	0.40	Kn/m^2
Services, W_6	0.25	Kn/m^2
Total	0.75	Kn/m^2
Dead Load on Plan, W_{DL}	1.00	Kn/m^2

Imposed Load

Roof Imposed Load on Plan, W_{IL} 0.75 Kn/m^2

Load Factors

Dead Load Factor	1.4
Live Load Factor	1.6
Wind Load Factor	1.2

Wind Loads

CP 3 : Chapter V : Part 2 : 1972

Height above Ground Level	5.83	m
Basic Wind Speed	45.0	m/sec
Topography factor 'S1'	Case a	$S1 = 1.0$
Factor 'S2' :- Ground roughness :	Class 2	
Cladding and Buildg size :	Class B	$S2 = 0.764$
Statistical factor 'S3'		1.0
Design Wind Speed, V_s		34.4 m/sec
Dynamic Pressure, q	$0.613 \cdot k \cdot V_s^2$ But Min. 1Kn/m^2	1.0 Kn/m^2

Design of Purlins

Dead Load, W_{TDL}	$(W_1 + W_4) \cdot L_P / 2$	0.06 Kn/m
Live Load, W_{TLL}	$W_{IL} \cdot L_P / 2$	0.45 Kn/m
Wind Load, W_{TWL}	$q \cdot L_P / 2$	0.61 Kn/m
Factored Load Downward, W_D	$1.4 \cdot W_{TDL} + 1.6 \cdot W_{TLL}$	0.81 Kn/m
Factored Load Upward, W_U	$1.0 \cdot W_{TDL} - 1.4 \cdot W_{TWL}$	0.79 Kn/m
Max Factored Load, W		0.81 Kn/m
Max Shear Force, V	$0.6 \cdot W \cdot L_{RS}$	0.89 Kn
Max +ve Bending Moment, M	$W \cdot L_{RS}^2 / 8$	0.34 Knm

Provide RSA 60X60X5 @ 4.57Kg/m

Depth (D)	60	mm
Thickness (t)	5	mm
Flange width bottom (F2)	60	mm

Elastic modulus (Zx)	4.45	cm ³
Moment of Inertia (Ix)	19.4	cm ⁴
Youngs modulus (E)	205	Kn/mm ²
Radius of gyration	1.82	cm
Area (Ag)	5.82	cm ²
Design strength Py from Table 6 is	275	N/mm ²

Elastic Modulus

Elastic Modulus of purlin about its axis parallel to the plane of cladding (Zx)	4.45	cm ³		
From BS 5950, Table 29				
$Z_{min} = W_p \cdot L_{RS} / 1800$				
Wp = Max. of Total unfactored load of $W_{TDL} + W_{TLL}$ or $W_{TWL} - W_{TDL}$	0.99	KN		
Zmin	1.00	cm ³		
Check, $Z_{min} < Z_x$	1.00	<	4.45	cm ³ Safe

Dimensions

The dimension of member perpendicular to the plan of the cladding (D)	60	mm		
From BS 5950, Table 29				
$D_{min} = L_{RS} / 45$	40.4			
Check, $D_{min} < D$	40.4	<	60	mm Safe
The dimension of member parallel to the plan of the cladding (B)	60	mm		
From BS 5950, Table 29				
$B_{min} = L_{RS} / 60$	30.3	mm		
Check, $B_{min} < B$	30.3	<	60	mm Safe

CHECK SECTION FOR SHEAR BUCKLING

Since $d/t < 63\epsilon$ no check required for shear buckling

CHECK SECTION FOR COMBINED MOMENT AND SHEAR

1) Maximum moment and co-existent shear (at the centre)

Shear area (Av)	0.9*A	524	mm ²	
Shear capacity (Pv)	0.6*Py*Av	86.43	kn	
	0.6*Pv	51.86	kn	
Design shear force $F_v = F_{vc}$	Fv	0.89	kn	
Check, $F_v < 0.6P_v$	0.89	<	51.86	kn Safe
Moment capacity for compact section and $z =$ elastic modulus = $1.2 \cdot P_y \cdot Z$		1.47	knm	
	Mc	1.47	knm	
Design moment at centre, (Mm)		0.81	knm	
Check, $M_m < M_c$	0.81	<	1.47	knm Safe

2) Maximum shear and co-existent moment (at the end)

Shear capacity (Pv)		86.43	kn	
Maximum shear force $F_v = F_{ve}$		0.89	kn	
Check, $F_v < P_v$	0.89	<	86.43	kn Safe
Moment capacity for compact section		1.47	knm	
	Mc	1.47	knm	
Design moment at end (Ms)		0.00	knm	
Check, $M_m < M_c$	0.00	<	1.47	knm —

CHECK FOR DEFLECTION UNDER SERVICEABILITY LOADS

This case will take the serviceability loads as unfactored imposed loads

Actual deflection	$Udl = 5WL^4 / 384EI$	1.6	mm	
Limiting deflection	Span/200	9.1	mm	
Actual < Limiting	1.6	<	9.1	mm Safe

Design of Diagonal Support

Self Weight of the Member		0.10	Kn/m
Dead Load, W_{TDL}	$(W_1 + W_4) \cdot L_p / 2$	0.16	Kn/m
Live Load, W_{TLL}	$W_{LL} \cdot L_p / 2$	0.61	Kn/m
Wind Load, W_{TWL}	$q \cdot L_p / 2$	0.81	Kn/m
Factored Load Downward, W_D	$1.4 \cdot W_{TDL} + 1.6 \cdot W_{TLL}$	1.20	Kn/m
Factored Load Upward, W_U	$1.0 \cdot W_{TDL} - 1.4 \cdot W_{TWL}$	0.97	Kn/m
Max Factored Load, W		1.20	Kn/m
Max Shear Force, V	$0.6 \cdot W \cdot L_s$	1.16	Kn
Max +ve Bending Moment, M	$W \cdot L_s^2 / 8$	0.39	Knm
Provide RSA 60X60X5 @ 4.57Kg/m			
Depth (D)		60	mm
Thickness (t)		5	mm
Flange width bottom (F2)		60	mm
Elastic modulus (Zx)		4.45	cm ³
Moment of Inertia (Ix)		19.4	cm ⁴
Youngs modulus (E)		205	Kn/mm ²
Radius of gyration		1.82	cm
Area (Ag)		5.82	cm ²
Design strength P_y from Table 6 is		275	N/mm ²

Elastic Modulus

Elastic Modulus of purlin about its axis parallel to the plane of cladding (Zx)		4.45	cm ³
From BS 5950, Table 29			
$Z_{min} = W_p \cdot L_{RS} / 1800$			
$W_p = \text{Max. of Total unfactored load of } W_{TDL} + W_{TLL} \text{ or } W_{TWL} - W_{TDL}$		1.24	KN
Z_{min}		1.11	cm ³
Check, $Z_{min} < Z_x$	1.11	<	4.45 cm ³ Safe

Dimensions

The dimension of member perpendicular to the plan of the cladding (D)		60	mm
From BS 5950, Table 29			
$D_{min} = L_s / 45$		35.9	
Check, $D_{min} < D$	35.9	<	60 mm Safe
The dimension of member parallel to the plan of the cladding (B)		60	mm
From BS 5950, Table 29			
$B_{min} = L_s / 60$		26.9	mm
Check, $B_{min} < B$	26.9	<	60 mm Safe

CHECK SECTION FOR SHEAR BUCKLING

Since $d/t < 63\epsilon$ no check required for shear buckling

CHECK SECTION FOR COMBINED MOMENT AND SHEAR

1) Maximum moment and co-existent shear (at the centre)

Shear area (A_v)	$0.9 \cdot A$	524	mm ²
Shear capacity (P_v)	$0.6 \cdot P_y \cdot A_v$	86.43	kn
	$0.6 \cdot P_v$	51.86	kn
Design shear force $F_v = F_{vc}$	F_v	1.16	kn
Check, $F_v < 0.6 P_v$	1.16	<	51.86 kn Safe
Moment capacity for compact section and $z = \text{elastic modulus} = 1.2 \cdot P_y \cdot Z$		1.47	knm
	M_c	1.47	knm
Design moment at centre, (Mm)		0.39	knm
Check, $M_m < M_c$	0.39	<	1.47 knm Safe

2) Maximum shear and co-existent moment (at the end)

Shear capacity (Pv)				86.43	kn	
Maximum shear force $F_v = F_{ve}$				1.16	kn	
Check, $F_v < P_v$	1.16	<		86.43	kn	Safe
Moment capacity for compact section						
M_c				1.47	knm	
Design moment at end (M_s)				0.00	knm	
Check, $M_m < M_c$	0.00	<		1.47	knm	—

CHECK FOR DEFLECTION UNDER SERVICEABILITY LOADS

This case will take the serviceability loads as unfactored imposed loads						
				1.3	mm	
Actual deflection	Udl- $5WL^4/384EI$			8.1	mm	
Limiting deflection	Span/200					
Actual < Limiting	1.3	<		8.1	mm	Safe

Design of Truss

Truss Loadings

Point Load on Truss

Dead Load	$W_{DL} * L_P * L_{RS}$	1.7	Kn
Live Load	$W_{LL} * L_P * L_{RS}$	1.7	Kn
Wind Load	$q * L_P * L_{RS}$	2.2	Kn

Please refer the Staad Analysis as attached

```

*****
*
*          S T A A D - III
*          Revision 22.3
*          Proprietary Program of
*          Research Engineers, Inc.
*          Date=    15:35: 25006
*          Time=    15:35:22
*
*          USER ID: BEL-YOAHAAH CONSULTANTS
*****

```

```

1. STAAD PLANE TRUSS T2 FOR SHUNT REACTOR ROOM - MPS
2. INPUT WIDTH 72
3. UNIT METER KNS
4. JOINT COORDINATES
5.      1      .000      .000      .000
6.      2      8.000      .000      .000
7.      3      8.000      1.000      .000
8.      4      .000      1.000      .000
9.      5      2.000      1.000      .000
10.     6      2.000      .000      .000
11.     7      4.000      1.000      .000
12.     8      4.000      .000      .000
13.     9      6.000      1.000      .000
14.    10      6.000      .000      .000
15. MEMBER INCIDENCES
16.      1      1      6
17.      2      2      3
18.      3      3      9
19.      4      4      1
20.      6      5      4
21.      7      6      8
22.      9      7      5
23.     10      8      10
24.     16      9      7
25.     18     10      2
26.     19      4      6
27.     20      6      5
28.     21      5      8
29.     22      8      7
30.     23      8      9
31.     24      9     10
32.     25     10      3
33. MEMBER PROPERTY BRITISH
34. 1 TO 4 6 7 9 10 16 18 TO 25 TABLE ST UA60X60X6
35. CONSTANT
36. E STEEL ALL
37. DENSITY STEEL ALL
38. POISSON STEEL ALL
39. SUPPORT
40. 1 2 PINNED
41. LOAD 1 DL

```

42. SELFWEIGHT Y -1.
 43. JOINT LOAD
 44. 5 TO 10 FY -1.6
 45. 1 TO 4 FY -.8
 46. LOAD 2 LL
 47. JOINT LOAD
 48. 5 TO 10 FY -1.6
 49. 1 TO 4 FY -.8
 50. LOAD 3 WL1
 51. JOINT LOAD
 52. 5 TO 10 FY 2.2
 53. 1 TO 4 FY 1.1
 54. LOAD COMB 4 ULT DL+LL
 55. 1 1.4 2 1.6
 56. LOAD COMB 5 SERVICE DL+LL
 57. 1 1. 2 1.
 58. LOAD COMB 6 ULT DL+WL1
 59. 1 1. 2 1.4
 60. LOAD COMB 7 ULT DL+LL+WL
 61. 1 1.2 2 1.2 3 1.2
 62. PERFORM ANALYSIS

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 10/ 17/ 2
 ORIGINAL/FINAL BAND-WIDTH = 8/ 3
 TOTAL PRIMARY LOAD CASES = 3, TOTAL DEGREES OF FREEDOM = 26
 SIZE OF STIFFNESS MATRIX = 312 DOUBLE PREC. WORDS
 REQD/AVAIL. DISK SPACE = 12.03/ 2047.7 MB, EXMEM = 1969.2 MB

++ Processing Element Stiffness Matrix. 15:35:25
 ++ Processing Global Stiffness Matrix. 15:35:25
 ++ Processing Triangular Factorization. 15:35:25
 ++ Calculating Joint Displacements. 15:35:25
 ++ Calculating Member Forces. 15:35:25

63. PRINT ANALYSIS RESULTS

SUPPORT REACTIONS -UNIT KNS METE STRUCTURE TYPE = PLANE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
1	1	5.44	7.19	.00	.00	.00	.00
	2	4.81	6.40	.00	.00	.00	.00
	3	-6.62	-8.80	.00	.00	.00	.00
	4	15.31	20.31	.00	.00	.00	.00
	5	10.25	13.59	.00	.00	.00	.00
	6	12.17	16.15	.00	.00	.00	.00
	7	4.36	5.75	.00	.00	.00	.00
2	1	-5.44	7.19	.00	.00	.00	.00
	2	-4.81	6.40	.00	.00	.00	.00
	3	6.62	-8.80	.00	.00	.00	.00
	4	-15.31	20.31	.00	.00	.00	.00
	5	-10.25	13.59	.00	.00	.00	.00
	6	-12.17	16.15	.00	.00	.00	.00
	7	-4.36	5.75	.00	.00	.00	.00

***** END OF LATEST ANALYSIS RESULT *****

64. PARAMETER
65. CODE BS5950
66. TRACK 1. ALL
67. CHECK CODE ALL

STAAD-III CODE CHECKING - (BSI)

PROGRAM CODE REVISION V2.1_5950_90/2

□

ALL UNITS ARE - KNS METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
--------	-------	---------------	----------------------	--------------	----------------------

□

1 ST UA60X60X6	PASS	BS-4.7 (C)	.249	4
15.18 C		.00	.06	.00

CALCULATED CAPACITIES FOR MEMB 1 UNIT - kN,m SECTION CLASS 3				
MCZ= 1.1	MCY= 2.4	PC= 60.9	PT= 190.0	MB= 1.7 PV= 35.6
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000				
PZ= 190.02	FX/PZ = .08	MRZ= 1.1	MRY= 2.4	

2 ST UA60X60X6	PASS	BS-4.7 (C)	.175	4
17.80 C		.00	.06	.00

CALCULATED CAPACITIES FOR MEMB 2 UNIT - kN,m SECTION CLASS 3				
MCZ= 1.1	MCY= 2.4	PC= 101.8	PT= 190.0	MB= 1.9 PV= 35.6
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000				
PZ= 190.02	FX/PZ = .09	MRZ= 1.1	MRY= 2.4	

3 ST UA60X60X6	PASS	BS-4.7 (C)	.499	4
30.40 C		.00	.04	.00

CALCULATED CAPACITIES FOR MEMB 3 UNIT - kN,m SECTION CLASS 3				
MCZ= 1.1	MCY= 2.4	PC= 60.9	PT= 190.0	MB= 1.7 PV= 35.6
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000				
PZ= 190.02	FX/PZ = .16	MRZ= 1.1	MRY= 2.4	

4 ST UA60X60X6	PASS	BS-4.7 (C)	.175	4
17.80 C		.00	-.06	1.00

CALCULATED CAPACITIES FOR MEMB 4 UNIT - kN,m SECTION CLASS 3				
MCZ= 1.1	MCY= 2.4	PC= 101.8	PT= 190.0	MB= 1.9 PV= 35.6
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000				
PZ= 190.02	FX/PZ = .09	MRZ= 1.1	MRY= 2.4	

6 ST UA60X60X6	PASS	BS-4.7 (C)	.499	4
30.40 C		.00	.00	.00

CALCULATED CAPACITIES FOR MEMB 6 UNIT - kN,m SECTION CLASS 3				
MCZ= 1.1	MCY= 2.4	PC= 60.9	PT= 190.0	MB= 1.7 PV= 35.6
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000				
PZ= 190.02	FX/PZ = .16	MRZ= 1.1	MRY= 2.4	

7 ST UA60X60X6	PASS	BS-4.7 (C)	.108	3
6.58 C		.00	.00	.00

CALCULATED CAPACITIES FOR MEMB 7 UNIT - kN,m SECTION CLASS 3				
MCZ= 1.1	MCY= 2.4	PC= 60.9	PT= 190.0	MB= 1.7 PV= 35.6
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000				
PZ= 190.02	FX/PZ = .03	MRZ= 1.1	MRY= 2.4	

ALL UNITS ARE - KNS METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
9 ST UA60X60X6	PASS	BS-4.7 (C)	.666	4	
	40.56 C	.00	.00	.00	
CALCULATED CAPACITIES FOR MEMB 9 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 60.9 PT= 190.0 MB= 1.7 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .21 MRZ= 1.1 MRY= 2.4					
10 ST UA60X60X6	PASS	BS-4.7 (C)	.108	3	
	6.58 C	.00	.01	.00	
CALCULATED CAPACITIES FOR MEMB 10 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 60.9 PT= 190.0 MB= 1.7 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .03 MRZ= 1.1 MRY= 2.4					
16 ST UA60X60X6	PASS	BS-4.7 (C)	.666	4	
	40.56 C	.00	.02	.00	
CALCULATED CAPACITIES FOR MEMB 16 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 60.9 PT= 190.0 MB= 1.7 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .21 MRZ= 1.1 MRY= 2.4					
18 ST UA60X60X6	PASS	BS-4.7 (C)	.249	4	
	15.18 C	.00	-.02	.00	
CALCULATED CAPACITIES FOR MEMB 18 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 60.9 PT= 190.0 MB= 1.7 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .08 MRZ= 1.1 MRY= 2.4					
19 ST UA60X60X6	PASS	BS-4.7 (C)	.284	3	
	14.68 C	.00	-.01	.00	
CALCULATED CAPACITIES FOR MEMB 19 UNIT - kN,m SECTION CLASS 3					
MCZ= 1.1 MCY= 2.4 PC= 51.8 PT= 190.0 MB= 1.4 PV= 35.6					
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ= 190.02 FX/PZ = .08 MRZ= 1.1 MRY= 2.4					
20 ST UA60X60X6	PASS	BS-4.7 (C)	.100	4	
	10.14 C	.00	-.04	.00	

ALL UNITS ARE - KNS METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
--------	-------	---------------	----------------------	--------------	----------------------

CALCULATED CAPACITIES FOR MEMB 20 UNIT - kN,m SECTION CLASS 3					
MCZ=	1.1	MCY= 2.4	PC= 101.8	PT= 190.0	MB= 1.9 PV= 35.6
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ=	190.02	FX/PZ = .05	MRZ= 1.1	MRY= 2.4	

21 ST UA60X60X6	PASS	BS-4.7 (C)	.094	3	
	4.89 C	.00	.00	.00	

CALCULATED CAPACITIES FOR MEMB 21 UNIT - kN,m SECTION CLASS 3					
MCZ=	1.1	MCY= 2.4	PC= 51.8	PT= 190.0	MB= 1.4 PV= 35.6
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ=	190.02	FX/PZ = .03	MRZ= 1.1	MRY= 2.4	

22 ST UA60X60X6	PASS	BS-4.7 (C)	.049	4	
	5.00 C	.00	.00	.00	

CALCULATED CAPACITIES FOR MEMB 22 UNIT - kN,m SECTION CLASS 3					
MCZ=	1.1	MCY= 2.4	PC= 101.8	PT= 190.0	MB= 1.9 PV= 35.6
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ=	190.02	FX/PZ = .03	MRZ= 1.1	MRY= 2.4	

23 ST UA60X60X6	PASS	BS-4.7 (C)	.094	3	
	4.89 C	.00	.01	.00	

CALCULATED CAPACITIES FOR MEMB 23 UNIT - kN,m SECTION CLASS 3					
MCZ=	1.1	MCY= 2.4	PC= 51.8	PT= 190.0	MB= 1.4 PV= 35.6
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ=	190.02	FX/PZ = .03	MRZ= 1.1	MRY= 2.4	

24 ST UA60X60X6	PASS	BS-4.7 (C)	.100	4	
	10.14 C	.00	.04	1.00	

CALCULATED CAPACITIES FOR MEMB 24 UNIT - kN,m SECTION CLASS 3					
MCZ=	1.1	MCY= 2.4	PC= 101.8	PT= 190.0	MB= 1.9 PV= 35.6
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ=	190.02	FX/PZ = .05	MRZ= 1.1	MRY= 2.4	

25 ST UA60X60X6	PASS	BS-4.7 (C)	.284	3	
	14.68 C	.00	.01	.00	

CALCULATED CAPACITIES FOR MEMB 25 UNIT - kN,m SECTION CLASS 3					
MCZ=	1.1	MCY= 2.4	PC= 51.8	PT= 190.0	MB= 1.4 PV= 35.6
BUCKLING CO-EFFICIENTS m AND n : m = 1.000 n = 1.000					
PZ=	190.02	FX/PZ = .08	MRZ= 1.1	MRY= 2.4	

***** END OF TABULATED RESULT OF DESIGN *****

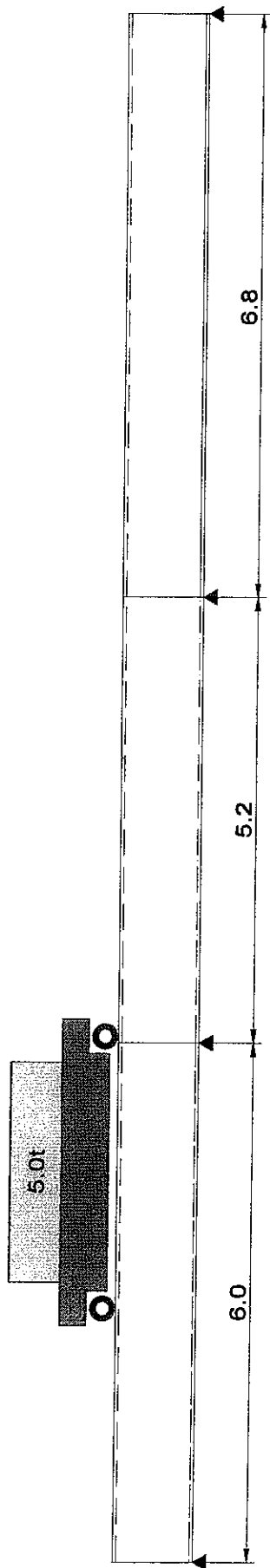
68. FINISH

***** END OF STAAD-III *****

**** DATE= MAY 23,2006 TIME= 15:35:25 ****

* For questions on STAAD-III, contact: *
* Research Engineers(Europe) Ltd *
* 10 St Mary Street, Thornbury, Bristol, BS12 2AB, UK *
* Tel: +44 1454 281080 Fax: +44 1454 415866 *

5T OVERHEAD CRANE



TITLE :5T OVER HEAD CRANE FOR SUBSTATIONS OF MPS 1 2 & 3

Data file : D:\Sadiq\My Doc..\IB-Y Structures..\DUBAI..\1225 MPS 123 CRANE DESIGN\1225CR5T.S05
Created on: 4/24/2006 12:42:19 PM

S05

**INPUT DATA**

Girder continuous [Y]/N	N
Modulus of Elasticity E (kN/m ²)	206e6
Rail height (mm)	30
Effective length factor typical span	1
Crane step size (m)	0.2
Dead Load factor	1.4
Live Load factor	1.6
Combined Load factor	1.4
Minimum spacing between cranes (m)	0
Fixity left	Pinned
Fixity right	Pinned

Dynamic load factors					Deflection limits	
Class	Vertical	Horizontal surge	Misalignment	Skewing	L/d vertical	L/d Horizontal
1	1.10	0.00	0.05	0.20	600	500
2	1.30	0.00	0.12	0.20	600	500
3	1.40	0.10	0.15	0.20	600	500
4	1.50	0.10	0.20	0.20	600	500

Section name	Designation	h (mm)	b top (mm)	tf top (mm)	b bot (mm)	tf bot (mm)	tw (mm)	r1 (mm)	r2 (mm)	β (°)	fy (MPa)
1	457x191x98 I1	467.4	192.8	19.6	192.8	19.6	11.4	10.2	0.0	0.0	275.000

Section name	Designation	h (mm)	b top (mm)	tf (mm)	tw (mm)	r1 (mm)	r2 (mm)	β (°)	fy (MPa)

Span no.	Section length (m)	Main section	Capping section
1	6	1	
2	5.2	1	
3	6.8	1	

Crane No.	Capacity (t)	Class	Weight of bridge (kN)	Weight of crab (kN)	Wheel spacing (m)	Maximum wheel load (kN)
1	5	3	67.2	3.6	3.15	42
2						



Software Consultants (Pty) Ltd
Internet: <http://www.prokon.com>
E-Mail: mail@prokon.com

Job Number

Sheet

Job Title

Client

Calcs by

Checked by

Date

DESIGN OPTION SELECTED : BS5950 - 1990
ALL OUTPUT BELOW IS FOR THE SELECTED DESIGN SECTION(S)
CHOSEN BY THE PROGRAM

CALCULATED SECTION PROPERTIES:

All combinations of Main and Capping beam encountered are listed:

Properties for section no 2

Main beam: 610x229x113 I section

Capping beam: none

Cross sectional area	A	: 14.4E3 mm ²
Second Moment of Area about major axis:	Ixx	: 874E6 mm ⁴
Second Moment of Area about minor axis:	Iyy	: 34.3E6 mm ⁴
Second Moment of Area Top flange only :	Iyt	: 17.1E6 mm ⁴
Elastic Section Modulus about major axis top	Zxt	: 2.88E6 mm ³
Elastic Section Modulus about major axis bot	Zxb	: 2.88E6 mm ³
Top Flange Section Modulus about minor axis:	Zyt	: 150E3 mm ³
Plastic Section Modulus about major axis:	Zpl	: 3.29E6 mm ³
Top Flange Plastic Modulus about minor axis:	Zpt	: 225E3 mm ³
Radius of gyration about minor axis:	ry	: 48.8 mm
St. Venant's Torsional constant	J	: 1.13E6 mm ⁴
Warping Torsional constant	Cw	: 2.98E12 mm ⁴

Web buckling capacity of this section	: 707 kN	
Maximum applied wheel load	: 94 kN	OK
Web crippling capacity of this section	: 647 kN	
Maximum applied wheel load	: 94 kN	OK

DESIGN CHECKS FOR SPAN 1

Flange Class : 1
Web Class : 1
Section Class : 1

Span length : 6.000 m
Eff Length : 6.000 m

Vertical forces only with full load factors:

BS 5950 : Part 1 : 1990 4.8.3

a) Local Capacity check

$$L_{\alpha} = \frac{F}{A_g \cdot P_y} + \frac{M_x}{M_{cx}} + \frac{M_y}{M_{cy}}$$

$$= \frac{0.000}{14\,448.320 \times 0.275} + \frac{153.630}{903.876} + \frac{0.000}{49.550}$$

$$= 0.170$$

OK

b) 4.8.3.3 Overall buckling check**4.8.3.3.1 Simplified approach**

$$O_{bc} = \frac{F}{A_g \cdot P_c} + \frac{m_x \cdot M_x}{M_b} + \frac{m_y \cdot M_y}{P_y \cdot Z_y}$$

$$= \frac{0.000}{14\,448.320 \times 0.275} + \frac{1.000 \times 153.630}{432.743} + \frac{1.000 \times 0.000}{0.275 \times 150.150}$$

$$= 0.355$$

OK

4.8.3.3.2 More exact approach

$$O_{bce} = \frac{m_x \cdot M_x}{M_{ax}} + \frac{m_y \cdot M_y}{M_{ay}}$$

$$= \frac{1.000 \times 153.630}{432.743} + \frac{1.000 \times 0.000}{49.550}$$

$$= 0.355$$

OK

Combined Vertical & Horizontal forces with combined load factors:

BS 5950 : Part 1 : 1990 4.8.3**a) Local Capacity check**

$$L_{\alpha} = \frac{F}{A_g \cdot P_y} + \frac{M_x}{M_{\alpha}} + \frac{M_y}{M_{\alpha y}}$$

$$= \frac{0.000}{14\,448.320 \times 0.275} + \frac{127.447}{903.876} + \frac{12.403}{49.550}$$

$$= 0.391$$

OK

b) 4.8.3.3 Overall buckling check**4.8.3.3.1 Simplified approach**

$$O_{bc} = \frac{F}{A_g \cdot P_c} + \frac{m_x \cdot M_x}{M_b} + \frac{m_y \cdot M_y}{P_y \cdot Z_y}$$

$$= \frac{0.000}{14\,448.320 \times 0.275} + \frac{1.000 \times 127.447}{432.743} + \frac{1.000 \times 12.403}{0.275 \times 150.150}$$

$$= 0.595$$

OK

4.8.3.3.2 More exact approach

$$O_{bcx} = \frac{m_x \cdot M_x}{M_{ax}} + \frac{m_y \cdot M_y}{M_{ay}}$$

$$= \frac{1.000 \times 127.447}{432.743} + \frac{1.000 \times 12.403}{49.550}$$

$$= 0.545$$

OK

DEFLECTION CHECKS:**At Serviceability Limit State loads with no impact**Vertical Span/deflection ratio $L/d = 3898 \geq 600$

OK

Horizontal Span/deflection ratio $L/d = 793 \geq 500$

OK

DESIGN CHECKS FOR SPAN 2

Flange Class : 1

Web Class : 1

Section Class : 1

Span length : 5.200 m

Eff Length : 5.200 m

Vertical forces only with full load factors:

BS 5950 : Part 1 : 1990 4.8.3**a) Local Capacity check**

$$I_{cx} = \frac{F}{A_g \cdot P_y} + \frac{M_x}{M_{cx}} + \frac{M_y}{M_{cy}}$$

$$= \frac{0.000}{14\,448.320 \times 0.275} + \frac{122.168}{903.876} + \frac{0.000}{49.550}$$

$$= 0.135$$

OK

b) 4.8.3.3 Overall buckling check**4.8.3.3.1 Simplified approach**

$$O_{bc} = \frac{F}{A_g \cdot P_c} + \frac{m_x \cdot M_x}{M_b} + \frac{m_y \cdot M_y}{P_y \cdot Z_y}$$

$$= \frac{0.000}{14\,448.320 \times 0.275} + \frac{1.000 \times 122.168}{503.735} + \frac{1.000 \times 0.000}{0.275 \times 150.150}$$

$$= 0.243$$

OK

4.8.3.3.2 More exact approach

$$O_{bx} = \frac{m_x \cdot M_x}{M_{ax}} + \frac{m_y \cdot M_y}{M_{ay}}$$

$$= \frac{1.000 \times 122.168}{503.735} + \frac{1.000 \times 0.000}{49.550}$$

$$= 0.243$$

OK

Combined Vertical & Horizontal forces with combined load factors:

BS 5950 : Part 1 : 1990 4.8.3**a) Local Capacity check**

$$L_{cx} = \frac{F}{A_g \cdot P_y} + \frac{M_x}{M_{cx}} + \frac{M_y}{M_{cy}}$$

$$= \frac{0.000}{14\,448.320 \times 0.275} + \frac{106.993}{903.876} + \frac{10.861}{49.550}$$

$$= 0.338$$

OK

b) 4.8.3.3 Overall buckling check**4.8.3.3.1 Simplified approach**

$$O_{bc} = \frac{F}{A_g \cdot P_c} + \frac{m_x \cdot M_x}{M_b} + \frac{m_y \cdot M_y}{P_y \cdot Z_y}$$

$$= \frac{0.000}{14\,448.320 \times 0.275} + \frac{1.000 \times 106.993}{503.735} + \frac{1.000 \times 10.861}{0.275 \times 150.150}$$

$$= 0.475$$

OK

4.8.3.3.2 More exact approach

$$O_{bx} = \frac{m_x \cdot M_x}{M_{ax}} + \frac{m_y \cdot M_y}{M_{ay}}$$

$$= \frac{1.000 \times 106.993}{503.735} + \frac{1.000 \times 10.861}{49.550}$$

$$= 0.432$$

OK

DEFLECTION CHECKS:**At Serviceability Limit State loads with no impact**Vertical Span/deflection ratio $L/d = 6192 \geq 600$

OK

Horizontal Span/deflection ratio $L/d = 1061 > 500$

OK

DESIGN CHECKS FOR SPAN 3

Flange Class : 1
Web Class : 1
Section Class : 1

Span length : 6.800 m
Eff Length : 6.800 m

Vertical forces only with full load factors:

BS 5950 : Part 1 : 1990 4.8.3**a) Local Capacity check**

$$L_{cx} = \frac{F}{A_g \cdot P_y} + \frac{M_x}{M_{cx}} + \frac{M_y}{M_{cy}}$$

$$= \frac{0.000}{14\,448.320 \times 0.275} + \frac{189.304}{903.876} + \frac{0.000}{49.550}$$

$$= 0.209$$

OK

b) 4.8.3.3 Overall buckling check**4.8.3.3.1 Simplified approach**

$$O_{bc} = \frac{F}{A_g \cdot P_c} + \frac{m_x \cdot M_x}{M_b} + \frac{m_y \cdot M_y}{P_y \cdot Z_y}$$

$$= \frac{0.000}{14\,448.320 \times 0.275} + \frac{1.000 \times 189.304}{375.760} + \frac{1.000 \times 0.000}{0.275 \times 150.150}$$

$$= 0.504$$

OK

4.8.3.3.2 More exact approach

$$O_{bce} = \frac{m_x \cdot M_x}{M_{ax}} + \frac{m_y \cdot M_y}{M_{ay}}$$

$$= \frac{1.000 \times 189.304}{375.760} + \frac{1.000 \times 0.000}{49.550}$$

$$= 0.504$$

OK

Combined Vertical & Horizontal forces with combined load factors:

BS 5950 : Part 1 : 1990 4.8.3**a) Local Capacity check**

$$L_{cc} = \frac{F}{A_g \cdot P_y} + \frac{M_x}{M_{cx}} + \frac{M_y}{M_{cy}}$$

$$= \frac{0.000}{14\,448.320 \times 0.275} + \frac{160.665}{903.876} + \frac{13.991}{49.550}$$

$$= 0.460$$

OK

b) 4.8.3.3 Overall buckling check**4.8.3.3.1 Simplified approach**

$$O_{bc} = \frac{F}{A_g \cdot P_c} + \frac{m_x \cdot M_x}{M_b} + \frac{m_y \cdot M_y}{P_y \cdot Z_y}$$

$$= \frac{0.000}{14\,448.320 \times 0.275} + \frac{1.000 \times 160.665}{375.760} + \frac{1.000 \times 13.991}{0.275 \times 150.150}$$

$$= 0.766$$

OK

4.8.3.3.2 More exact approach

$$O_{bce} = \frac{m_x \cdot M_x}{M_{cx}} + \frac{m_y \cdot M_y}{M_{cy}}$$

$$= \frac{1.000 \times 160.665}{375.760} + \frac{1.000 \times 13.991}{49.550}$$

$$= 0.710$$

OK

DEFLECTION CHECKS:**At Serviceability Limit State loads with no impact**Vertical Span/deflection ratio $L/d = 2807 \geq 600$

OK

Horizontal Span/deflection ratio $L/d = 579 \geq 500$

OK

MAXIMUM SUPPORT REACTIONS

Supp no. Rvert (kN) Rhoriz (kN)

1	125.3	8.4
2	145.2	9.7
3	133.8	8.3
4	129.7	8.7



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VERTICAL MOMENT ,SHEAR FORCE AND DEFLECTION ENVELOPES:

Span No.	Pos. (m)	M-Max (kNm)	M-Min (kNm)	V-Max (kN)	V-Min (kN)	SLS deflection (mm)
1	0.00	-0.00	-0.00	125.31	-7.15	0.000
	0.07	-0.02	-9.79	125.30	-7.15	-0.044
	0.14	-0.05	-18.67	118.08	-7.16	-0.105
	0.22	-0.07	-28.01	118.07	-7.17	-0.166
	0.29	-0.09	-37.35	118.07	-7.18	-0.226
	0.36	-0.11	-44.42	110.96	-7.19	-0.287
	0.43	-0.13	-53.31	110.95	-7.19	-0.346
	0.50	-0.15	-61.81	103.97	-7.20	-0.405
	0.58	-0.17	-67.46	103.96	-7.21	-0.464
	0.65	-0.19	-75.89	103.95	-7.22	-0.521
	0.72	-0.21	-82.43	97.10	-7.23	-0.578
	0.79	-0.23	-87.78	97.09	-7.23	-0.634
	0.86	-0.24	-95.75	97.08	-7.24	-0.689
	0.94	-0.26	-100.34	90.38	-7.25	-0.742
	1.01	-0.28	-105.38	90.37	-7.26	-0.795
	1.08	-0.29	-112.91	90.36	-7.26	-0.846
	1.15	-0.31	-115.54	83.81	-7.27	-0.895
	1.22	-0.32	-120.27	83.81	-7.28	-0.944
	1.30	-0.33	-127.34	83.80	-7.29	-0.991
	1.37	-0.35	-128.01	77.42	-10.29	-1.036
	1.44	-0.36	-132.45	77.41	-10.30	-1.079
	1.51	-0.37	-137.94	71.22	-16.68	-1.121
	1.58	-0.38	-137.78	71.21	-16.68	-1.161
	1.66	-0.40	-141.91	71.20	-16.69	-1.198
	1.73	-0.41	-145.44	65.20	-22.89	-1.234
	1.80	-0.42	-144.83	65.19	-22.90	-1.268
	1.87	-0.42	-148.66	65.18	-22.90	-1.299
	1.94	-0.43	-150.23	59.38	-28.91	-1.329
	2.02	-0.44	-149.16	59.37	-28.92	-1.357
	2.09	-0.45	-152.69	59.37	-28.92	-1.383
	2.16	-0.46	-152.30	53.78	-34.72	-1.406
	2.23	-0.46	-150.78	53.78	-34.73	-1.428
	2.30	-0.47	-153.63	48.42	-40.31	-1.448
	2.38	-0.47	-151.66	48.41	-40.32	-1.466
	2.45	-0.48	-149.69	48.41	-40.33	-1.482
	2.52	-0.48	-150.73	43.29	-45.68	-1.496
	2.59	-0.49	-148.30	43.29	-45.69	-1.508
	2.66	-0.49	-145.88	43.28	-45.70	-1.518
	2.74	-0.49	-145.11	38.42	-50.81	-1.526
	2.81	-0.49	-142.23	38.41	-50.82	-1.532
	2.88	-0.49	-140.48	38.40	-50.83	-1.536
	2.95	-0.49	-139.09	33.80	-55.69	-1.539
	3.02	-0.49	-138.00	33.79	-55.70	-1.539
	3.10	-0.49	-141.28	33.78	-55.70	-1.537
	3.17	-0.49	-141.54	30.88	-60.31	-1.534
	3.24	-0.49	-144.53	30.87	-60.31	-1.529
	3.31	-0.49	-145.56	25.34	-64.66	-1.522
	3.38	-0.49	-148.10	25.33	-64.66	-1.513
	3.46	-0.48	-150.07	25.33	-70.44	-1.502
	3.53	-0.48	-149.86	21.49	-70.45	-1.489
	3.60	-0.47	-151.94	21.48	-70.46	-1.475
	3.67	-0.47	-151.96	21.48	-77.57	-1.458
	3.74	-0.46	-151.44	17.89	-77.57	-1.439
	3.82	-0.46	-153.07	17.88	-77.58	-1.419
	3.89	-0.45	-151.13	17.88	-84.57	-1.396



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4.10	-0.43	-147.59	11.44	-91.43	-1.316
4.18	-0.42	-146.47	11.43	-91.44	-1.286
4.25	-0.41	-147.19	11.43	-91.45	-1.253
4.32	-0.40	-141.33	8.58	-98.15	-1.219
4.39	-0.39	-139.91	8.57	-98.16	-1.182
4.46	-0.38	-138.86	8.56	-104.71	-1.143
4.54	-0.36	-132.36	5.96	-104.72	-1.103
4.61	-0.35	-130.63	5.95	-104.73	-1.060
4.68	-0.34	-127.63	5.94	-111.10	-1.016
4.75	-0.33	-120.67	4.12	-111.11	-0.970
4.82	-0.31	-118.65	4.11	-111.12	-0.923
4.90	-0.30	-113.68	4.10	-117.32	-0.874
4.97	-0.28	-106.27	4.09	-117.32	-0.823
5.04	-0.27	-103.94	4.09	-117.33	-0.771
5.11	-0.25	-97.02	4.08	-123.34	-0.718
5.18	-0.23	-89.16	4.07	-123.34	-0.663
5.26	-0.21	-85.96	4.06	-129.14	-0.607
5.33	-0.20	-77.64	4.06	-129.15	-0.550
5.40	-0.18	-69.33	4.05	-129.16	-0.492
5.47	-0.16	-64.32	4.04	-134.74	-0.433
5.54	-0.14	-55.55	4.03	-134.75	-0.373
5.62	-0.12	-46.78	4.02	-134.76	-0.313
5.69	-0.10	-39.97	4.02	-140.11	-0.252
5.76	-0.08	-30.75	4.01	-140.12	-0.190
5.83	-0.05	-21.52	4.00	-140.13	-0.129
5.90	-0.03	-12.90	3.99	-145.24	-0.066
5.98	-0.01	-3.23	3.98	-145.25	-0.005
6.00	-0.00	-0.00	127.30	-26.48	-0.002
6.07	-0.02	-9.20	127.29	-26.49	-0.018
6.14	-0.04	-17.37	120.52	-26.49	-0.058
6.22	-0.06	-26.05	120.51	-26.50	-0.097
6.29	-0.08	-34.73	120.51	-26.51	-0.137
6.36	-0.10	-40.80	113.59	-26.52	-0.176
6.43	-0.11	-48.96	113.58	-26.53	-0.215
6.50	-0.13	-56.74	106.56	-26.53	-0.253
6.58	-0.15	-61.11	106.55	-26.54	-0.290
6.65	-0.16	-68.75	106.54	-26.55	-0.327
6.72	-0.18	-74.50	99.45	-26.56	-0.363
6.79	-0.19	-78.29	99.44	-26.57	-0.398
6.86	-0.21	-85.40	99.43	-26.57	-0.432
6.94	-0.22	-89.12	92.29	-26.58	-0.465
7.01	-0.23	-92.33	92.29	-26.59	-0.497
7.08	-0.24	-98.92	92.28	-26.60	-0.528
7.15	-0.26	-100.62	88.17	-26.60	-0.557
7.22	-0.27	-103.24	88.16	-26.61	-0.585
7.30	-0.28	-109.31	85.13	-26.62	-0.611
7.37	-0.29	-108.98	83.83	-26.63	-0.636
7.44	-0.30	-111.02	83.82	-26.64	-0.659
7.51	-0.31	-115.44	79.20	-26.64	-0.681
7.58	-0.31	-114.21	79.19	-26.65	-0.702
7.66	-0.32	-115.67	74.29	-26.66	-0.721
7.73	-0.33	-118.06	74.28	-26.67	-0.739
7.80	-0.34	-116.31	74.28	-26.68	-0.755
7.87	-0.34	-117.19	69.10	-26.68	-0.770
7.94	-0.35	-117.55	69.10	-30.03	-0.783
8.02	-0.35	-115.28	69.09	-30.04	-0.795
8.09	-0.36	-117.46	63.66	-30.05	-0.806
8.16	-0.36	-115.86	63.65	-36.81	-0.815
8.23	-0.36	-119.49	63.64	-36.82	-0.822
8.30	-0.37	-120.88	57.96	-43.40	-0.829



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8.52	-0.37	-121.59	52.02	-49.81	-0.839
8.59	-0.37	-119.95	52.01	-49.82	-0.840
8.66	-0.37	-121.96	45.86	-49.83	-0.839
8.74	-0.37	-120.73	45.85	-56.02	-0.837
8.81	-0.37	-119.76	45.84	-56.03	-0.834
8.88	-0.37	-120.21	39.47	-56.04	-0.830
8.95	-0.36	-118.31	39.46	-62.01	-0.825
9.02	-0.36	-117.99	39.46	-62.02	-0.818
9.10	-0.36	-117.99	32.88	-62.03	-0.809
9.17	-0.35	-115.80	32.87	-67.78	-0.800
9.24	-0.35	-118.20	32.86	-67.78	-0.788
9.31	-0.34	-115.45	26.09	-73.29	-0.776
9.38	-0.34	-117.34	26.09	-73.30	-0.762
9.46	-0.33	-118.65	20.14	-73.31	-0.746
9.53	-0.32	-114.39	20.14	-78.57	-0.729
9.60	-0.32	-115.75	20.13	-78.58	-0.711
9.67	-0.31	-115.04	20.12	-78.59	-0.691
9.74	-0.30	-110.18	20.11	-83.59	-0.670
9.82	-0.29	-111.02	20.10	-83.59	-0.647
9.89	-0.28	-108.28	20.10	-83.60	-0.623
9.96	-0.27	-102.85	20.09	-88.33	-0.597
10.03	-0.26	-103.17	20.08	-88.34	-0.570
10.10	-0.25	-98.40	20.07	-92.78	-0.541
10.18	-0.23	-92.39	20.06	-92.79	-0.511
10.25	-0.22	-92.18	20.06	-92.79	-0.479
10.32	-0.21	-85.39	20.05	-96.94	-0.447
10.39	-0.20	-78.80	20.04	-96.95	-0.413
10.46	-0.18	-76.75	20.03	-103.38	-0.377
10.54	-0.17	-69.24	20.03	-103.39	-0.341
10.61	-0.15	-62.07	20.02	-103.40	-0.304
10.68	-0.13	-57.99	20.01	-110.50	-0.267
10.75	-0.12	-49.96	20.00	-110.50	-0.228
10.82	-0.10	-42.21	19.99	-110.51	-0.189
10.90	-0.08	-36.11	19.99	-117.53	-0.149
10.97	-0.06	-27.56	19.98	-117.54	-0.109
11.04	-0.04	-19.22	19.97	-117.55	-0.069
11.11	-0.02	-11.09	19.96	-124.46	-0.028
11.18	-0.00	-2.02	19.95	-124.47	0.007
11.20	-0.00	-0.00	153.72	-2.32	0.002
11.27	-0.03	-10.24	153.71	-2.33	-0.037
11.34	-0.05	-19.68	149.49	-2.34	-0.069
11.42	-0.08	-29.51	149.48	-2.34	-0.204
11.49	-0.10	-39.35	149.47	-2.35	-0.288
11.56	-0.13	-47.20	145.04	-2.36	-0.372
11.63	-0.15	-56.63	145.03	-2.37	-0.455
11.70	-0.17	-65.69	140.40	-2.37	-0.538
11.78	-0.20	-72.32	140.39	-2.38	-0.620
11.85	-0.22	-81.35	140.38	-2.39	-0.701
11.92	-0.24	-88.51	135.55	-2.40	-0.782
11.99	-0.26	-95.04	135.54	-2.41	-0.861
12.06	-0.28	-103.68	135.53	-2.41	-0.940
12.14	-0.30	-108.93	130.52	-2.42	-1.017
12.21	-0.32	-115.38	130.51	-2.43	-1.093
12.28	-0.34	-123.61	130.50	-2.44	-1.167
12.35	-0.36	-126.96	125.31	-2.45	-1.241
12.42	-0.38	-133.31	125.30	-2.45	-1.312
12.50	-0.39	-141.15	125.30	-4.33	-1.382
12.57	-0.41	-142.59	119.94	-4.34	-1.450
12.64	-0.42	-148.85	119.93	-4.35	-1.517
12.71	-0.44	-155.16	114.41	-6.44	-1.581



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12.93	-0.48	-166.39	108.72	-8.76	-1.761
13.00	-0.49	-166.66	108.71	-8.76	-1.816
13.07	-0.51	-172.74	108.71	-11.26	-1.869
13.14	-0.52	-175.24	102.89	-11.27	-1.920
13.22	-0.53	-175.10	102.89	-11.28	-1.968
13.29	-0.54	-181.09	102.88	-13.99	-2.015
13.36	-0.55	-181.68	96.93	-13.99	-2.059
13.43	-0.56	-181.21	96.92	-14.00	-2.100
13.50	-0.57	-186.67	90.85	-16.92	-2.139
13.58	-0.58	-185.73	90.84	-16.93	-2.176
13.65	-0.59	-185.17	90.84	-16.93	-2.210
13.72	-0.59	-188.72	84.65	-20.06	-2.241
13.79	-0.60	-187.39	84.64	-20.07	-2.270
13.86	-0.61	-186.73	84.63	-23.41	-2.296
13.94	-0.61	-188.38	78.33	-23.42	-2.320
14.01	-0.62	-186.65	78.33	-23.43	-2.342
14.08	-0.62	-185.90	78.32	-26.99	-2.360
14.15	-0.62	-185.65	71.92	-27.00	-2.377
14.22	-0.63	-183.51	71.91	-27.01	-2.391
14.30	-0.63	-182.67	71.91	-30.79	-2.402
14.37	-0.63	-180.51	65.42	-30.80	-2.411
14.44	-0.63	-177.97	65.41	-30.80	-2.417
14.51	-0.63	-175.92	58.84	-34.81	-2.421
14.58	-0.64	-172.98	58.83	-34.81	-2.422
14.66	-0.64	-174.38	58.82	-39.04	-2.421
14.73	-0.63	-177.02	50.61	-39.04	-2.418
14.80	-0.63	-179.66	50.60	-39.05	-2.412
14.87	-0.63	-180.79	45.97	-43.48	-2.403
14.94	-0.63	-183.03	45.96	-43.49	-2.392
15.02	-0.63	-185.27	45.95	-43.50	-2.378
15.09	-0.62	-184.81	41.12	-48.14	-2.363
15.16	-0.62	-186.65	41.11	-48.14	-2.344
15.23	-0.61	-188.49	41.11	-48.15	-2.323
15.30	-0.61	-186.43	36.09	-52.98	-2.300
15.38	-0.60	-187.87	36.08	-52.99	-2.274
15.45	-0.60	-189.30	36.08	-53.00	-2.246
15.52	-0.59	-185.65	30.88	-58.01	-2.215
15.59	-0.58	-186.69	30.88	-58.02	-2.182
15.66	-0.57	-186.41	25.52	-63.21	-2.146
15.74	-0.56	-182.48	25.51	-63.22	-2.108
15.81	-0.56	-183.11	25.50	-63.23	-2.067
15.88	-0.55	-180.93	19.98	-68.59	-2.024
15.95	-0.53	-176.91	19.97	-68.60	-1.979
16.02	-0.52	-177.15	19.96	-68.60	-1.931
16.10	-0.51	-173.05	14.29	-74.12	-1.880
16.17	-0.50	-168.94	14.29	-74.13	-1.827
16.24	-0.49	-168.78	14.28	-74.14	-1.772
16.31	-0.47	-162.78	8.47	-79.81	-1.714
16.38	-0.46	-158.58	8.46	-79.82	-1.654
16.46	-0.45	-157.45	5.85	-85.63	-1.592
16.53	-0.43	-150.12	5.84	-85.64	-1.527
16.60	-0.42	-145.82	5.83	-85.65	-1.461
16.67	-0.40	-142.79	5.83	-91.59	-1.392
16.74	-0.38	-135.06	5.82	-91.60	-1.321
16.82	-0.37	-130.67	5.81	-91.61	-1.249
16.89	-0.35	-125.73	5.80	-97.68	-1.175
16.96	-0.33	-117.60	5.79	-97.69	-1.099
17.03	-0.31	-113.12	5.79	-97.70	-1.022
17.10	-0.29	-106.28	5.78	-103.89	-0.944
17.18	-0.27	-97.74	5.77	-103.89	-0.864



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17.39	-0.21	-75.49	5.75	-110.21	-0.619
17.46	-0.18	-69.52	5.74	-116.60	-0.535
17.54	-0.16	-60.18	5.73	-116.61	-0.450
17.61	-0.14	-50.85	5.72	-116.62	-0.364
17.68	-0.11	-43.28	5.71	-123.11	-0.278
17.75	-0.09	-33.54	5.71	-123.11	-0.192
17.82	-0.06	-23.81	5.70	-123.12	-0.105
17.90	-0.04	-14.64	5.69	-129.69	-0.019
17.97	-0.01	-4.51	5.68	-129.70	-0.012

HORIZONTAL MOMENT ,SHEAR FORCE AND DEFLECTION ENVELOPES:

Span No.	Pos. (m)	M-Max (kNm)	M-Min (kNm)	V-Max (kN)	V-Min (kN)	SLS deflection (mm)
1	0.00	0.00	0.00	8.36	-5.27	0.000
	0.07	0.32	-0.65	8.36	-5.27	-0.215
	0.14	0.63	-1.25	7.88	-5.20	-0.511
	0.22	0.95	-1.87	7.88	-5.20	-0.805
	0.29	1.27	-2.49	7.88	-5.20	-1.099
	0.36	1.59	-2.96	7.41	-5.12	-1.390
	0.43	1.90	-3.56	7.41	-5.12	-1.679
	0.50	2.22	-4.12	6.94	-5.03	-1.966
	0.58	2.54	-4.50	6.94	-5.03	-2.250
	0.65	2.85	-5.06	6.94	-5.03	-2.529
	0.72	3.17	-5.50	6.48	-4.93	-2.805
	0.79	3.49	-5.86	6.48	-4.93	-3.076
	0.86	3.81	-6.39	6.48	-4.93	-3.341
	0.94	4.12	-6.69	6.03	-4.82	-3.601
	1.01	4.44	-7.03	6.03	-4.82	-3.855
	1.08	4.76	-7.53	6.03	-4.82	-4.102
	1.15	5.07	-7.71	5.60	-4.71	-4.342
	1.22	5.39	-8.02	5.60	-4.71	-4.578
	1.30	5.71	-8.49	5.60	-4.71	-4.805
	1.37	6.03	-8.54	5.17	-4.58	-5.024
	1.44	6.34	-8.83	5.17	-4.58	-5.235
	1.51	6.66	-9.20	4.76	-4.45	-5.437
	1.58	6.98	-9.19	4.76	-4.45	-5.629
	1.66	7.29	-9.46	4.76	-4.45	-5.812
	1.73	7.61	-9.70	4.35	-4.31	-5.985
	1.80	7.93	-9.66	4.35	-4.31	-6.148
	1.87	8.25	-9.91	4.35	-4.31	-6.302
	1.94	8.56	-10.02	4.16	-4.16	-6.446
	2.02	8.88	-9.95	4.16	-4.16	-6.580
	2.09	9.20	-10.18	4.16	-4.16	-6.705
	2.16	9.51	-10.16	4.23	-4.23	-6.821
	2.23	9.83	-10.05	4.23	-4.23	-6.927
	2.30	10.15	-10.24	4.38	-4.38	-7.023
	2.38	10.47	-10.47	4.38	-4.38	-7.114
	2.45	10.78	-10.78	4.38	-4.38	-7.211
	2.52	11.10	-11.10	4.55	-4.55	-7.296
	2.59	11.42	-11.42	4.55	-4.55	-7.370
	2.66	11.73	-11.73	4.55	-4.55	-7.432
	2.74	11.86	-11.86	4.72	-4.72	-7.482
	2.81	12.17	-12.17	4.72	-4.72	-7.521
	2.88	12.48	-12.48	4.72	-4.72	-7.546
	2.95	12.36	-12.36	4.90	-4.90	-7.559
	3.02	12.26	-12.26	4.90	-4.90	-7.562
	3.10	12.55	-12.55	4.90	-4.90	-7.560
	3.17	12.28	-12.28	5.09	-5.09	-7.545



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3.38	12.07	-12.07	5.29	-5.29	-7.424
3.46	12.08	-12.08	5.29	-5.29	-7.359
3.53	12.10	-12.10	5.49	-5.49	-7.282
3.60	11.75	-11.75	5.49	-5.49	-7.194
3.67	11.81	-11.81	5.49	-5.49	-7.095
3.74	11.67	-11.67	5.69	-5.69	-6.990
3.82	11.30	-11.30	5.69	-5.69	-6.881
3.89	11.42	-11.42	5.69	-5.69	-6.771
3.96	11.12	-11.12	5.90	-5.90	-6.652
4.03	10.73	-10.73	5.90	-5.90	-6.523
4.10	10.87	-10.87	6.11	-6.11	-6.385
4.18	10.46	-10.46	6.11	-6.11	-6.236
4.25	10.10	-10.10	6.11	-6.11	-6.078
4.32	10.10	-10.10	6.32	-6.55	-5.910
4.39	9.67	-9.67	6.32	-6.55	-5.733
4.46	9.36	-9.36	6.32	-6.99	-5.546
4.54	9.21	-9.21	6.53	-6.99	-5.349
4.61	8.76	-8.76	6.53	-6.99	-5.142
4.68	8.51	-8.51	6.53	-7.41	-4.928
4.75	8.20	-8.20	6.74	-7.41	-4.707
4.82	7.73	-7.91	6.74	-7.41	-4.477
4.90	7.53	-7.58	6.74	-7.83	-4.239
4.97	7.07	-7.09	6.94	-7.83	-3.994
5.04	6.58	-6.93	6.94	-7.83	-3.741
5.11	6.33	-6.47	7.14	-8.23	-3.482
5.18	5.82	-5.95	7.14	-8.23	-3.216
5.26	5.31	-5.73	7.14	-8.62	-2.945
5.33	4.98	-5.18	7.33	-8.62	-2.668
5.40	4.45	-4.62	7.33	-8.62	-2.386
5.47	3.91	-4.29	7.33	-8.99	-2.101
5.54	3.51	-3.71	7.52	-8.99	-1.811
5.62	2.95	-3.12	7.52	-8.99	-1.518
5.69	2.40	-2.67	7.52	-9.35	-1.222
5.76	1.91	-2.05	7.70	-9.35	-0.924
5.83	1.34	-1.44	7.70	-9.35	-0.624
5.90	0.79	-0.86	7.87	-9.69	-0.335
5.98	0.20	-0.22	7.87	-9.69	-0.063
6.00	0.00	0.00	8.50	-7.10	-0.024
6.07	0.58	-0.61	8.50	-6.84	-0.202
6.14	1.15	-1.16	8.04	-6.84	-0.398
6.22	1.73	-1.74	8.04	-6.84	-0.602
6.29	2.21	-2.32	8.04	-6.56	-0.807
6.36	2.76	-2.76	7.58	-6.56	-1.012
6.43	3.31	-3.31	7.58	-6.56	-1.216
6.50	3.70	-3.79	7.11	-6.28	-1.417
6.58	4.23	-4.23	7.11	-6.28	-1.616
6.65	4.76	-4.76	7.11	-6.28	-1.813
6.72	5.05	-5.05	6.64	-6.00	-2.006
6.79	5.56	-5.56	6.64	-6.00	-2.195
6.86	5.95	-5.95	6.64	-5.71	-2.381
6.94	6.27	-6.27	6.16	-5.71	-2.564
7.01	6.75	-6.75	6.16	-5.71	-2.743
7.08	6.98	-6.98	6.16	-5.55	-2.917
7.15	7.34	-7.34	5.89	-5.54	-3.085
7.22	7.80	-7.80	5.89	-5.54	-3.248
7.30	7.87	-7.87	5.69	-5.54	-3.406
7.37	8.28	-8.28	5.60	-5.50	-3.556
7.44	8.71	-8.71	5.60	-5.50	-3.701
7.51	8.66	-8.66	5.43	-5.43	-3.838
7.58	9.07	-9.07	5.43	-5.43	-3.968
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7.80	9.73	-9.73	5.34	-5.34	-4.309
7.87	9.93	-9.93	5.34	-5.34	-4.406
7.94	9.88	-9.88	5.22	-5.22	-4.499
8.02	10.25	-10.25	5.22	-5.22	-4.584
8.09	10.44	-10.44	5.22	-5.22	-4.660
8.16	10.30	-10.30	5.08	-5.08	-4.726
8.23	10.62	-10.62	5.08	-5.08	-4.782
8.30	10.75	-10.75	4.91	-4.91	-4.828
8.38	10.54	-10.54	4.91	-4.91	-4.863
8.45	10.86	-10.86	4.91	-4.91	-4.888
8.52	10.81	-10.81	4.73	-4.73	-4.901
8.59	10.66	-10.66	4.73	-4.73	-4.903
8.66	10.84	-10.84	4.73	-4.73	-4.903
8.74	10.73	-10.73	4.72	-4.72	-4.892
8.81	10.65	-10.65	4.72	-4.72	-4.871
8.88	10.69	-10.69	4.91	-4.91	-4.838
8.95	10.52	-10.52	4.91	-4.91	-4.795
9.02	10.49	-10.49	4.91	-4.91	-4.742
9.10	10.49	-10.49	5.08	-5.08	-4.679
9.17	10.16	-10.16	5.08	-5.08	-4.606
9.24	9.93	-9.93	5.08	-5.08	-4.524
9.31	10.05	-10.05	5.22	-5.22	-4.433
9.38	9.67	-9.67	5.22	-5.22	-4.334
9.46	9.48	-9.48	5.34	-5.34	-4.231
9.53	9.44	-9.44	5.34	-5.34	-4.119
9.60	9.03	-9.03	5.34	-5.34	-3.998
9.67	8.89	-8.89	5.43	-5.43	-3.870
9.74	8.69	-8.69	5.43	-5.58	-3.734
9.82	8.26	-8.26	5.43	-5.58	-3.591
9.89	8.15	-8.15	5.50	-5.58	-3.441
9.96	7.80	-7.80	5.50	-5.90	-3.285
10.03	7.35	-7.35	5.50	-5.90	-3.122
10.10	7.25	-7.25	5.54	-6.19	-2.954
10.18	6.77	-6.77	5.54	-6.19	-2.780
10.25	6.30	-6.30	5.54	-6.19	-2.601
10.32	6.10	-6.10	5.64	-6.47	-2.417
10.39	5.61	-5.61	5.64	-6.47	-2.229
10.46	5.11	-5.12	5.64	-6.90	-2.038
10.54	4.82	-4.82	5.92	-6.90	-1.844
10.61	4.30	-4.30	5.92	-6.90	-1.645
10.68	3.78	-3.87	5.92	-7.37	-1.444
10.75	3.40	-3.40	6.19	-7.37	-1.239
10.82	2.85	-2.85	6.19	-7.37	-1.032
10.90	2.37	-2.41	6.19	-7.84	-0.822
10.97	1.83	-1.84	6.46	-7.84	-0.611
11.04	1.26	-1.28	6.46	-7.84	-0.403
11.11	0.72	-0.74	6.72	-8.30	-0.203
11.18	0.13	-0.13	6.72	-8.30	-0.117
11.20	0.00	0.00	10.25	-8.00	-0.026
11.27	0.58	-0.68	10.25	-7.86	-0.213
11.34	1.16	-1.31	9.97	-7.86	-0.440
11.42	1.75	-1.97	9.97	-7.86	-0.988
11.49	2.26	-2.62	9.97	-7.72	-1.396
11.56	2.82	-3.15	9.67	-7.72	-1.802
11.63	3.38	-3.78	9.67	-7.72	-2.206
11.70	3.82	-4.38	9.36	-7.57	-2.608
11.78	4.37	-4.82	9.36	-7.57	-3.006
11.85	4.92	-5.43	9.36	-7.57	-3.400
11.92	5.29	-5.90	9.04	-7.41	-3.791
11.99	5.81	-6.34	9.04	-7.41	-4.176



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12.21	7.15	-7.69	8.71	-7.25	-5.300
12.28	7.41	-8.24	8.71	-7.08	-5.662
12.35	7.89	-8.47	8.36	-7.08	-6.017
12.42	8.38	-8.89	8.36	-7.08	-6.364
12.50	8.55	-9.41	8.36	-6.91	-6.703
12.57	9.03	-9.51	8.00	-6.91	-7.034
12.64	9.50	-9.93	8.00	-6.91	-7.355
12.71	9.61	-10.35	7.63	-6.74	-7.667
12.78	10.06	-10.39	7.63	-6.74	-7.968
12.86	10.47	-10.80	7.63	-6.56	-8.259
12.93	10.55	-11.10	7.25	-6.56	-8.539
13.00	10.99	-11.11	7.25	-6.56	-8.807
13.07	11.25	-11.52	7.25	-6.39	-9.064
13.14	11.39	-11.69	6.87	-6.39	-9.309
13.22	11.81	-11.81	6.87	-6.39	-9.546
13.29	11.92	-12.08	6.87	-6.21	-9.772
13.36	12.13	-12.13	6.47	-6.21	-9.985
13.43	12.53	-12.53	6.47	-6.21	-10.186
13.50	12.48	-12.48	6.06	-6.04	-10.375
13.58	12.75	-12.75	6.06	-6.04	-10.551
13.65	13.14	-13.14	6.06	-6.04	-10.716
13.72	12.94	-12.94	5.87	-5.87	-10.868
13.79	13.27	-13.27	5.87	-5.87	-11.008
13.86	13.52	-13.52	5.70	-5.70	-11.136
13.94	13.33	-13.33	5.70	-5.70	-11.251
14.01	13.68	-13.68	5.70	-5.70	-11.355
14.08	13.78	-13.78	5.54	-5.54	-11.446
14.15	13.66	-13.66	5.54	-5.54	-11.525
14.22	13.99	-13.99	5.54	-5.54	-11.593
14.30	13.94	-13.94	5.38	-5.38	-11.648
14.37	13.67	-13.67	5.38	-5.38	-11.690
14.44	13.75	-13.75	5.38	-5.38	-11.721
14.51	13.56	-13.56	5.23	-5.23	-11.740
14.58	13.28	-13.28	5.23	-5.23	-11.747
14.66	13.43	-13.43	5.09	-5.09	-11.741
14.73	13.48	-13.48	5.09	-5.09	-11.724
14.80	13.15	-13.15	5.09	-5.09	-11.694
14.87	12.83	-12.83	4.95	-4.95	-11.653
14.94	12.50	-12.50	4.95	-4.95	-11.600
15.02	12.18	-12.35	4.95	-4.95	-11.534
15.09	11.85	-12.32	4.81	-4.81	-11.457
15.16	11.53	-12.44	4.81	-4.81	-11.367
15.23	11.21	-12.57	4.81	-4.81	-11.266
15.30	10.88	-12.43	4.69	-4.69	-11.152
15.38	10.56	-12.52	4.69	-4.69	-11.027
15.45	10.23	-12.62	4.69	-4.69	-10.890
15.52	9.91	-12.38	4.56	-4.56	-10.740
15.59	9.59	-12.45	4.56	-4.56	-10.579
15.66	9.26	-12.43	4.45	-4.45	-10.406
15.74	8.94	-12.17	4.45	-4.45	-10.221
15.81	8.61	-12.21	4.45	-4.45	-10.024
15.88	8.29	-12.06	4.34	-4.58	-9.816
15.95	7.96	-11.80	4.34	-4.58	-9.595
16.02	7.68	-11.81	4.34	-4.58	-9.363
16.10	7.40	-11.54	4.24	-4.95	-9.118
16.17	7.12	-11.27	4.24	-4.95	-8.862
16.24	6.84	-11.26	4.24	-4.95	-8.594
16.31	6.56	-10.86	4.15	-5.33	-8.315
16.38	6.28	-10.58	4.15	-5.33	-8.023



Software Consultants (Pty) Ltd
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Job Number		Sheet
Job Title		
Client		
Calcs by	Checked by	Date

16.46	6.00	-10.50	4.25	-5.72	-7.721
16.53	5.72	-10.01	4.25	-5.72	-7.408
16.60	5.44	-9.73	4.25	-5.72	-7.084
16.67	5.16	-9.52	4.34	-6.11	-6.751

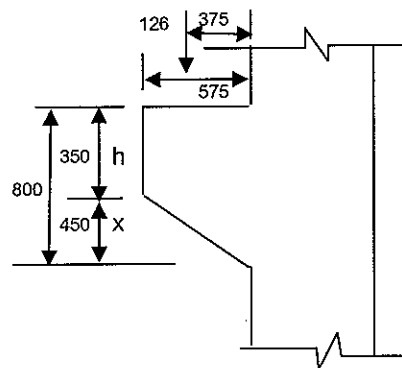
BEL-YOAHAH Architectural & Eng. Consultants

P.O Box : 827, Dubai, United Arab Emirates.

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CORBEL DESIGN**JOB No. : 1225****DATE : April 25, 2006****STRUCTURE : Substation - MPS****Design ref. B S 8110 : Part 1 : 1985****CORBEL REFERENCE : CR1**

Fcu	35	N/mm²	Fy	425	N/mm²	Cover	40	mm
Corbel size (B x H)	450	x	800	mm				
Corbel Projection from Column face, w	575	mm						
Load on corbel (Pu)	126	kn						
Distance between Column face and Load point	375	mm						
Depth of Corbel upto slope, h	350	mm						
Depth of Corbel at slope, x	450	mm						
Angle of slope	45 °							

**Design Of Corbel**

Effective Depth of Corbel, d	790	mm				
Min distance of "x" $x_{min} = 5 \cdot F_v / (\sin 2\theta \cdot B \cdot F_{cu})$	40.00	<	450		OK	
Effective length of Corbel, a_v	425	mm				
Distance " a_v " Shall not be greater than 0.6d	425	<	474		OK	
Distance "h" Shall be not less than 0.40H and not greater than 0.6H	320	<	350	<	480	OK

BM and Main Reinforcement

Max. Area of Steel (1.3%)	4622	mm ²			
Min. Area of Steel (0.4%)	1422	mm ²			
Max Bending Moment	53.55	Knm			
Area of Steel	256.3	mm ²			

Provide Main Bar : Y 20 mm 5 Nos

Check, A_{st} Provided	1571	mm ²	>	1422	<	4622	mm ²	OK
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Shear Reinforcement

Actual Shear Stress	0.35	N/mm ²			
$100 \cdot A_{st} / b \cdot d$	0.44				

From Table 3.9 of BS 8110-1 : 1985

Allowable Shear Stress		0.45	N/mm ²		
Shear Resistance of section	(2*B*d ² *vc)/a _v	594.7	Kn	>	126.0 Kn
	A _{sv} /S _v	0.49			Safe

Stirrups	2	L - Y	10	@	75	mm c/c	$A_{sv} / s_{v_{pro}}$	2.09	OK
-----------------	---	-------	----	---	----	--------	------------------------	------	----

Hence Provide

Corbel size	450	x	800		
Main bar	Y 20	mm	5	Nos.	
Stirrups	Y 10	mm @	75	mm c/c	

BEL-YOAHAAH Architectural & Eng. Consultants

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COLUMN BASE PLATE CONNECTION**Steel structures****JOB No. : 1225****DATE : April 25, 2006****STRUCTURE : MPS Building****Design ref. : B S 5950 : Part 1 : 1990****Column Section : UC 457x191x98 kg/m****Design : Subject to Axial Load, BM and Shear****LOAD COMBINATION CONSIDERED**

Load on Column (Nu) 146 kn

Size of Base plate (Hp x Bp) 400 x 400 mm

Base plate thickness (t) 20 mm

Concrete Grade C 35

Thickness of weld (t_w) 6 mmDesign strength of weld (P_w) in Table 36 215 N/mm²Design strength P_y from Table 6 265 N/mm²

Details of Anchor bolt (dp) Grade 8.8 Dia. 20 mm

Design Tension strength of bolt in Table 32 450 N/mm²

Edge Distance @ LHS 'el' for Major axis-XX 50 mm

Edge Distance @ RHS 'er' for Major axis-XX 50 mm

Edge Distance @ LHS 'el' for Minor axis-YY 50 mm

Edge Distance @ RHS 'er' for Minor axis-YY 50 mm

Provide, 2 Rows of bolts

2 Columns of bolts

Section UC 457x191x98 kg/m (Grade 43 Steel)

Depth (Dc) 467 mm

Breadth (Bc) 193 mm

Depth between fillets (dc) 407 mm

Web thickness (tc) 11.4 mm

Flang thickness (Tc) 19.6 mm

Check for Presence of Tension &**Compression under the Base plate :****Under BM and Axial Force**

$$M_{max} = 0.6 \cdot F_{cu} \cdot B_p \cdot X \cdot (H - X/2) - N \cdot (H - H_p/2)$$

Where, H = H_p-e_{r,xx}

Therefore, X = 7.5 or 692.5 mm

Under Compression C = 63.3 Kn

Under Tension T = C - N 0.0 Kn (Divided between 2 Bolts)

Tension due to B_{myy} B_{myy} / (B_p-e_r) 0.0 Kn (Divided between 2 Bolts)

Therefore, Tension Per Bolt 0.0 Kn

Use M 20 Grade 8.8 Anchor Bolts

Tension Capacity of each Bolt 108.9 Kn > 0.0 Safe

Design of Base Plate**Compression side Bending :**Bending Moment 'Mc' 0.6 * F_{cu} * X * (e - X/2) 6362.3 Nmm

Therefore, Thickness of Plate reqd. 9.8 mm > 20 Safe

Tension side Bending :Bending Moment 'Mc' T * m Where m = e_{r,xx} - t_w 0.0 Nmm

Therefore, Thickness of Plate reqd. 0.0 mm > 20 Safe

Provide Base plate 400 x 400 x 20 mm**Column to Base plate Welds**

Assuming that the moment is carried by the column

flanges, the force in each flange

$$V_f = M/(D_c - T_c)$$

0.00 kn

Length of weld resisting tensile force

$$L_w = 2 \cdot B_c$$

386 mm

Use 6 mm fillet weld

Leg length (s)

6 mm

Stress on the weld due to tensile force (fwt)

$$f_{wt} = v_f / (l_w \cdot 0.7 \cdot s)$$

0.00 N/mm²

Design strength of weld (Pw) in Table 36

215 N/mm²

Check, fwt < Pw 0 <

215 N/mm² Safe

Length of weld resisting shear (Lwv)

$$L_{wv} = 2 \cdot d_c$$

814 mm

Stress on the weld due to shear force

$$F_{wv} = F_v / (L_{wv} \cdot 0.7 \cdot s)$$

0.000 N/mm²

Check, fvw < Pw 0 <

215 N/mm² Safe

Use 6 mm fillet weld

Programme created by Eng M. I. SADIQ, B.E (Struc.)

COLUMNS & WALLS

COLUMN LOADS

JOB No. : 1225

DATE : March 13, 2006

STRUCTURE : SUBSTATION - MPS1, 2 & 3

Note :- All Loads are Ultimate load in KN

Col. Grid Ref.	At Tie Beam Reactions (a)	At 1st Flr Beam Reactions (b)	At 2nd Flr Beam Reactions (c)	No. of Typ. Floors	At Roof Beam Reactions (d)	Total Load (a+b+c+d)	Self Wt. of Col	Gross Load on Column
B/4	176	586	191		188	1141	230	1371
B/5	88	388	542		517	1535	101	1636
B/8	231	720	303		377	1631	230	1861
D/5	974					974	20	994
E/2	184	281	299		323	1087	135	1222
E/4	723	437	1215		1084	3459	180	3639
E/8	105	100	711		951	1867	180	2047
E/5	1104					1104	20	1124
F/2	217	541	519		508	1784	180	1964
F/4	847	625	1468		1297	4238	180	4418
F/8	105	100	909		964	2077	180	2257
F/4	712					712	20	732
F/5	1066					1066	20	1086
G/2	229	417	366		367	1379	101	1480
G/4	607	554	666		526	2353	180	2533
G/5	743	344	900		684	2670	135	2805
G/8	300	363	536		517	1717	180	1897
H/1-2		668	418		208	1294	417	1711
H-1/4	358	792	337		284	1770	230	2000
I/5	452	576	435		321	1785	101	1886
I/8	331	382	244		234	1191	230	1421
J/4	168	644	196			1008	76	1084
J/5	91	236				327	47	374
J/7	24	227				251	78	329
K/1	223	603	233			1059	47	1106
K/3	100	990	344			1434	47	1481

Programme created by Eng. M. I. SADIQ., B.E (Struc)

COLUMN DESIGN

JOB No. : 1225

DATE : March 14, 2006

STRUCTURE :

MPS 1, 2 & 3

Design ref.

B S 8110 : Part 1 : 1985

COLUMN REFERENCE : C1 GRID I/5 : GF-1F

Fcu	35	N/mm²	Fy	415	N/mm²	Cover	40	mm
Column size (B x H)			450	x	450	mm		
Load on column (Pu)					1885	kn		
Clear height of the column on XX axis(Lx)					4.4	m		
Clear height of the column on YY axis(Ly)					4.4	m		
<u>Effective length of column on XX axis</u>								
End condition at top	1	and at bottom	1					
<u>As per B S 8110, Part 1, Table 3.21 & 3.22</u>								
Effective length of column (Lex)	0.75	* L			3.30	m		
The ratio of Lex / H					7.33	<	15	Short Column
<u>Effective length of column on YY axis</u>								
End condition at top	1	and at bottom	1					
<u>As per B S 8110, Part 1, Table 3.21 & 3.22</u>								
Effective length of column (Ley)	0.75	* L			3.30	m		
The ratio of Ley / B					7.33	<	15	Short Column
<u>Column design</u>								
Area of concrete(Ac)	B*H				202.5 *10 ³	mm ²		
Area of steel, Asc	(1.2 %)				2513	mm ²		
Minimum area of reinf.	(0.4 %)				810	mm ²		
Maximum area of reinf.	(4 %)				8100	mm ²		
The ultimate load carrying capacity of column								
$N = 0.35 * Fcu * (Ac - Asc) + 0.67 * Fy * Asc$					3149	kn		
Since Pu	<	N	1885	<	3149	kn	Safe	
<u>Provide Main Bar : Y 20 mm 8 Nos</u>								
Check, Asc Provided	2513	>	810	&	<	8100	mm ²	OK
Max. No. of bars provided in one face	3	Nos						
Spacing of main bar					185	mm		
Maximum allowable spacing of main bar					250	mm		
Check, Provided < Allowed	185	<	250	mm			OK	
<u>Design of links</u>								
Minimum size of links								
Lmin = 1/4 of the largest comp. bar but					5	mm		
not less than 6mm					6	mm		
<u>Provide Y 8 mm bars</u>	8	>	6	mm			OK	
<u>Spacing of links</u>								
Maximum spacing of links								
Maximum spacing should not exceed					250	mm		
12 * size of smallest comp. bar					240	mm		
Smallest c/s size of column					450	mm		
<u>Provide links @</u>	150	mm c/c	150	<	240	mm	OK	
<u>Hence Provide</u>								
Column size	450	x	450					
Main bar Y	20	mm	8	Nos.				
Links Y	8	mm @	150	mm c/c				

Programme created by Eng. M. I. SADIQ., B.E. (Struct.)

COLUMN DESIGN

JOB No. : 1225

DATE : March 14, 2006

STRUCTURE : MPS 1, 2 & 3 Design ref. B S 8110 : Part 1 : 1985

COLUMN REFERENCE : C1 GRID I/5 : 2F-RF

Fcu 35 N/mm² Fy 415 N/mm² Cover 40 mm

Column size (B x H) 450 x 450 mm

Load on column (Pu) 756 kn

Clear height of the column on XX axis(Lx) 4.6 m

Clear height of the column on YY axis(Ly) 4.6 m

Effective length of column on XX axis

End condition at top 1 and at bottom 1

As per B S 8110, Part 1, Table 3.21 & 3.22

Effective length of column (Lex) 0.75 * L 3.45 m

The ratio of Lex / H 7.67 < 15 **Short Column**

Effective length of column on YY axis

End condition at top 1 and at bottom 1

As per B S 8110, Part 1, Table 3.21 & 3.22

Effective length of column (Ley) 0.75 * L 3.45 m

The ratio of Ley / B 7.67 < 15 **Short Column**

Column design

Area of concrete(Ac) B*H 202.5 *10³ mm²

Area of steel, Asc (0.8 %) 1608 mm²

Minimum area of reinf. (0.4 %) 810 mm²

Maximum area of reinf. (4 %) 8100 mm²

The ultimate load carrying capacity of column

$$N = 0.35 * F_{cu} * (A_c - A_{sc}) + 0.67 * F_y * A_{sc}$$

Since Pu < N 756 < 2908 kn **Safe**

Provide Main Bar : Y 16 mm 8 Nos

Check, Asc Provided 1608 > 810 & < 8100 mm² **OK**

Max. No. of bars provided in one face 3 Nos

Spacing of main bar 185 mm

Maximum allowable spacing of main bar 250 mm

Check, Provided < Allowed 185 < 250 mm **OK**

Design of links

Minimum size of links

Lmin = 1/4 of the largest comp. bar but 4 mm

not less than 6mm 6 mm

Provide Y 8 mm bars 8 > 6 mm OK

Spacing of links

Maximum spacing of links

Maximum spacing should not exceed 250 mm

12 * size of smallest comp. bar 192 mm

Smallest c/s size of column 450 mm

Provide links @ 150 mm c/c 150 < 192 mm OK

Hence Provide

Column size 450 x 450

Main bar Y 16 mm 8 Nos.

Links Y 8 mm @ 150 mm c/c

COLUMN DESIGN

JOB No. : 1225

DATE : March 14, 2006

STRUCTURE : MPS 1, 2 & 3 Design ref. B S 8110 : Part 1 : 1985

COLUMN REFERENCE : C2 GRID G/5 : 2F-RF

Fcu	35	N/mm ²	Fy	415	N/mm ²	Cover	43	mm
-----	----	-------------------	----	-----	-------------------	-------	----	----

Column size (B x H) 450 x 600 mm

Load on column (Pu) 1584 kn

Clear height of the column on XX axis(Lx) 4.6 m

Clear height of the column on YY axis(Ly) 4.6 m

Effective length of column on XX axis

End condition at top 1 and at bottom 1

As per B S 8110, Part 1, Table 3.21 & 3.22

Effective length of column (Lex) 0.75 * L 3.45 m

The ratio of Lex / H 5.75 < 15 **Short Column**

Effective length of column on YY axis

End condition at top 1 and at bottom 1

As per B S 8110, Part 1, Table 3.21 & 3.22

Effective length of column (Ley) 0.75 * L 3.45 m

The ratio of Ley / B 7.67 < 15 **Short Column**

Column design

Area of concrete(Ac) B*H 270 *10³ mm²

Area of steel, Asc (0.7 %) 2011 mm²

Minimum area of reinf. (0.4 %) 1080 mm²

Maximum area of reinf. (4 %) 10800 mm²

The ultimate load carrying capacity of column

$$N = 0.35 * F_{cu} * (A_c - A_{sc}) + 0.67 * F_y * A_{sc} \quad 3842 \text{ kn}$$

Since Pu < N 1584 < 3842 kn **Safe**

Provide Main Bar : Y 16 mm 10 Nos

Check, Asc Provided 2011 > 1080 **&** < 10800 mm² **OK**

Max. No. of bars provided in one face 4 **Nos**

Spacing of main bar 171 mm

Maximum allowable spacing of main bar 250 mm

Check, Provided < Allowed 171.333 < 250 mm **OK**

Design of links

Minimum size of links

Lmin = 1/4 of the largest comp. bar but 4 mm

not less than 6mm 6 mm

Provide Y 8 mm bars 8 > 6 mm **OK**

Spacing of links

Maximum spacing of links

Maximum spacing should not exceed 250 mm

12 * size of smallest comp. bar 192 mm

Smallest c/s size of column 450 mm

Provide links @ 150 mm c/c 150 < 192 mm **OK**

Hence Provide

Column size 450 x 600

Main bar Y 16 mm 10 Nos.

Links Y 8 mm @ 150 mm c/c

COLUMN DESIGN

(Short Column (Braced) with Biaxial Moments)

JOB No. : 1225

DATE : April 21, 2006

STRUCTURE : MPS

Design ref. : B S 8110 : Part 1 : 1985

COLUMN REFERENCE : C3

B S 8110 : Part 3 : 1985

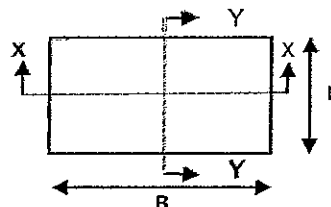
COLUMN ON GRID : F/4 GF to 1st Floor

Fcu 35 N/mm²

Fy 415 N/mm²

Cover 40 mm

Column Width, B	800	mm
Column Depth, H	450	mm
Dia. Of Main bar	20	mm
Dia. Of Links	8	mm
Unsupported Length about Width, L _{ox}	3.4	m
Unsupported Length about Depth, L _{oy}	3.3	m
Effective Length factor about Width, β_x	0.75	
Effective Length factor about Depth, β_y	0.75	
Load from Upper Floor	3122	Kn



Refer Table 3.19 & 3.20, BS 8110:Part 1: 1997

CALCULATIONS

Effective Width, B'	742	mm	B'/B =	0.93
Effective Depth, H'	392	mm	H'/H =	0.87
Effective Length of Column about Width, L _{ex}	2.55	m		
Effective Length of Column about Depth, L _{ey}	2.48	m		

Slenderness Check

Slenderness about Width, L _{ex} /D	5.7	<	15	OK Short Column
Slenderness about Depth, L _{ey} /B	3.1	<	15	OK Short Column

Total load on column (Nu)	4534.0	kn
Minimum area of steel (0.4%)	1440	mm ²
Maximum area of steel (6%)	21600	mm ²
Area of steel at laps (10%)	36000	mm ²

Note : 1. Input Loads & Moments must be Ultimate unless otherwise noted.
2. LF denotes "Lower Floor" & UF denotes "Upper Floor"

COLUMN MOMENTS

Floor	LHS Beam				RHS Beam				Upper Column			Lower Column		
	Beam size		Length	load	Beam size		length	load	size		length	size		length
	mm	mm	m	kn/m	mm	mm	m	kn/m	mm	mm	m	mm	mm	m
UF-xx	250	600	6.00	74.00	0	0	0.00	0.00	800	450	4.00	800	450	5.00
UF-yy	250	700	5.10	63.00	250	700	7.00	63.00	450	800	4.00	450	800	5.00
LF-xx	300	600	6.00	93.00	1000	200	1.00	20.00	800	450	4.00	800	450	3.22
LF-yy	300	800	5.10	121.00	300	800	7.00	60.00	450	800	4.00	450	800	3.22

Floors		column Stiffness factors		column moments (knm)	
		upper	lower	upper	lower
UF	xx	0.489	0.391	108.5	86.8
	yy	0.487	0.390	58.8	47.0
LF	xx	0.363	0.450	100.6	124.9
	yy	0.371	0.461	6.4	8.0

Floor	Load	Max. Moments, Knm	
	kn	Mx	My
Upper Flr Load	3122.0		
@ UF	3726.0	21.7	11.8
@ LF	4534.0	24.4	1.6

Max load on column (N_u)

4534 kn

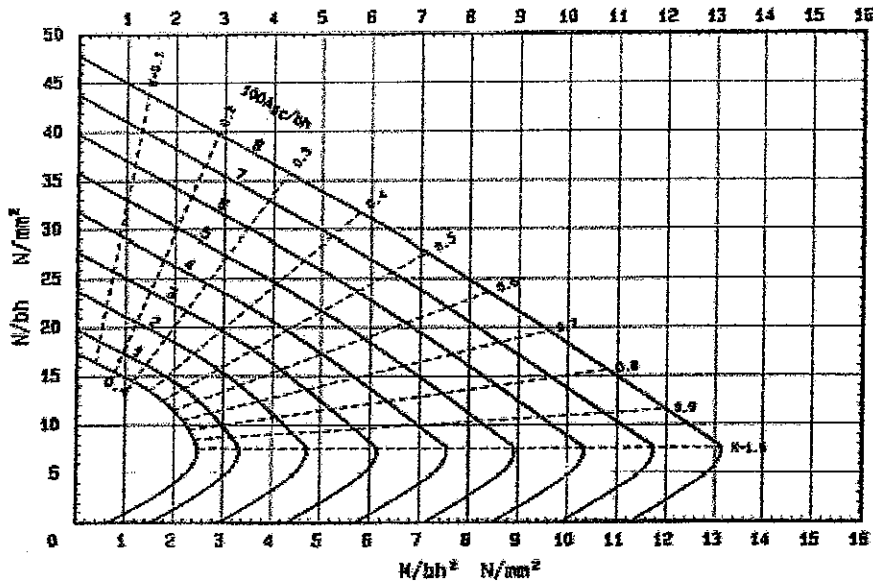
Moment about Width, M_x 24.4 knm
 Moment about Depth, M_y 11.8 knm
 Therefore Max. Moment 24.4 Knm

Design of Column

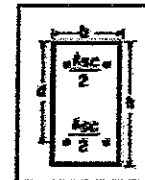
Ratios for Chart entry for M_x

		UF	LF
N ratio	$N/(B \cdot H)$	10.35	12.59
M_x ratio	$M_x/(B \cdot H^2)$	0.13	0.15
	H'/H	0.87	

Chart No. 33

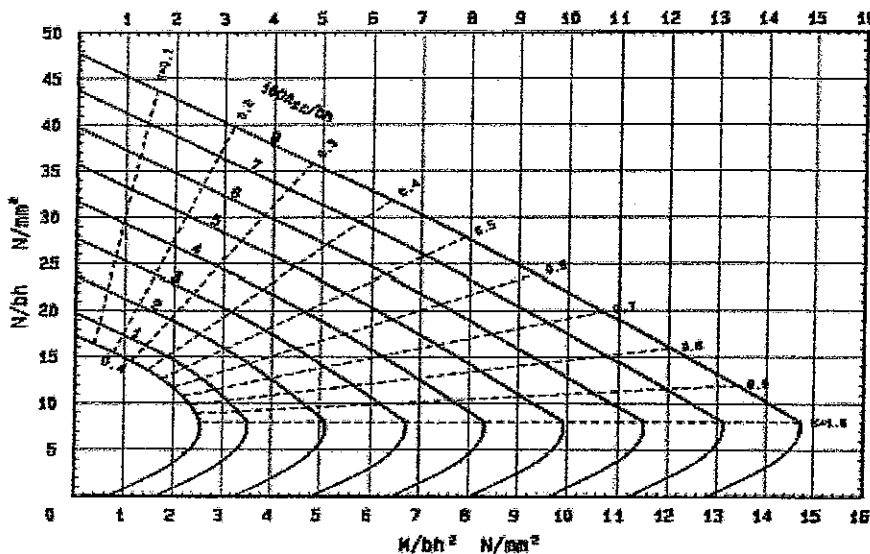


Rectangular columns

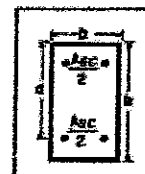


f_{cu} 35
 f_y 460
 d/h 0.85

Chart No. 34



Rectangular columns



f_{cu} 35
 f_y 460
 d/h 0.90

Refer Chart 33 of BS 8110:P3:1985 (f_{cu} 35 Vs f_y 460 Vs H'/H 0.88)
 Actual Percentage of Steel req. 0.40 % = 1440 mm²
 Ratios for Chart entry for M_y UF LF
 N ratio $N/(H \cdot B)$ 10.35 12.59
 M_y ratio $M_y/(H \cdot B^2)$ 0.04 0.01

	B/B	0.93				
Refer Chart	34 of BS 8110:P3:1985	(Fcu 35	Vs Fy 460	Vs H/H 0.93)		
Actual Percentage of Steel req.		0.40	% = 1440	mm ²		
Therefore Provide Min. Asc		1440	mm ²			
Provided Percentage of Steel		1.91	%			
Provide Main Bar : Y 25 mm 14 Nos						
Check, Asc Provided	6872	1440	&	<	21600 mm ²	OK
The ultimate load carrying capacity of column						
$N = 0.35 \cdot F_{cu} \cdot (A_c - A_{sc}) + 0.67 \cdot F_y \cdot A_{sc}$					6237 kn	> 4534 OK
Max. No. of bars provided in one face	6	Nos				
Spacing of main bar					144 mm	
Maximum allowable spacing of main bar					250 mm	
Check, Provided < Allowed	144	<			250 mm	OK

Design of links

Minimum size of links

Lmin = 1/4 of the largest comp. bar

6.25 mm

But, not less than 6mm

6 mm

Provide Y 8 mm bars

8

>

6.25 mm

OK

Spacing of links

Maximum spacing of links

Maximum spacing should not exceed

250 mm

12 * size of smallest comp. bar

300 mm

Smallest c/s size of column

450 mm

Provide links @

175 mm c/c

175

<

250 mm

OK

Hence Provide

Column size 800 x 450

Main bar Y 25 mm 14 Nos.

Links Y 8 mm @ 175 mm c/c

Programme created by Eng. M. I. SADIQ., B.E (Struct.)

COLUMN DESIGN

(Short Column (Braced) with Biaxial Moments)

JOB No. : 1225

DATE : April 21, 2006

STRUCTURE : MPS

Design ref. : B S 8110 : Part 1 : 1985

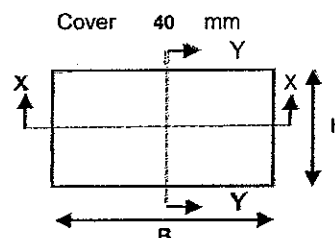
COLUMN REFERENCE : C3

B S 8110 : Part 3 : 1985

COLUMN ON GRID : F/4 2nd Flr to Roof

Fcu 35 N/mm² Fy 415 N/mm²

Column Width, B 800 mm
 Column Depth, H 450 mm
 Dia. Of Main bar 20 mm
 Dia. Of Links 8 mm
 Unsupported Length about Width, L_{ox} 4.6 m
 Unsupported Length about Depth, L_{oy} 4.6 m
 Effective Length factor about Width, β_x 0.75
 Effective Length factor about Depth, β_y 0.75
 Load from Upper Floor 0 Kn



Refer Table 3.19 & 3.20, BS 8110:Part 1: 1997

CALCULATIONS

Effective Width, B' 742 mm B'/B = 0.93
 Effective Depth, H' 392 mm H'/H = 0.87
 Effective Length of Column about Width, L_{ex} 3.45 m
 Effective Length of Column about Depth, L_{ey} 3.45 m

Slenderness Check

Slenderness about Width, L_{ex}/D 7.7 < 15 OK Short Column
 Slenderness about Depth, L_{ey}/B 4.3 < 15 OK Short Column

Total load on column (N_u) 3122.0 kn
 Minimum area of steel (0.4%) 1440 mm²
 Maximum area of steel (6%) 21600 mm²
 Area of steel at laps (10%) 36000 mm²

Note : 1. Input Loads & Moments must be Ultimate unless otherwise noted.
 2. LF denotes "Lower Floor" & UF denotes "Upper Floor"

COLUMN MOMENTS

Floor	LHS Beam				RHS Beam				Upper Column			Lower Column		
	Beam size		Length	load	Beam size		length	load	size		length	size		length
	mm	mm	m	kn/m	mm	mm	m	kn/m	mm	mm	m	mm	mm	m
UF-xx	800	650	6.00	83.00	800	650	12.50	131.00	0	0	0.00	800	450	5.20
UF-yy	250	600	5.10	62.00	250	700	7.00	55.00	0	0	0.00	450	800	5.20
LF-xx	500	400	6.00	68.00	550	1200	12.50	156.00	800	450	5.20	800	450	4.00
LF-yy	300	500	5.10	75.00	500	500	7.00	95.00	450	800	5.20	450	800	4.00

Floors		column Stiffness factors		column moments (knm)	
		upper	lower	upper	lower
UF	xx	0.000	0.341	0.0	496.7
	yy	0.000	0.795	0.0	71.7
LF	xx	0.192	0.250	351.3	456.6
	yy	0.403	0.523	90.7	118.0

Floor	Load	Max. Moments, Knm	
	kn	Mx	My
Upper Flr Load	0.0		
@ UF	1419.0	496.7	71.7
@ LF	3122.0	105.4	27.2

Max load on column (N_u) 3122 kn

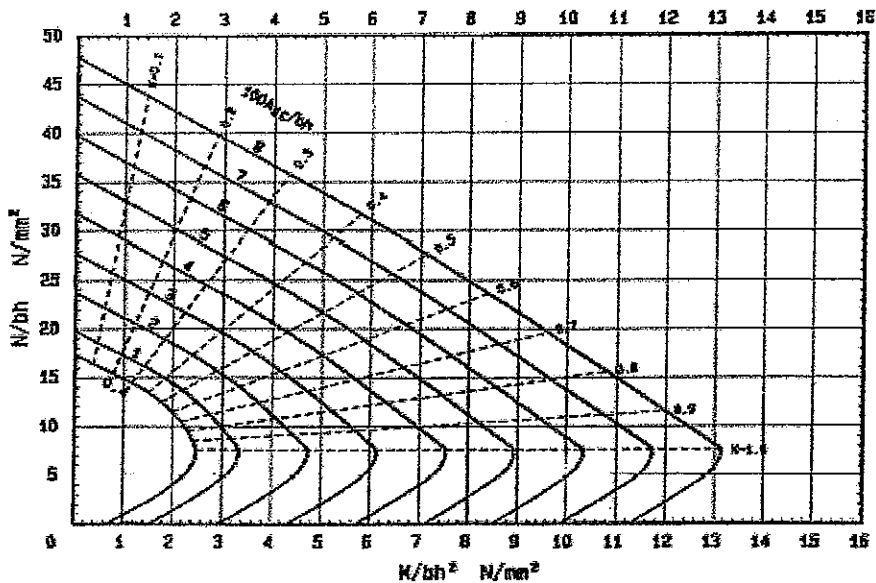
Moment about Width, Mx 496.7 knm
 Moment about Depth, My 71.7 knm
 Therefore Max. Moment 496.7 Knm

Design of Column

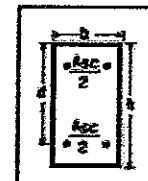
Ratios for Chart entry for Mx

		UF	LF
N ratio	$N/(B \cdot H)$	3.94	8.67
Mx ratio	$Mx/(B \cdot H^2)$	3.07	0.65
	H'/H	0.87	

Chart No. 33

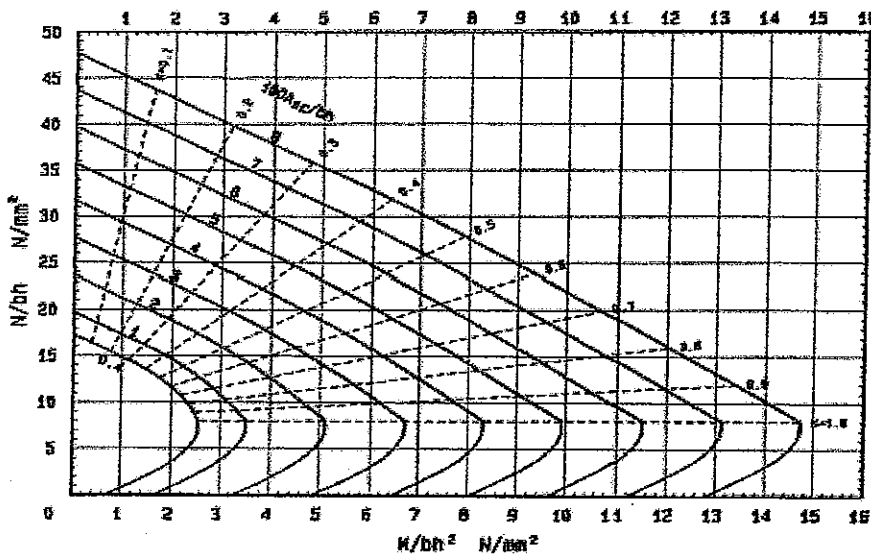


Rectangular columns

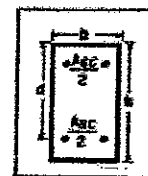


f_{cu}	35
f_y	460
d/h	0.85

Chart No. 34



Rectangular columns



f_{cu}	35
f_y	460
d/h	0.90

Refer Chart 33 of BS 8110:P3:1985 (f_{cu} 35 Vs f_y 460 Vs H'/H 0.87)
 Actual Percentage of Steel req. 1.40 % = 5040 mm²
 Ratios for Chart entry for My

N ratio	$N/(H \cdot B)$	3.94	8.67
My ratio	$My/(H \cdot B^2)$	0.25	0.09

COLUMN DESIGN

JOB No. : 1225

DATE : March 14, 2006

STRUCTURE : MPS 1, 2 & 3 Design ref. B S 8110 : Part 1 : 1985

COLUMN REFERENCE : C4 GRID G/4 : GF-1F

Fcu	35	N/mm²	Fy	415	N/mm²	Cover	40	mm
Column size (B x H)			450	x	800	mm		
Load on column (Pu)					2533	kn		
Clear height of the column on XX axis(Lx)					4.4	m		
Clear height of the column on YY axis(Ly)					4.3	m		
<u>Effective length of column on XX axis</u>								
End condition at top	1							
and at bottom	1							
<u>As per B S 8110, Part 1, Table 3.21 & 3.22</u>								
Effective length of column (Lex)	0.75		* L		3.30	m		
The ratio of Lex / H					4.13	<	15	Short Column
<u>Effective length of column on YY axis</u>								
End condition at top	1							
and at bottom	1							
<u>As per B S 8110, Part 1, Table 3.21 & 3.22</u>								
Effective length of column (Ley)	0.75		* L		3.23	m		
The ratio of Ley / B					7.17	<	15	Short Column
<u>Column design</u>								
Area of concrete(Ac)	B*H				360 *10 ³	mm ²		
Area of steel, Asc (0.8 %)					2815	mm ²		
Minimum area of reinf. (0.4 %)					1440	mm ²		
Maximum area of reinf. (4 %)					14400	mm ²		
The ultimate load carrying capacity of column								
$N = 0.35 * Fcu * (Ac - Asc) + 0.67 * Fy * Asc$					5158	kn		
Since Pu < N			2533	<	5158	kn		Safe
<u>Provide Main Bar : Y 16 mm 14 Nos</u>								
Check, Asc Provided	2815	>	1440	&	<	14400	mm ²	OK
Max. No. of bars provided in one face	6		Nos					
Spacing of main bar					144	mm		
Maximum allowable spacing of main bar					250	mm		
Check, Provided < Allowed	144	<	250	mm				OK
<u>Design of links</u>								
Minimum size of links								
Lmin = 1/4 of the largest comp. bar but not less than 6mm					4	mm		
					6	mm		
<u>Provide Y 8 mm bars</u>	8	>	6	mm				OK
<u>Spacing of links</u>								
Maximum spacing of links								
Maximum spacing should not exceed					250	mm		
12 * size of smallest comp. bar					192	mm		
Smallest c/s size of column					450	mm		
<u>Provide links @ 150 mm c/c</u>	150	<	192	mm				OK
<u>Hence Provide</u>								
Column size	450	x	800					
Main bar Y	16	mm	14	Nos.				
Links Y	8	mm @	150	mm c/c				

Programme created by Eng. M. I. SADIQ., B.E.(Struc.)

COLUMN DESIGN

JOB No. : 1225

DATE : March 14, 2006

STRUCTURE :

MPS 1, 2 & 3

Design ref.

B S 8110 : Part 1 : 1985

COLUMN REFERENCE : C5 GRID F/2 : GF-1F

Fcu	35	N/mm ²	Fy	415	N/mm ²	Cover	40	mm
-----	----	-------------------	----	-----	-------------------	-------	----	----

Column size (B x H) 450 x 800 mm

Load on column (Pu) 1980 kn

Clear height of the column on XX axis(Lx) 4.4 m

Clear height of the column on YY axis(Ly) 4.4 m

Effective length of column on XX axis

End condition at top 1 and at bottom 1

As per B S 8110, Part 1, Table 3.21 & 3.22

Effective length of column (Lex) 0.75 * L 3.30 m

The ratio of Lex / H 4.13 < 15 **Short Column**

Effective length of column on YY axis

End condition at top 1 and at bottom 1

As per B S 8110, Part 1, Table 3.21 & 3.22

Effective length of column (Ley) 0.75 * L 3.30 m

The ratio of Ley / B 7.33 < 15 **Short Column**

Column design

Area of concrete(Ac) B*H 360 *10³ mm²

Area of steel, Asc (0.8 %) 2815 mm²

Minimum area of reinf. (0.4 %) 1440 mm²

Maximum area of reinf. (4 %) 14400 mm²

The ultimate load carrying capacity of column

$N = 0.35 * F_{cu} * (A_c - A_{sc}) + 0.67 * F_y * A_{sc}$ 5158 kn

Since Pu < N 1980 < 5158 kn **Safe**

Provide Main Bar : Y 16 mm 14 Nos

Check, Asc Provided 2815 > 1440 & < 14400 mm² **OK**

Max. No. of bars provided in one face 6 Nos

Spacing of main bar 144 mm

Maximum allowable spacing of main bar 250 mm

Check, Provided < Allowed 144 < 250 mm **OK**

Design of links

Minimum size of links

Lmin = 1/4 of the largest comp. bar but 4 mm

not less than 6mm 6 mm

Provide Y 8 mm bars 8 > 6 mm OK

Spacing of links

Maximum spacing of links

Maximum spacing should not exceed 250 mm

12 * size of smallest comp. bar 192 mm

Smallest c/s size of column 450 mm

Provide links @ 150 mm c/c 150 < 192 mm OK

Hence Provide

Column size 450 x 800

Main bar Y 16 mm 14 Nos.

Links Y 8 mm @ 150 mm c/c

Programme created by Eng. M. I. SADIQ, B.E (Struct.)

BEL-YOAHAH Architectural & Eng. Consultants

P.O Box : 827, Dubai, United Arab Emirates.

Ph : 04 - 2232311 Fax : 04 - 2216422

BASEMENT WALL DESIGN**JOB No. : 1225****DATE : April 1, 2006****STRUCTURE****MPS****Design ref. B S 8110 : Part 1 : 1985****WALL REFERENCE****W1****GRID****A/4-8****Fcu****35****N/mm²****Fy****415****N/mm²****Cover****46****mm****Wall Details**

Height of Wall upto Top Slab, Hw

3.22 m

Height of Soil

2.00 m

Wall Thickness

250 mm

Density of Soil

17 Kn/m³

Angle of Internal Friction

33 Deg

Coefficient of Active Pressure

0.45

Depth of Base Slab

0.70 m

Note :

Wall Designed as Propped
Cantilever**Load Details**

Load on Wall

62.00 Kn/m

Additional Load on earth - UDL

20.00 Kn/m²

Water Load

9.81 Kn/m³ @ Height from Bottom of Wall, D_w 2.00 m**Design of Wall**

Max. Dry Soil pressure @ Bottom of the Wall

0.00 Kn/m²

Max. Pressure due to UDL @ Bottom of the Wall

9.00 Kn/m²

Max. Wet soil pressure @ Bottom of the Wall

28.09 Kn/m²

Total Pressure @ Bottom of the Wall, p

35.09 Kn/m²**Consider 1m Run. Of Wall****Bending Moment**

Max BM (SLS) Producing Tension

near Outer face, M_{os} p*Hw²/15

29.82 Knm

Max BM (SLS) Producing Tension

near Inner face, M_{is} p*Hw²/33.5

13.35 Knm

Max BM (ULS) Producing Tension

near Outer face, M_{os}*1.4

41.7 Knm

Max BM (ULS) Producing Tension

near Inner face, M_{is}*1.4

18.7 Knm

Area of Steel

Thickness of Wall

250 mm

Effective depth

204 mm

$$Ast = M*10^6 / (0.87*Fy*Z)$$

K**Z****Ast****mm****mm²****Provide****Ast_{pro}****mm²**

Outer Face 0.029 193.8 596.6

Y 12 @ 150 mm c/c 754 OK

Inner Face 0.013 193.8 325.0

Y 12 @ 150 mm c/c 754 OK

Ultimate Load Carrying Capacity of Wall / M run

$$N = 0.35*Fcu*(Ac-Asc)+0.67*Fy*Ast$$

3463 kn

Since Pu < N

62.00

<

3463 kn

Safe

Transverse Bars (0.13%)

325 mm²

Provide

Y

10

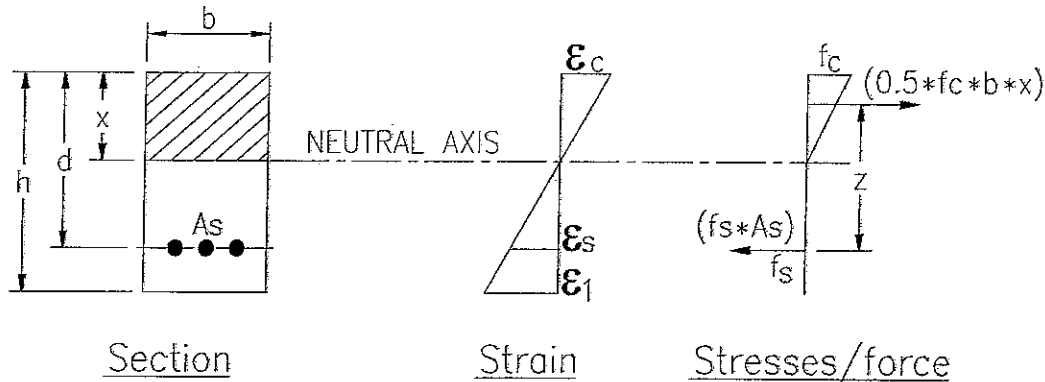
@

200 mm c/c on EF

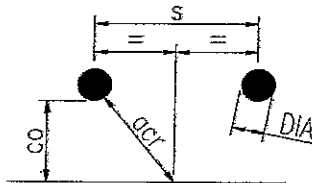
393 OK

Project	METRO POWER STATION		BEL-YOAH AH Architectural & Eng. Consultants		
Client	DUBAI RAIL		Made by	Date	Page
Location	MPS 1 2 & 3		SADIQ	21-Apr-2006	1
Crack Width Calculations to BS8110: 1997/ BS8007:1987			Checked	Revision	Job No
Originated from RCC14.xls on CD © 1999 BCA for RCC			NG	-	1225

CRACK WIDTH CALCULATIONS - FLEXURE



INPUT

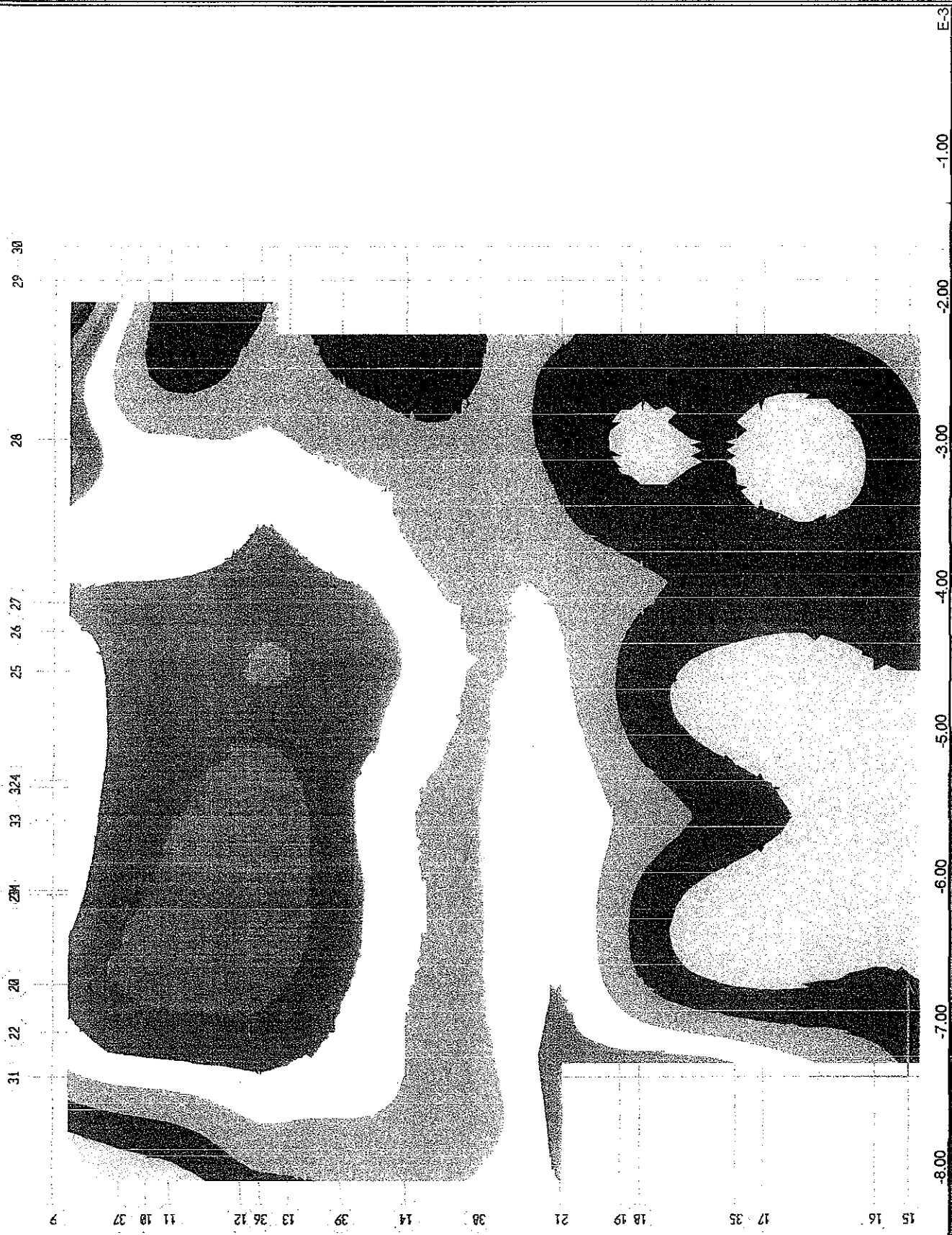


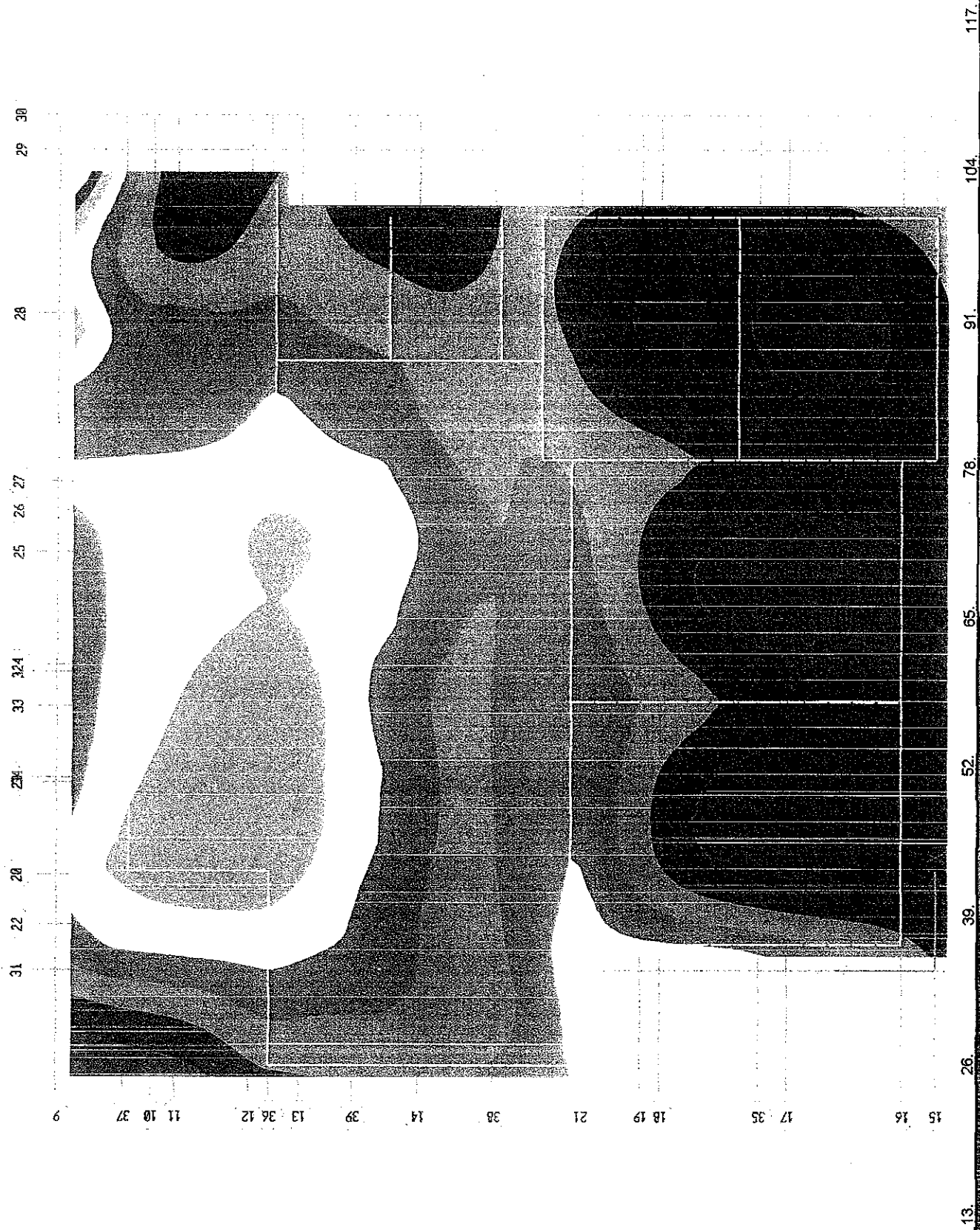
$f_{cu} =$	35	N/mm ²
$f_y =$	425	N/mm ²
Area of reinforcement " A_s " =	754	mm ²
$b =$	1000	mm
$h =$	250	mm
$d =$	204	mm
Minimum cover to tension reinforcement " CO " =	46	mm
Maximum bar spacing " S " =	150	mm
Bar dia " DIA " =	12	mm
" a_{cr} " = ((($S/2$) ² + ($CO + DIA/2$) ²) ^{1/2} - $DIA/2$) as default or enter other value =	85.3	mm
Applied service moment " M_s " =	30.0	KNm

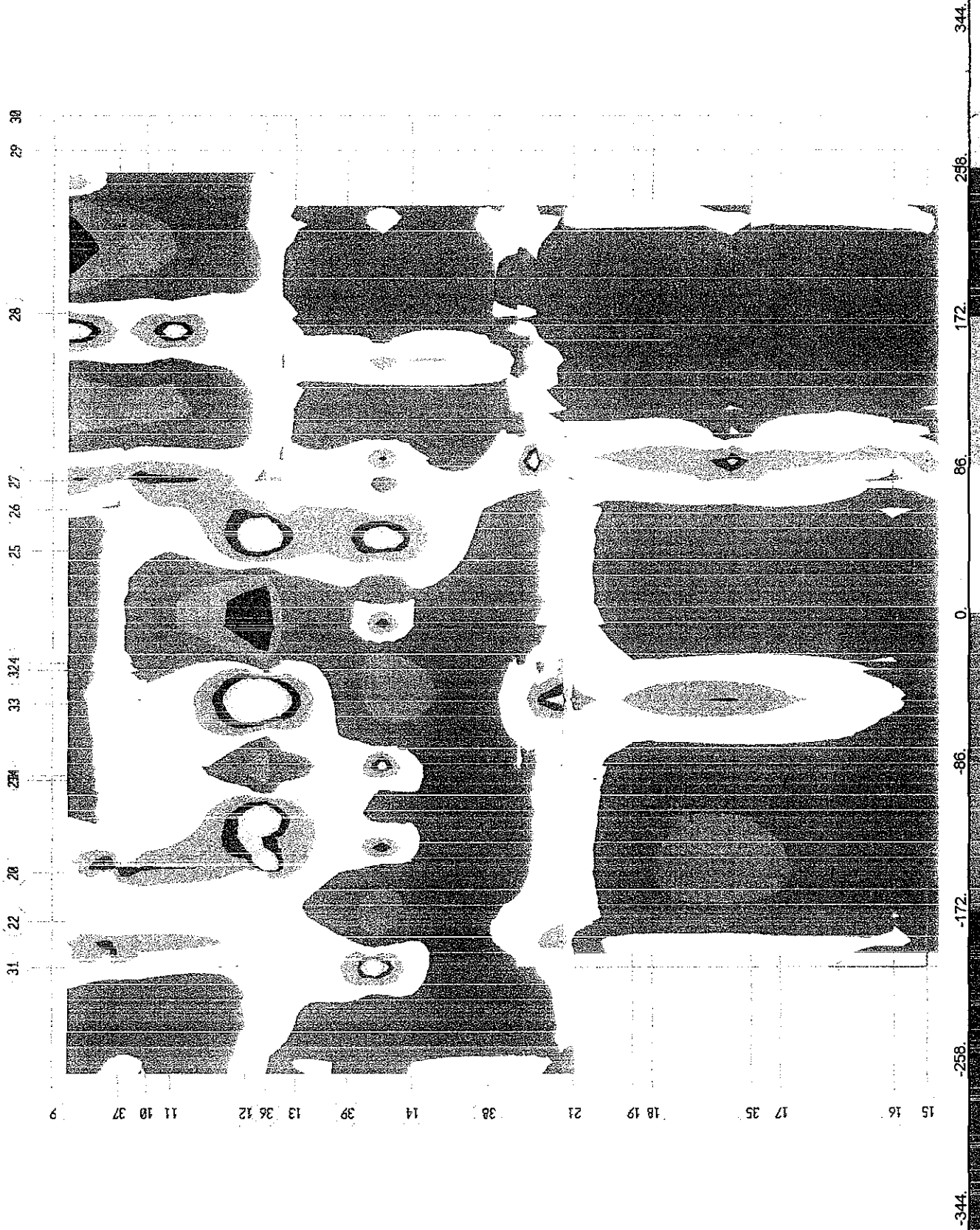
CALCULATIONS

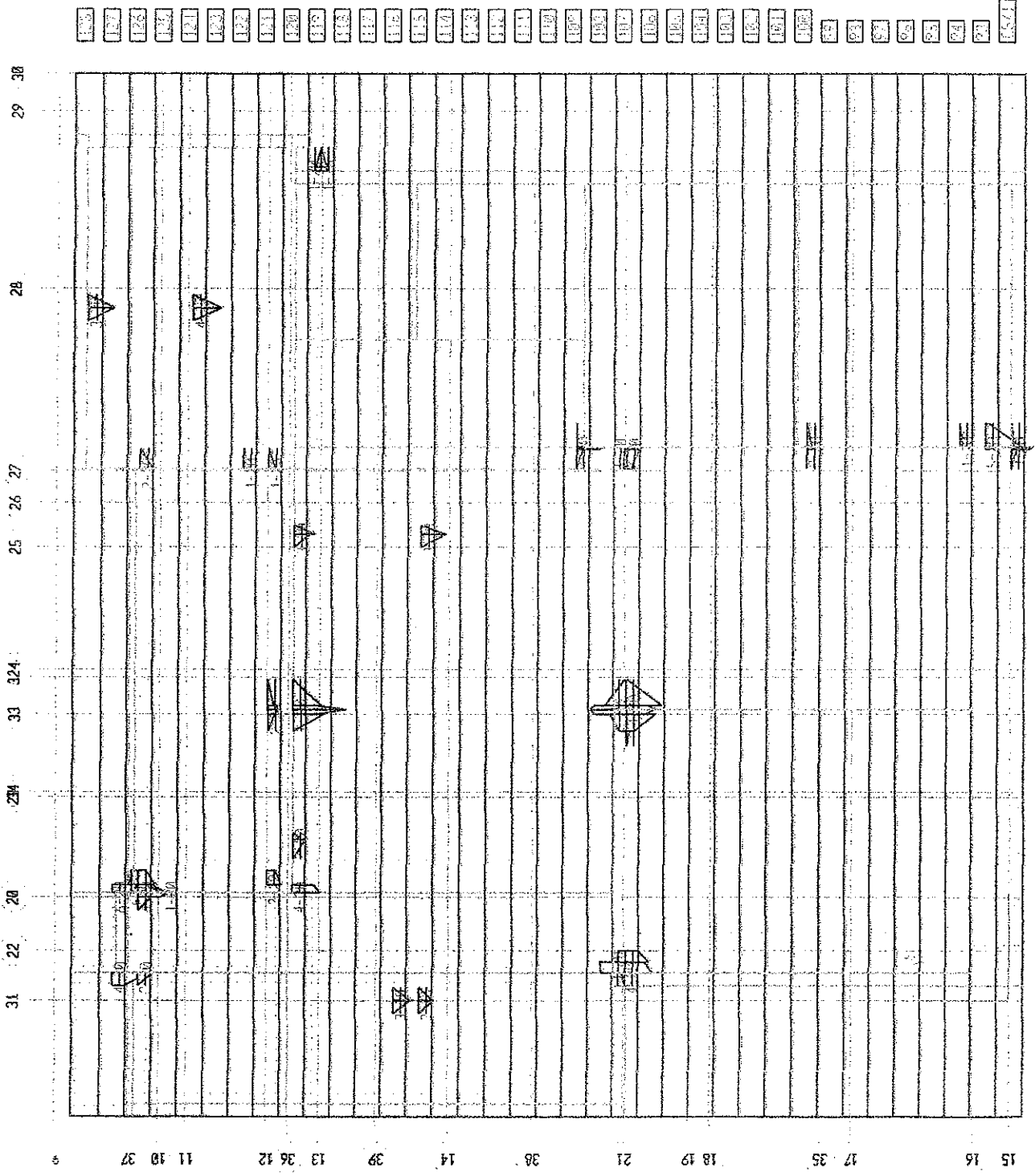
moduli of elasticity of concrete " E_c " = $(1/2) \cdot (20 + 0.2 \cdot f_{cu}) =$	13.5	KN/mm ²
moduli of elasticity of steel " E_s " =	200.0	KN/mm ²
Modular ratio " α " = $(E_s/E_c) =$	14.81	
" ρ " = $A_s/bd =$	0.004	
depth to neutral axis, " x " = $(-\alpha \cdot \rho + ((\alpha \cdot \rho)^2 + 2 \cdot \alpha \cdot \rho)^{0.5}) \cdot d =$	57	mm
" Z " = $d - (x/3) =$	185	
Reinforcement stress " f_s " = $M_s/(A_s \cdot Z) =$	215	N/mm ²
Concrete stress " f_c " = $(f_s \cdot A_s)/(0.5 \cdot b \cdot x) =$	5.67	N/mm ²
Strain at soffit of concrete beam/slab " ϵ_1 " = $(f_s/E_s) \cdot (h-x)/(d-x) =$	0.001413	
Strain due to stiffening effect of concrete between cracks " ϵ_2 " =		
$\epsilon_2 = b \cdot (h-x)^2 / (3 \cdot E_s \cdot A_s \cdot (d-x))$ for crack widths of 0.2 mm	Used	
$\epsilon_2 = 1.5 \cdot b \cdot (h-x)^2 / (3 \cdot E_s \cdot A_s \cdot (d-x))$ for crack widths of 0.1 mm	n/a	
$\epsilon_2 =$	0.000560	
Average strain for calculation of crack width " ϵ_m " = $\epsilon_1 - \epsilon_2 =$	0.000853	
Calculated crack width, " w " = $3 \cdot a_{cr} \cdot \epsilon_m / (1 + 2 \cdot (a_{cr} - c)/(h-x))$		
CALCULATED CRACK WIDTH, " w " =	0.16	mm

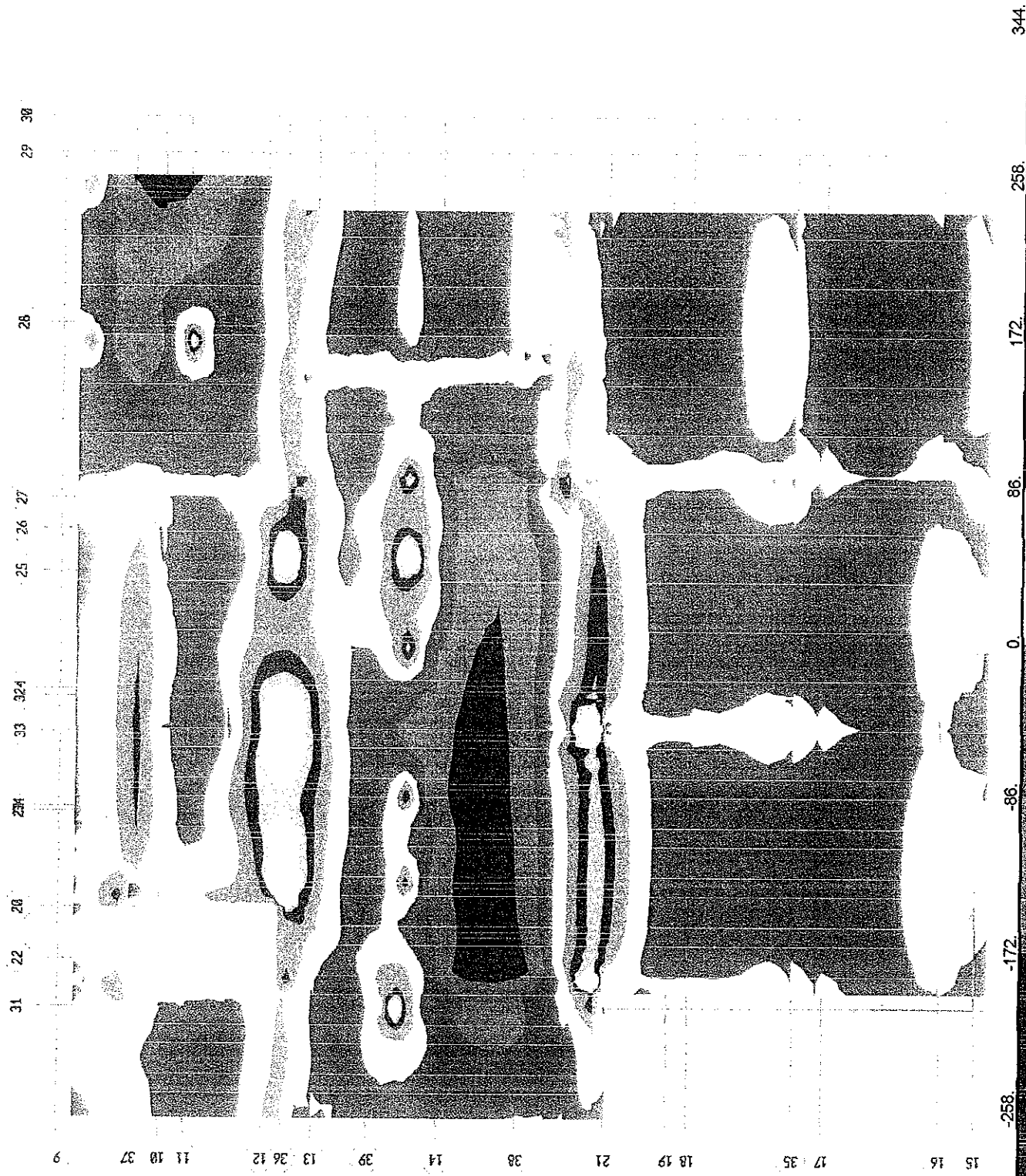
FOUNDATION











-344.

-258.

-172.

-86.

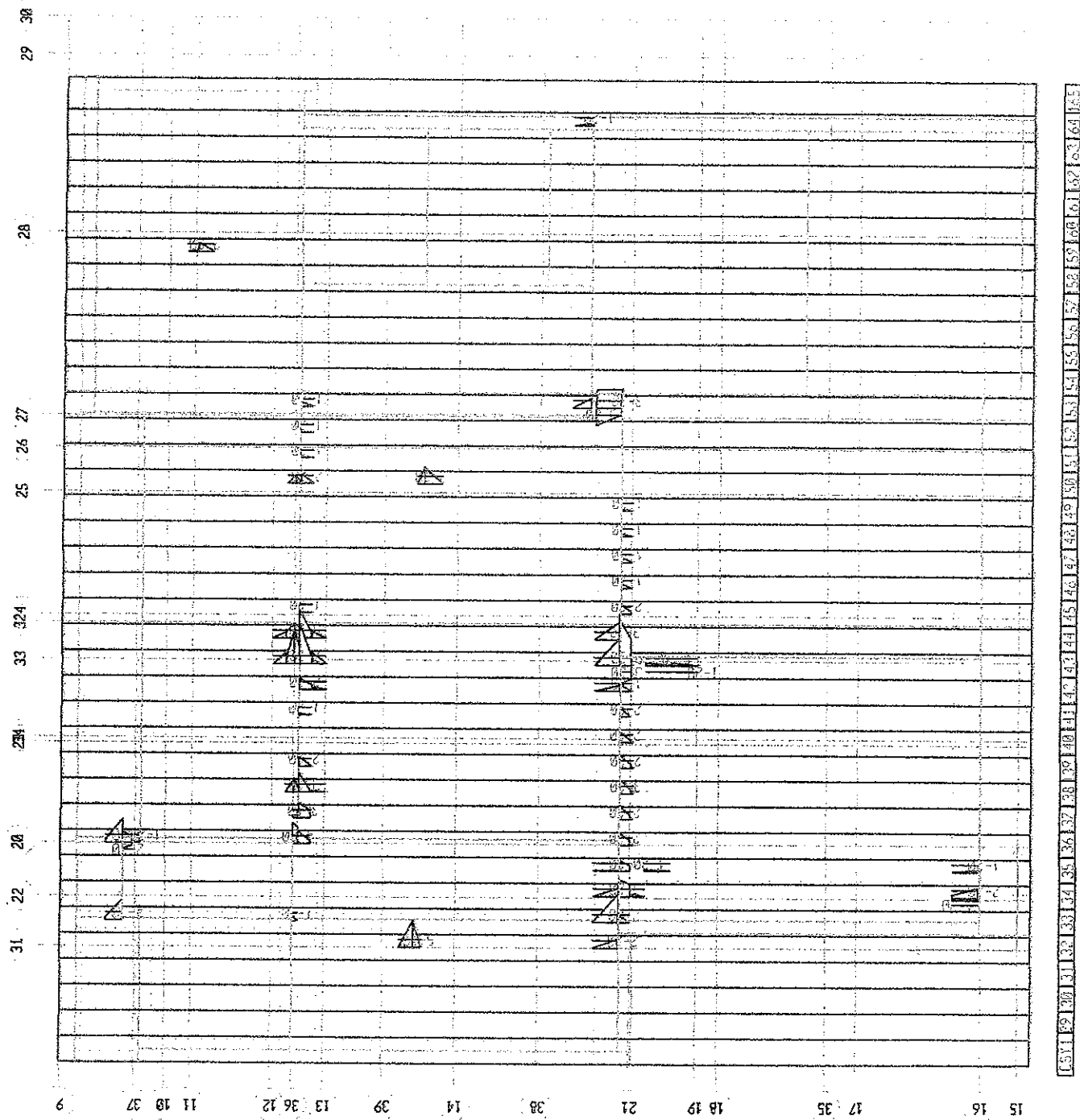
0.

86.

172.

258.

344.



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PUNCHING SHEAR CHECK FOR FOOTINGS**JOB No. : 1225****DATE: March 30, 2006****STRUCTURE : MPS****Design ref. : B S 8110 : Part 1 : 1985****FOOTING REFERENCE : RAFT****B S 8004****FOOTINGS ON GRID : F/4****FOOTING TYPE : RAFT**Fcu 35 N/mm²Fy 415 N/mm²

Footing : Cover 60 mm

Column & Footing Details**Column Size****Column Load (Pu)****Footing Depth**

mm x mm

Kn

mm

450 x 800

4418

700

Footing Side Distances from Centre of Column**Left Side****Right Side****Top Side****Bottom Side**

m

m

m

m

5.13

5.80

6.10

4.70

Upward Pressure @ Punching Area**Provided Reinforcement**152.0 Kn/m²

Y 16 @

150 mm c/c

Extra @ Bottom

Y 20 @

150 mm c/c

3435 mm²**Effective depth**

640 mm

100Asp/bd

0.54

Punching Shear Check @ 1.5d from Column face**Critical
Perimetre****Punching
Shear Area****Punching
Shear Force**

mm

m²

Kn

10180

6.45

3438.1

**Punching Shear
Stress****Allowable Shear
Stress**N/mm²N/mm²

0.53

0.57

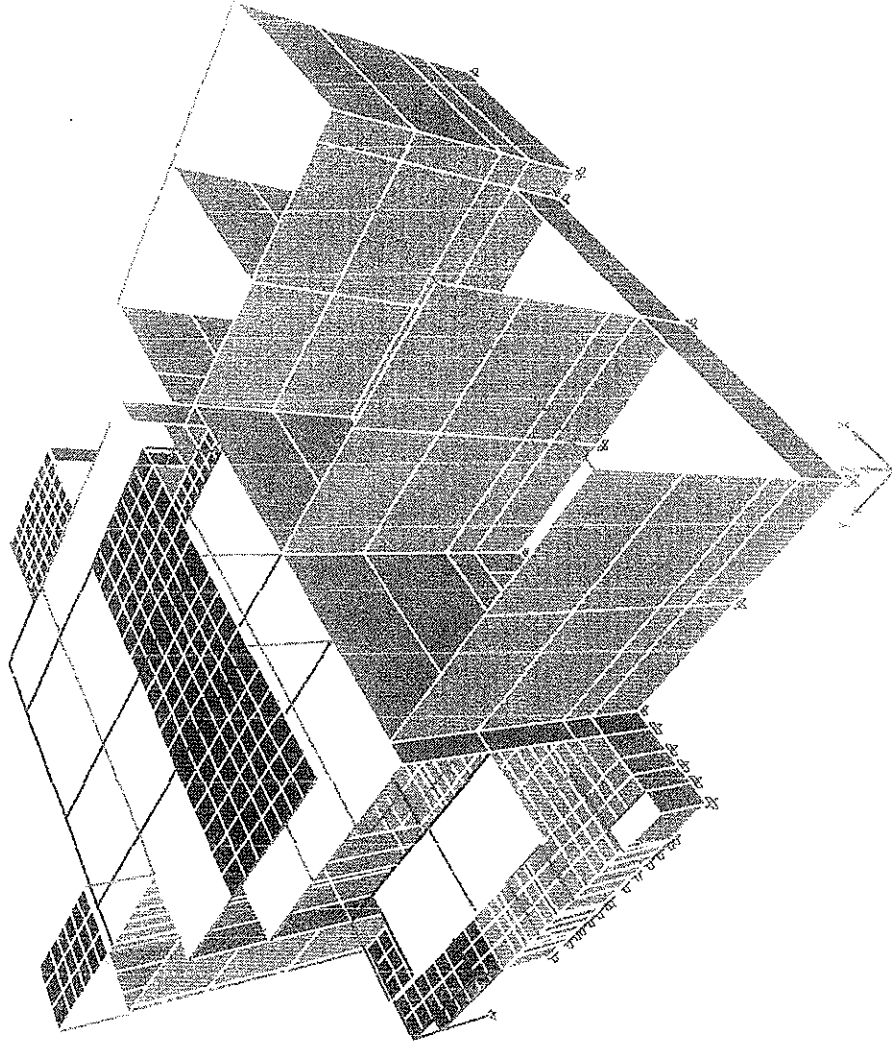
>

0.53

Safe

Programme created by Eng. M. I. SADIQ, B.E (Struct.)

MODEL BY ETABS - SEISMIC ANALYSIS



PROJECT INFORMATION

Company Name = Bel-yoahah

S T O R Y D A T A

STORY	SIMILAR TO	HEIGHT	ELEVATION
TOP ROOF	None	3.400	21.900
ROOF	TOP ROOF	4.000	18.500
2ND	TOP ROOF	4.550	14.500
1ST	None	5.000	9.950
LVL3.15	1ST	1.800	4.950
GROUND	TOP ROOF	3.150	3.150
BASE	None		0.000

S T A T I C L O A D C A S E S

STATIC CASE	CASE TYPE	AUTO LAT LOAD	SELF WT MULTIPLIER
DEAD	DEAD	N/A	1.0000
LIVE	LIVE	N/A	0.0000
QX	QUAKE	UBC97	0.0000
QY	QUAKE	UBC97	0.0000
WX	WIND	BS6399-95	0.0000
WY	WIND	BS6399-95	0.0000

R E S P O N S E S P E C T R U M C A S E S

RESP SPEC CASE: SPEC1

BASIC RESPONSE SPECTRUM DATA

MODAL COMBO	DIRECTION COMBO	MODAL DAMPING	SPECTRUM ANGLE	TYPICAL ECCEN
CQC	SRSS	0.0500	0.0000	0.0000

RESPONSE SPECTRUM FUNCTION ASSIGNMENT DATA

DIRECTION	FUNCTION	SCALE FACT
U1	FUNC1	4.1500
U2	----	N/A
UZ	----	N/A

RESP SPEC CASE: SPEC2

BASIC RESPONSE SPECTRUM DATA

MODAL COMBO	DIRECTION COMBO	MODAL DAMPING	SPECTRUM ANGLE	TYPICAL ECCEN
CQC	SRSS	0.0500	0.0000	0.0000

RESPONSE SPECTRUM FUNCTION ASSIGNMENT DATA

DIRECTION	FUNCTION	SCALE FACT
U1	----	N/A
U2	FUNC1	4.4200
UZ	----	N/A

ETABS v8.5.7 File:MPS1 Units:KN-m June 1, 2006 9:04 PAGE 5

A U T O W I N D B S 6 3 9 9 - 9 5
Case: WX

AUTO WIND INPUT DATA

Exposure From: Rigid diaphragm extents
Direction Angle = 0 degrees
Windward Cp = 0.8
Leeward Cp = 0.3

Top Story: TOP ROOF
Bottom Story: BASE

A parapet is included
Parapet height = 2.1

Effective Wind Speed, $V_e = 45$ m/s
Size Effect Factor, $C_a = 1$
Dynamic Augmentation Factor, $C_r = 0.25$

AUTO WIND EXPOSURE WIDTH INFORMATION (Exposure widths are from diaphragm extents)

STORY	DIAPHRAGM	WIDTH	X	Y
TOP ROOF	D1	6.675	2.637	31.513
ROOF	D1	18.700	10.800	24.800
2ND	D1	36.100	16.625	18.050
1ST	D1	37.400	14.612	18.700
LVL3.15	D1	36.100	14.612	18.050
GROUND	D1	36.100	14.612	18.050

AUTO WIND CALCULATION FORMULAS

q_s = dynamic pressure in Pa ($1 \text{ Pa} = 1 \text{ N/m}^2$)
 $q_s = 0.613 V_e^2$

p_e = pressure acting on external surface of building
 $p_e = q_s C_p C_a$

P_{surface} = net load on building surface

Asurface = area of building surface
 Psurface = pe Asurface

P = overall load on building
 $P = 0.85 * \text{SUM}(\text{Psurface}) * (1 + \text{Cr})$

AUTO WIND CALCULATION RESULTS

qs = 1241.3250 Pa

AUTO WIND STORY FORCES

STORY	MY	MZ	FX	FY	FZ	MX
TOP ROOF			36.80	0.00	0.00	0.000
0.000	0.000					
ROOF			70.72	0.00	0.00	0.000
0.000	-12.652					
2ND			173.41	0.00	0.00	0.000
0.000	835.506					
1ST			254.80	0.00	0.00	0.000
0.000	850.467					
LVL3.15			182.79	0.00	0.00	0.000
0.000	-82.574					
GROUND			129.63	0.00	0.00	0.000
0.000	468.242					

ETABS v8.5.7 File:MPS1 Units:KN-m June 1, 2006 9:04 PAGE 6

A U T O W I N D B S 6 3 9 9 - 9 5
 Case: WY

AUTO WIND INPUT DATA

Exposure From: Rigid diaphragm extents
 Direction Angle = 90 degrees
 Windward Cp = 0.8
 Leeward Cp = 0.3

Top Story: TOP ROOF
 Bottom Story: BASE

A parapet is included
 Parapet height = 2.1

Effective Wind Speed, Ve = 45 m/s
 Size Effect Factor, Ca = 1
 Dynamic Augmentation Factor, Cr = 0.25

AUTO WIND EXPOSURE WIDTH INFORMATION (Exposure widths are from diaphragm extents)

STORY	DIAPHRAGM	WIDTH	X	Y
TOP ROOF	D1	3.125	2.637	31.513

ROOF	D1	21.550	10.800	24.800
2ND	D1	33.200	16.625	18.050
1ST	D1	37.225	14.612	18.700
LVL3.15	D1	37.225	14.612	18.050
GROUND	D1	37.225	14.612	18.050

AUTO WIND CALCULATION FORMULAS

qs = dynamic pressure in Pa (1 Pa = 1 N/m²)
 $qs = 0.613 Ve^2$

pe = pressure acting on external surface of building
 $pe = qs Cp Ca$

Psurface = net load on building surface
 Asurface = area of building surface
 $Psurface = pe Asurface$

P = overall load on building
 $P = 0.85 * SUM(Psurface) * (1 + Cr)$

AUTO WIND CALCULATION RESULTS

qs = 1241.3250 Pa

AUTO WIND STORY FORCES

STORY	FX	FY	FZ	MX
MY MZ				
TOP ROOF	0.00	17.23	0.00	0.000
0.000 0.000				
ROOF	0.00	70.24	0.00	0.000
0.000 -25.499				
2ND	0.00	172.11	0.00	0.000
0.000 441.811				
1ST	0.00	244.59	0.00	0.000
0.000 -806.867				
LVL3.15	0.00	183.62	0.00	0.000
0.000 -247.961				
GROUND	0.00	133.66	0.00	0.000
0.000 294.340				

ETABS v8.5.7 File:MPS1 Units:KN-m June 1, 2006 9:04 PAGE 7

A U T O S E I S M I C U B C 9 7
 Case: QX

AUTO SEISMIC INPUT DATA

Direction: X + EccY
 Typical Eccentricity = 5%
 Eccentricity Overrides: No

Period Calculation: Program Calculated
 Ct = 0.02 (in feet units)

Top Story: TOP ROOF
Bottom Story: BASE

R = 5.5
I = 1
hn = 21.900 (Building Height)

Soil Profile Type = SC
Z = 0.15
Ca = 0.1800
Cv = 0.2500

AUTO SEISMIC CALCULATION FORMULAS

$$T_a = C_t (h_n^{3/4})$$

If Z >= 0.35 (Zone 4) then: If Tetabs <= 1.30 T_a then T = Tetabs, else T = T_a
If Z < 0.35 (Zone 1, 2 or 3) then: If Tetabs <= 1.40 T_a then T = Tetabs, else T = T_a

$$V = (C_v I W) / (R T) \quad (\text{Eqn. 1})$$

$$V \leq 2.5 C_a I W / R \quad (\text{Eqn. 2})$$

$$V \geq 0.11 C_a I W \quad (\text{Eqn. 3})$$

If T <= 0.7 sec, then $F_t = 0$

If T > 0.7 sec, then $F_t = 0.07 T V \leq 0.25 V$

AUTO SEISMIC CALCULATION RESULTS

$T_a = 0.4936$ sec
T Used = 0.0565 sec
W Used = 36772.09

V (Eqn 1) = 0.8040W
V (Eqn 2) = 0.0818W
V (Eqn 3) = 0.0198W
V (Eqn 4) = 0.0349W

V Used = 0.0818W = 3008.63

F_t Used = 0.00

AUTO SEISMIC STORY FORCES

STORY	FX	FY	FZ	MX
MY MZ				
TOP ROOF	25.77	0.00	0.00	0.000
0.000 -8.602				
ROOF	903.82	0.00	0.00	0.000
0.000 -920.237				
2ND	875.47	0.00	0.00	0.000
0.000 -1700.623				
1ST	742.38	0.00	0.00	0.000
0.000 -1707.655				
LVL3.15	165.61	0.00	0.00	0.000

0.000 -348.672
GROUND 295.57 0.00 0.00 0.000
0.000 -603.774

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AUTO SEISMIC UBC 97
Case: QY

AUTO SEISMIC INPUT DATA

Direction: Y + EccX
Typical Eccentricity = 5%
Eccentricity Overrides: No

Period Calculation: Program Calculated
Ct = 0.02 (in feet units)

Top Story: TOP ROOF
Bottom Story: BASE

R = 5.5
I = 1
hn = 21.900 (Building Height)

Soil Profile Type = SC
Z = 0.15
Ca = 0.1800
Cv = 0.2500

AUTO SEISMIC CALCULATION FORMULAS

$T_a = C_t (h_n)^{(3/4)}$

If $Z \geq 0.35$ (Zone 4) then: If $T_{etabs} \leq 1.30 T_a$ then $T = T_{etabs}$, else $T = T_a$
If $Z < 0.35$ (Zone 1, 2 or 3) then: If $T_{etabs} \leq 1.40 T_a$ then $T = T_{etabs}$, else $T = T_a$

$V = (C_v I W) / (R T)$ (Eqn. 1)
 $V \leq 2.5 C_a I W / R$ (Eqn. 2)
 $V \geq 0.11 C_a I W$ (Eqn. 3)

If $T \leq 0.7$ sec, then $F_t = 0$
If $T > 0.7$ sec, then $F_t = 0.07 T V \leq 0.25 V$

AUTO SEISMIC CALCULATION RESULTS

$T_a = 0.4936$ sec
 $T_{Used} = 0.0741$ sec
 $W_{Used} = 36772.09$

V (Eqn 1) = 0.6134W
 V (Eqn 2) = 0.0818W
 V (Eqn 3) = 0.0198W
 V (Eqn 4) = 0.0349W

$V_{Used} = 0.0818W = 3008.63$

Ft Used = 0.00

AUTO SEISMIC STORY FORCES

STORY MY	MZ	FX	FY	FZ	MX
TOP ROOF		0.00	25.77	0.00	0.000
0.000	4.027				
ROOF		0.00	903.82	0.00	0.000
0.000	896.208				
2ND		0.00	875.47	0.00	0.000
0.000	1298.952				
1ST		0.00	742.38	0.00	0.000
0.000	1303.686				
LVL3.15		0.00	165.61	0.00	0.000
0.000	259.331				
GROUND		0.00	295.57	0.00	0.000
0.000	432.675				

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MASS SOURCE DATA

MASS	LATERAL	LUMP MASS
FROM	MASS ONLY	AT STORIES

Loads	Yes	Yes
-------	-----	-----

MASS SOURCE LOADS

LOAD	MULTIPLIER
------	------------

DEAD	1.0000
------	--------

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DIAPHRAGM MASS DATA

STORY M	DIAPHRAGM Y-M	MASS-X	MASS-Y	MMI	X-
TOP ROOF	D1	14.1298	14.1298	127.0273	
2.637	31.513				
ROOF	D1	581.3307	581.3307	39996.6528	
11.163	24.621				
2ND	D1	714.9151	714.9151	121758.3429	
14.058	22.868				
1ST	D1	863.8875	863.8875	202774.4581	
17.911	22.038				
LVL3.15	D1	393.8818	393.8818	93436.1807	
15.963	17.598				
GROUND	D1	1095.8159	1095.8159	160457.3767	
12.410	21.662				

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ASSEMBLED POINT MASSES

STORY RY	RZ	UX	UY	UZ	RX
TOP ROOF	14.129750	14.129750	0.000000	0.000000	
0.000000	127.027298				
ROOF	586.579289	586.579289	0.000000	0.000000	
0.000000	39996.65283				
2ND	724.919768	724.919768	0.000000	0.000000	
0.000000	121758.34287				
1ST	895.811681	895.811681	0.000000	0.000000	
0.000000	202774.45811				
LVL3.15	401.693425	401.693425	0.000000	0.000000	
0.000000	93436.18073				
GROUND	1126.575374	1126.575374	0.000000	0.000000	
0.000000	160457.37671				
BASE	256.799863	256.799863	0.000000	0.000000	
0.000000	0.000000				
Totals	4006.509149	4006.509149	0.000000	0.000000	
0.000000	618550.0386				

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C E N T E R S O F C U M U L A T I V E M A S S & C E N T E R S
O F R I G I D I T Y

STORY OF RIGIDITY--/ LEVEL X ORDINATE-Y	DIAPHRAGM NAME	/-----CENTER OF MASS-----//--CENTER MASS	ORDINATE-X	ORDINATE-Y	ORDINATE-
TOP ROOF	D1	14.1298	2.637	31.513	
2.649	31.486				
ROOF	D1	595.4604	10.961	24.785	
5.073	26.033				
2ND	D1	1310.3755	12.651	23.739	
12.128	15.828				
1ST	D1	2174.2630	14.741	23.063	
12.790	16.340				
LVL3.15	D1	2568.1448	14.928	22.225	
13.172	16.749				
GROUND	D1	3663.9606	14.175	22.057	
14.188	17.054				

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M O D A L P E R I O D S A N D F R E Q U E N C I E S

MODE CIRCULAR FREQ NUMBER (RADIANS/TIME)	PERIOD (TIME)	FREQUENCY (CYCLES/TIME)
Mode 1	0.27402	3.64937
22.92969		

Mode 2	0.17217	5.80830
36.49465		
Mode 3	0.16138	6.19649
38.93368		
Mode 4	0.13353	7.48905
47.05507		
Mode 5	0.12393	8.06885
50.69810		
Mode 6	0.09680	10.33024
64.90681		
Mode 7	0.07411	13.49371
84.78350		
Mode 8	0.05653	17.68852
111.14027		
Mode 9	0.03778	26.46824
166.30488		
Mode 10	0.03110	32.15238
202.01935		
Mode 11	0.02292	43.63241
274.15054		
Mode 12	0.02048	48.81827
306.73427		

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MODAL PARTICIPATING MASS RATIOS

MODE	X-TRANS	Y-TRANS	Z-TRANS	RX-
ROTN	RY-ROTN	RZ-ROTN		
NUMBER	%MASS <SUM>	%MASS <SUM>	%MASS <SUM>	%MASS
<SUM>	%MASS <SUM>	%MASS <SUM>		
Mode 1	0.01 < 0>	0.70 < 1>	0.00 < 0>	2.30 <
2>	0.02 < 0>	0.46 < 0>		
Mode 2	16.50 < 17>	7.67 < 8>	0.00 < 0>	17.80 <
20>	36.06 < 36>	1.64 < 2>		
Mode 3	8.75 < 25>	9.58 < 18>	0.00 < 0>	19.32 <
39>	17.58 < 54>	8.06 < 10>		
Mode 4	10.65 < 36>	2.68 < 21>	0.00 < 0>	4.43 <
44>	17.40 < 71>	5.26 < 15>		
Mode 5	0.02 < 36>	0.13 < 21>	0.00 < 0>	0.26 <
44>	0.01 < 71>	0.20 < 16>		
Mode 6	0.01 < 36>	22.26 < 43>	0.00 < 0>	34.94 <
79>	0.58 < 72>	22.05 < 38>		
Mode 7	0.30 < 36>	23.83 < 67>	0.00 < 0>	19.02 <
98>	1.80 < 73>	18.29 < 56>		
Mode 8	37.95 < 74>	0.17 < 67>	0.00 < 0>	0.04 <
98>	25.70 < 99>	4.98 < 61>		
Mode 9	0.02 < 74>	9.98 < 77>	0.00 < 0>	1.51
<100>	0.34 < 99>	11.66 < 73>		
Mode 10	3.20 < 77>	3.47 < 80>	0.00 < 0>	0.10
<100>	0.10 <100>	3.65 < 76>		
Mode 11	5.28 < 83>	15.77 < 96>	0.00 < 0>	0.16
<100>	0.11 <100>	0.00 < 76>		
Mode 12	16.68 < 99>	3.15 < 99>	0.00 < 0>	0.05
<100>	0.27 <100>	0.87 < 77>		

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MODAL LOAD PARTICIPATION RATIOS (STATIC AND DYNAMIC RATIOS ARE IN PERCENT)

TYPE	NAME	STATIC	DYNAMIC
Load	DEAD	0.1896	0.0000
Load	LIVE	0.1692	0.0000
Load	QX	99.9980	99.9266
Load	QY	99.9947	99.7923
Load	WX	99.8946	90.3230
Load	WY	99.8519	85.6704
Accel	UX	100.0000	99.3540
Accel	UY	100.0000	99.3866
Accel	UZ	0.0000	0.0000
Accel	RX	99.9966	99.9403
Accel	RY	99.9991	99.9661
Accel	RZ	48.1652	77.1144

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TOTAL REACTIVE FORCES (RECOVERED LOADS) AT ORIGIN

LOAD		FX	FY	FZ	MX
MY	MZ				
DEAD		0.00	0.00	39517.68	870660.849 -
557304.316	0.000				
LIVE		0.00	0.00	6653.27	173126.158 -
88876.337	0.000				
QX		-3008.63	0.00	0.00	0.000 -
39117.005	74052.751				
QY		0.00	-3008.63	0.00	39117.005
0.000	-46268.311				
WX		-848.14	0.00	0.00	0.000 -
8477.103	16447.414				
WY		0.00	-821.45	0.00	7935.932
0.000	-11875.752				
SPEC1		2815.93	808.08	0.00	11308.947
35336.280	68506.695				
SPEC2		857.49	2751.03	0.00	34794.401
12847.565	48811.293				

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S T O R Y F O R C E S

STORY	LOAD		P	VX	VY
T	MX	MY			
TOP ROOF	QX		0.00	-25.77	0.00
820.771	0.000	-87.628			
ROOF	QX		0.00	-929.60	0.00
23994.154	0.000	-3806.015			
2ND	QX		0.00	-1805.07	0.00
45715.190	0.000	-12019.086			
1ST	QX		0.00	-2547.45	0.00
63783.210	0.000	-24756.330			
LVL3.15	QX		0.00	-2713.06	0.00
67046.320	0.000	-29639.835			
GROUND	QX		0.00	-3008.63	0.00
74052.751	0.000	-39117.005			
TOP ROOF	QY		0.00	0.00	-25.77

72.002	87.628	0.000				
ROOF	QY		0.00	0.00	-929.60	-
11057.625	3806.015	0.000				
2ND	QY		0.00	0.00	-1805.07	-
24663.939	12019.086	0.000				
1ST	QY		0.00	0.00	-2547.45	-
39264.584	24756.330	0.000				
LVL3.15	QY		0.00	0.00	-2713.06	-
42167.523	29639.835	0.000				
GROUND	QY		0.00	0.00	-3008.63	-
46268.311	39117.005	0.000				
TOP ROOF	WX		0.00	-36.80	0.00	
1159.644	0.000	-125.118				
ROOF	WX		0.00	-107.52	0.00	
2913.570	0.000	-555.208				
2ND	WX		0.00	-280.93	0.00	
6043.626	0.000	-1833.451				
1ST	WX		0.00	-535.73	0.00	
10808.389	0.000	-4512.115				
LVL3.15	WX		0.00	-718.52	0.00	
14107.678	0.000	-5805.449				
GROUND	WX		0.00	-848.14	0.00	
16447.414	0.000	-8477.103				
TOP ROOF	WY		0.00	0.00	-17.23	-
45.440	58.577	0.000				
ROOF	WY		0.00	0.00	-87.47	-
803.998	408.439	0.000				
2ND	WY		0.00	0.00	-259.57	-
3665.297	1589.500	0.000				
1ST	WY		0.00	0.00	-504.17	-
7239.423	4110.338	0.000				
LVL3.15	WY		0.00	0.00	-687.79	-
9922.575	5348.356	0.000				
GROUND	WY		0.00	0.00	-821.45	-
11875.752	7935.932	0.000				
TOP ROOF	SPEC1		0.00	141.40	5.85	
4460.520	19.874	480.750				
ROOF	SPEC1		0.00	1169.64	487.63	
28874.235	1939.630	4907.335				
2ND	SPEC1		0.00	1681.08	580.40	
44916.105	4519.329	12031.200				
1ST	SPEC1		0.00	2368.18	693.38	
61051.852	7840.882	22908.394				
LVL3.15	SPEC1		0.00	2497.56	706.69	
63220.935	9079.527	27188.401				
GROUND	SPEC1		0.00	2815.93	808.08	
68506.695	11308.947	35336.280				
TOP ROOF	SPEC2		0.00	75.47	38.81	
2409.253	131.957	256.612				
ROOF	SPEC2		0.00	659.19	1069.99	
22109.836	4270.125	2641.378				
2ND	SPEC2		0.00	724.79	1671.95	
26691.594	11493.424	5741.174				
1ST	SPEC2		0.00	746.73	2332.10	
42719.646	22453.505	9191.685				
LVL3.15	SPEC2		0.00	768.75	2459.74	
45293.713	26724.026	10495.701				
GROUND	SPEC2		0.00	857.49	2751.03	
48811.293	34794.401	12847.565				

STORY DRIFTS

STORY	DIRECTION	LOAD	MAX DRIFT
TOP ROOF	X	QX	1/2458
TOP ROOF	Y	QX	1/90467
ROOF	X	QX	1/5333
ROOF	Y	QX	1/16051
2ND	X	QX	1/12478
2ND	Y	QX	1/18341
1ST	X	QX	1/19800
1ST	Y	QX	1/33163
LVL3.15	X	QX	1/25974
LVL3.15	Y	QX	1/45505
GROUND	X	QX	1/63363
GROUND	Y	QX	1/165347
TOP ROOF	X	QY	1/34253
TOP ROOF	Y	QY	1/936
ROOF	X	QY	1/6328
ROOF	Y	QY	1/3307
2ND	X	QY	1/46151
2ND	Y	QY	1/28870
1ST	X	QY	1/133022
1ST	Y	QY	1/33120
LVL3.15	X	QY	1/168859
LVL3.15	Y	QY	1/37238
GROUND	X	QY	1/460888
GROUND	Y	QY	1/73742
TOP ROOF	X	WX	1/2405
ROOF	X	WX	1/46485
ROOF	Y	WX	1/293646
2ND	X	WX	1/91240
2ND	Y	WX	1/170653
1ST	X	WX	1/137668
1ST	Y	WX	1/327186
LVL3.15	X	WX	1/153755
LVL3.15	Y	WX	1/437043
GROUND	X	WX	1/320057
TOP ROOF	X	WY	1/500587
TOP ROOF	Y	WY	1/1464
ROOF	X	WY	1/112652
ROOF	Y	WY	1/45087
2ND	X	WY	1/559000
2ND	Y	WY	1/187627
1ST	Y	WY	1/186711
LVL3.15	Y	WY	1/172949
GROUND	Y	WY	1/294448
TOP ROOF	X	SPEC1	1/622
TOP ROOF	Y	SPEC1	1/4343
ROOF	X	SPEC1	1/3391
ROOF	Y	SPEC1	1/3867
2ND	X	SPEC1	1/11725
2ND	Y	SPEC1	1/15739
1ST	X	SPEC1	1/19983
1ST	Y	SPEC1	1/27964
LVL3.15	X	SPEC1	1/26762
LVL3.15	Y	SPEC1	1/37905
GROUND	X	SPEC1	1/67928
GROUND	Y	SPEC1	1/143526
TOP ROOF	X	SPEC2	1/1058

TOP ROOF	Y	SPEC2	1/655
ROOF	X	SPEC2	1/3210
ROOF	Y	SPEC2	1/2222
2ND	X	SPEC2	1/20881
2ND	Y	SPEC2	1/24857
1ST	X	SPEC2	1/34222
1ST	Y	SPEC2	1/28245
LVL3.15	X	SPEC2	1/42921
LVL3.15	Y	SPEC2	1/33091
GROUND	X	SPEC2	1/94813
GROUND	Y	SPEC2	1/67743

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DISPLACEMENTS AT DIAPHRAGM CENTER OF MASS

STORY	DIAPHRAGM	LOAD	UX	UY	
RZ					
TOP ROOF	D1	QX	0.0025	0.0002	-
0.00001					
ROOF	D1	QX	0.0011	0.0001	-
0.00001					
2ND	D1	QX	0.0004	0.0000	-
0.00002					
1ST	D1	QX	0.0002	0.0000	-
0.00001					
LVL3.15	D1	QX	0.0001	0.0000	
0.00000					
GROUND	D1	QX	0.0000	0.0000	
0.00000					
TOP ROOF	D1	QY	-0.0002	0.0041	
0.00004					
ROOF	D1	QY	0.0001	0.0010	
0.00006					
2ND	D1	QY	0.0000	0.0003	
0.00000					
1ST	D1	QY	0.0000	0.0002	
0.00000					
LVL3.15	D1	QY	0.0000	0.0001	
0.00000					
GROUND	D1	QY	0.0000	0.0000	
0.00000					
TOP ROOF	D1	WX	0.0016	0.0000	
0.00000					
ROOF	D1	WX	0.0002	0.0000	
0.00000					
2ND	D1	WX	0.0001	0.0000	
0.00000					
1ST	D1	WX	0.0000	0.0000	
0.00000					
LVL3.15	D1	WX	0.0000	0.0000	
0.00000					
GROUND	D1	WX	0.0000	0.0000	
0.00000					
TOP ROOF	D1	WY	0.0000	0.0024	
0.00000					
ROOF	D1	WY	0.0000	0.0001	
0.00000					
2ND	D1	WY	0.0000	0.0001	
0.00000					

1ST	D1	WY	0.0000	0.0000
0.00000				
LVL3.15	D1	WY	0.0000	0.0000
0.00000				
GROUND	D1	WY	0.0000	0.0000
0.00000				
TOP ROOF	D1	SPEC1	0.0061	0.0006
0.00008				
ROOF	D1	SPEC1	0.0012	0.0006
0.00005				
2ND	D1	SPEC1	0.0004	0.0001
0.00002				
1ST	D1	SPEC1	0.0002	0.0001
0.00001				
LVL3.15	D1	SPEC1	0.0001	0.0000
0.00000				
GROUND	D1	SPEC1	0.0000	0.0000
0.00000				
TOP ROOF	D1	SPEC2	0.0031	0.0052
0.00020				
ROOF	D1	SPEC2	0.0006	0.0011
0.00010				
2ND	D1	SPEC2	0.0002	0.0003
0.00002				
1ST	D1	SPEC2	0.0001	0.0002
0.00001				
LVL3.15	D1	SPEC2	0.0000	0.0001
0.00000				
GROUND	D1	SPEC2	0.0000	0.0000
0.00000				

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STORY MAXIMUM AND AVERAGE LATERAL DISPLACEMENTS

STORY	LOAD	DIR	MAXIMUM	AVERAGE	RATIO
TOP ROOF	QX	X	0.0025	0.0025	1.020
ROOF	QX	X	0.0012	0.0011	1.080
2ND	QX	X	0.0007	0.0003	2.164
1ST	QX	X	0.0004	0.0002	2.014
LVL3.15	QX	X	0.0001	0.0001	1.775
GROUND	QX	X	0.0000	0.0000	1.508
TOP ROOF	QY	Y	0.0042	0.0041	1.015
ROOF	QY	Y	0.0016	0.0010	1.617
2ND	QY	Y	0.0004	0.0003	1.165
1ST	QY	Y	0.0002	0.0002	1.165
LVL3.15	QY	Y	0.0001	0.0001	1.169
GROUND	QY	Y	0.0000	0.0000	1.084
TOP ROOF	WX	X	0.0016	0.0016	1.003
ROOF	WX	X	0.0002	0.0002	1.097
2ND	WX	X	0.0001	0.0001	1.644
1ST	WX	X	0.0001	0.0000	1.471
LVL3.15	WX	X	0.0000	0.0000	1.303
GROUND	WX	X	0.0000	0.0000	1.165
TOP ROOF	WY	Y	0.0024	0.0024	1.001
ROOF	WY	Y	0.0002	0.0001	1.329
2ND	WY	Y	0.0001	0.0001	1.120
1ST	WY	Y	0.0000	0.0000	1.101
LVL3.15	WY	Y	0.0000	0.0000	1.094
GROUND	WY	Y	0.0000	0.0000	1.030

TOP ROOF	SPEC1	X	0.0062	0.0061	1.011
ROOF	SPEC1	X	0.0014	0.0013	1.054
2ND	SPEC1	X	0.0007	0.0004	1.664
1ST	SPEC1	X	0.0004	0.0002	1.560
LVL3.15	SPEC1	X	0.0001	0.0001	1.466
GROUND	SPEC1	X	0.0000	0.0000	1.354
TOP ROOF	SPEC1	Y	0.0006	0.0006	1.055
ROOF	SPEC1	Y	0.0011	0.0007	1.571
2ND	SPEC1	Y	0.0005	0.0003	1.701
1ST	SPEC1	Y	0.0002	0.0002	1.634
LVL3.15	SPEC1	Y	0.0001	0.0000	1.548
GROUND	SPEC1	Y	0.0000	0.0000	1.361
TOP ROOF	SPEC2	X	0.0035	0.0031	1.115
ROOF	SPEC2	X	0.0013	0.0009	1.457
2ND	SPEC2	X	0.0004	0.0002	1.577
1ST	SPEC2	X	0.0002	0.0001	1.620
LVL3.15	SPEC2	X	0.0001	0.0000	1.575
GROUND	SPEC2	X	0.0000	0.0000	1.454
TOP ROOF	SPEC2	Y	0.0052	0.0052	1.001
ROOF	SPEC2	Y	0.0020	0.0013	1.534
2ND	SPEC2	Y	0.0004	0.0004	1.191
1ST	SPEC2	Y	0.0003	0.0002	1.193
LVL3.15	SPEC2	Y	0.0001	0.0001	1.146
GROUND	SPEC2	Y	0.0000	0.0000	1.079